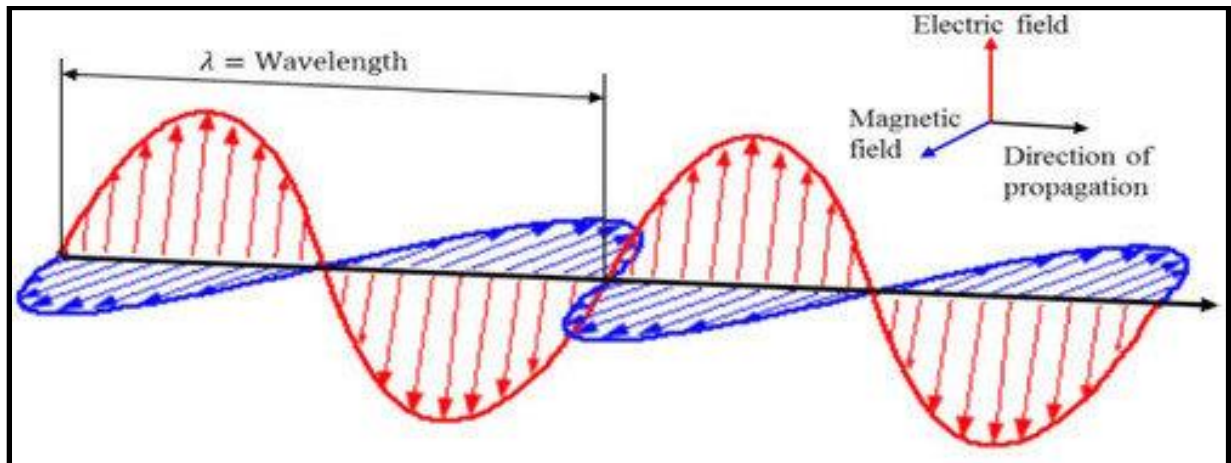


RADIO WAVE

PROPERTIES OF RADIO WAVE:

- ✓ Electromagnetic (EM) Wave is a radio wave which has two components Electric and Magnetic; both components are perpendicular to each other:

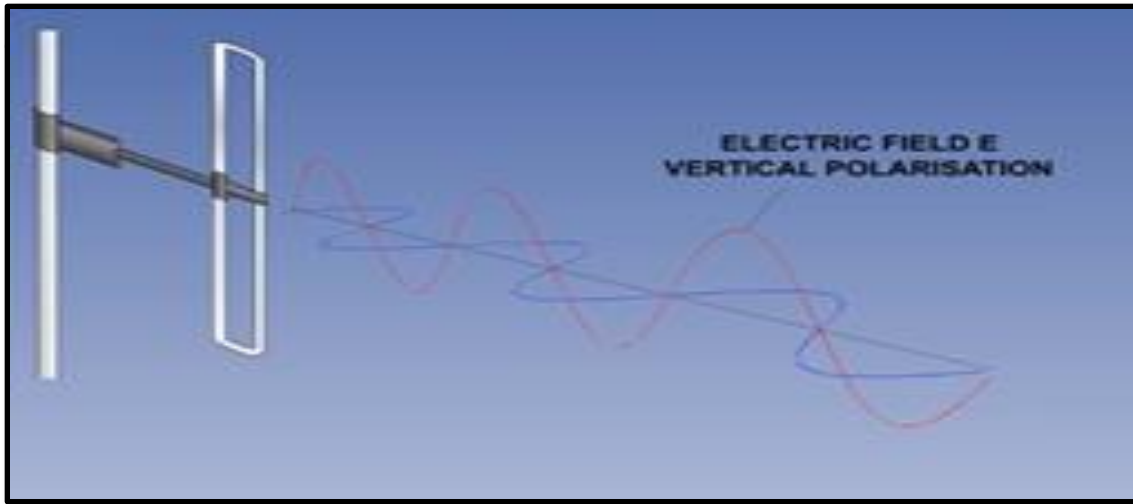


- ✓ Radio wave gets reflected, refracted (Change in medium), diffracted (Bending of beam), attenuated. (loss of energy)
- ✓ At higher frequencies most obstacles will cause reflection or absorption.
- ✓ If the medium through which the radio wave is travelling changes, wave will tend to curve towards high density medium.

PLANE OF POLARISATION-

- ✓ Polarization of radio wave is defined as per plane of electric field.
- ✓ Direction of wave oscillating electric field perpendicular to the direction of motion

Example: -



and
ane.

- ☞ **Reception of a signal is best when Tx & Rx are in same plane of electric field.** (Vertical Tx & Vertically polarized Rx will have best reception same goes to horizontally polarized Tx & Rx)
- ☞ Speed of EM wave is same as speed of light in vacuum.

186000 SM/SEC
162000 NM/SEC
 3×10^8 M/SEC
 3×10^5 KM/SEC
300 million METER/SEC

TOP CREW AVIATION

- ☞ Speed varies depending on density of medium they are travelling through faster in less dense medium.

DEFINITIONS:

FREQUENCY- Number of cycles per second expressed in hertz.

HERTZ- One hertz is one cycle per second. The number of cycles per second is expressed in hertz.

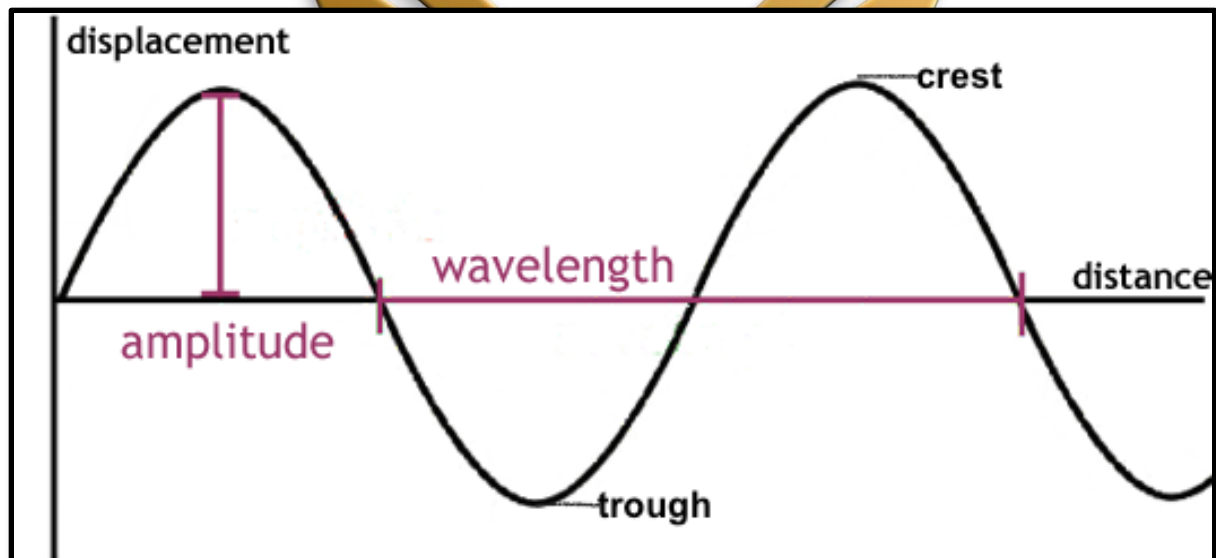
WAVELENGTH (λ): Distance travelled by the radio wave during one cycle or distance between successive crests or successive troughs.

AMPLITUDE: amplitude of a wave is the max displacement (or max value) it attains from the mean value.

$$1 \text{ Cycle/SEC} = 1 \text{ Hz}$$

$$1000 \text{ Cycle/SEC} = 1 \text{ KHZ} = 10^3 \text{ Hz}$$

$$1000 \text{ KHz} = 1 \text{ MHz} = 10^6 \text{ Hz}$$



RELATIONSHIP BETWEEN WAVE LENGTH AND FREQUENCY:

$$C = f \lambda$$

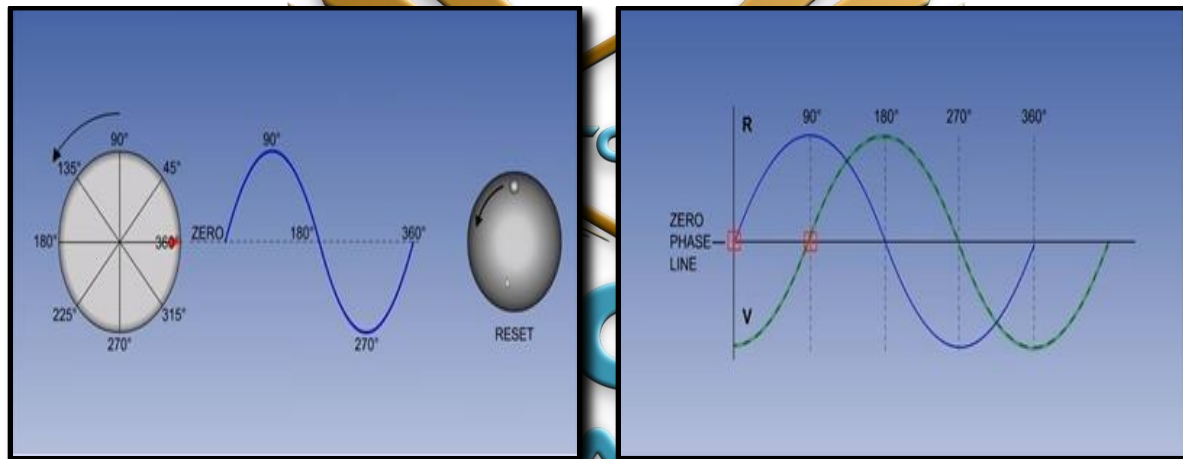
$$C = 3 \times 10^8 \text{ m/s}$$

$$\text{Wave length (distance)} = \frac{\text{Speed of radio wave}}{\text{Frequency (1/time)}}$$

PHASE- Position of a radio wave in terms of angle at any particular time

To compare two radio waves, they need to have in same frequency.

$$\text{Phase diff} = \text{Phase of (Reference Wave - Variable Wave)}$$



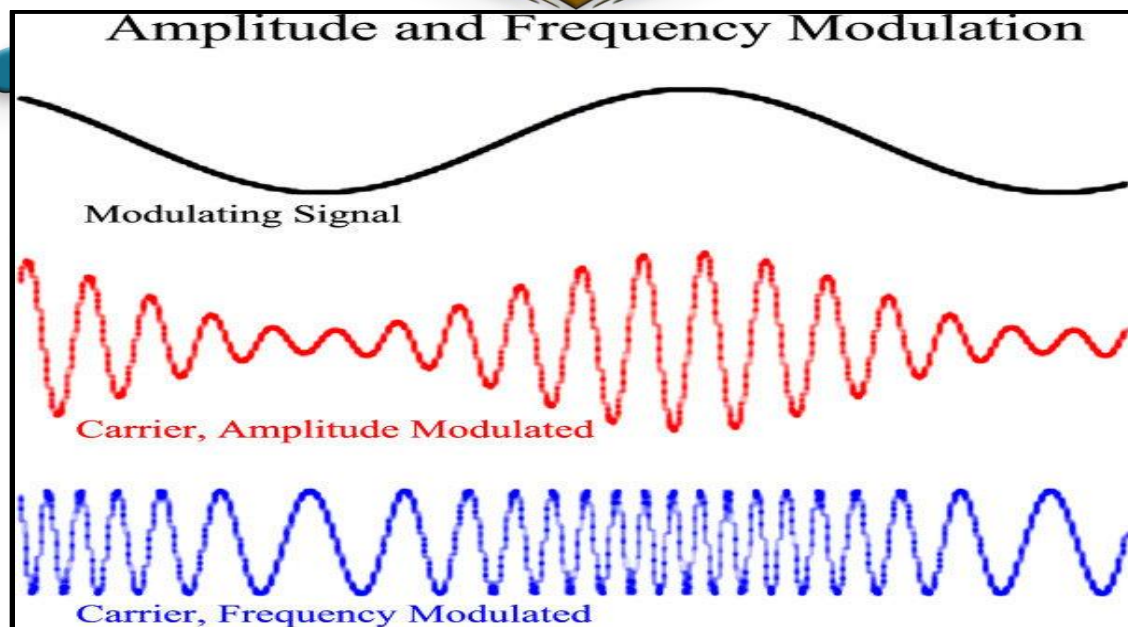
Problem - Phase of reference wave is 180 and phase of variable wave is 230 find phase difference?

MODULATION:

- ✓ Process of impressing intelligence on a radio wave is called modulation. In this the radio wave is called the carrier wave and the information impressed on the radio wave is called the modulation wave.

Types of Modulation:

- A. **Keying**: On and off or dots and dashes. Like in telegraphy
- B. **Amplitude Modulation**: When amplitude of the modulating signal is varied, in confirmation with the amplitude of the audio modulating signal, keeping the **FREQUENCY CONSTANT**.
- C. **Frequency Modulation**: When frequency of the carrier is varied in confirmation with the amplitude of the radio modulating signal, keeping the **AMPLITUDE CONSTANT**.
- D. **PULSE MODULATION**: the Modulating pulse amplitude modulates the carrier, giving it shape of a pulse. Used in radar.
- E. **PHASE MODULATION**: Used in VOR



SIDE BANDS:

- ✓ These are additional frequencies when a carrier wave is modulated by a freq lower than itself (normally done). So, there always 3 result of modulation
- A. Carrier freq
 - B. Carrier freq+ audio freq
 - C. Carrier freq – audio freq

DESIGNATION AND CLASSIFICATION OF EMISSIONS:

3 symbols are used:

EXAMPLES:

- ☞ NDB - NON A1A & NON A2A,
- ☞ HF – J3E
- ☞ VHF – A3E
- ☞ VDF – A3E
- ☞ ILS – A8W
- ☞ VOR – A9W
- ☞ DME – PON

**OTHER BAND:**

- | | |
|-------------------|------------------|
| 1. L Band: | 1000 – 2000 MHZ |
| 2. S Band: | 2000 – 4000 MHZ |
| 3. C Band: | 4000 – 8000 MHZ |
| 4. X Band: | 8000 – 12500 MHZ |



EMISSION CHARACTERISTICS					
First Symbol		Second Symbol		Third Symbol	
Type of modulation of the main carrier		Nature of signals modulating the main carrier		Type of information transmitted	
N	Emissions of an unmodulated carrier	0	No modulating signal	N	No information transmitted
A	Amplitude modulation - Double sideband	1	Single channel containing quantised or digital information without the use of a modulating sub-carrier, excluding time division multiplex	A	Telegraphy for aural reception
H	Amplitude modulation - Single sideband, full carrier	2	Single channel containing quantised or digital information with the use of a modulating sub-carrier, excluding time division multiplex	B	Telegraphy for automatic reception
J	Amplitude modulation - Single sideband – suppressed carrier	3	Single channel containing analogue information	C	Facsimile
				D	Data transmission, telemetry, telecommand
F	Frequency modulation	7	Two or more channels containing quantised or digital information	E	Telephony, including sound broadcasting
G	Phase modulation	8	Two or more channels containing analogue information	F	Television (video)
		9	Composite system with one or more channels containing quantised or digital information, together with one or more channels containing analogue information	W	Combinations of the above
P	Sequence of unmodulated pulses				
K	Sequence of pulses modulated in amplitude	X	Cases not otherwise covered	X	Cases not otherwise covered

Frequency Band	Frequencies	Wavelengths	Propagation Characteristics	Uses
Very low Frequency (VLF)	3-30 KHz	100 – 10 Km	Ducting	No Civil Uses
Low Frequency (LF)	30 – 300 KHz	10 -1 km	Surface wave (Sky wave)	ADF, DECCA, LORAN C
Medium Frequency (MF)	300 -3000 KHz	1000 -100 m	Surface wave (Sky wave)	ADF, Long range communications
High frequency (HF)	3 – 30 MHz	100 -10 m	Sky Wave (Surface wave)	Long rang communications
Very high frequency (VHF)	30 -300 MHz	10 – 1 m	Space wave	Short range communication VDF, VOR, ILS localizer, Marker Beacons
Ultra-high frequency (UHF)	300 -3000 MHz	100 – 10 cm	Space Wave	ILS Glide path, DME, SSR, SATCOM, SATNAV, long range radars

Super high frequency(SHF)	3 – 30 GHz	10 - 1 cm	Space wave	MLS, RAD ALT, AWR, Doppler Radars, Short
Extremely high frequency (EHF)	30 – 300 GHz	10 – 1 mm	Space Wave	No civil uses



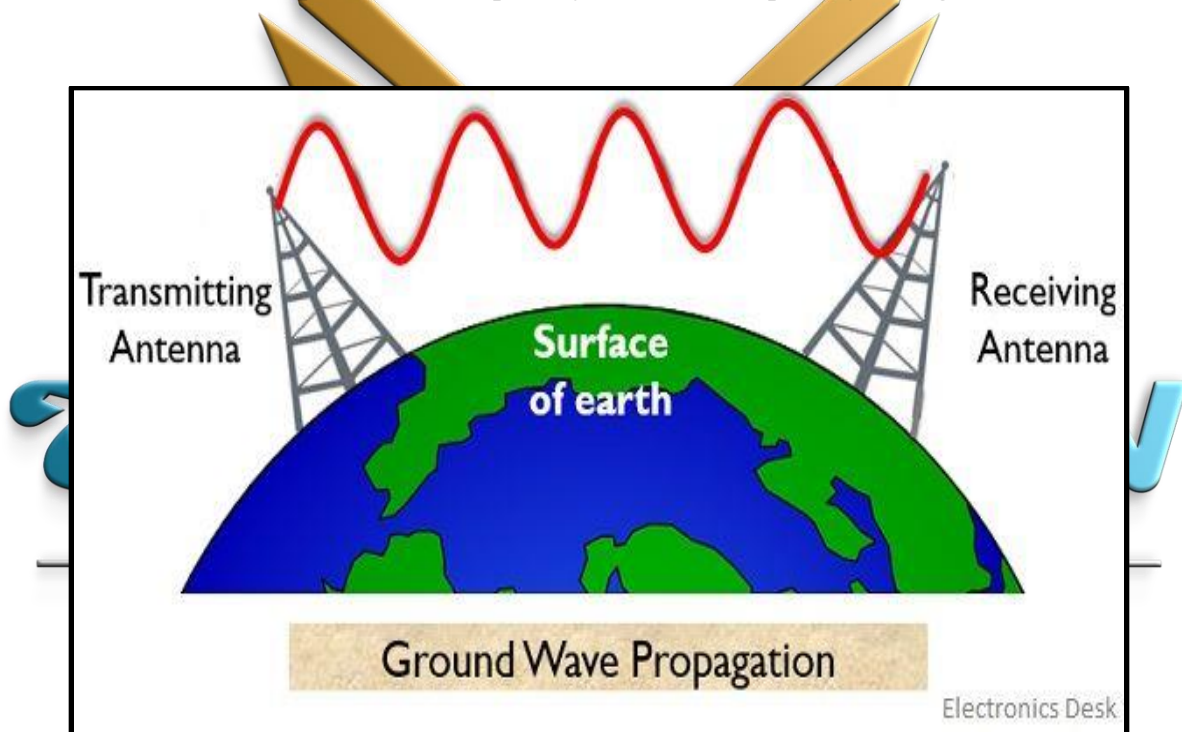
(<20 KHz)

NON-IONOSPHERIC PROPAGATION

SURFACE WAVES

✓ Which wave reaches receiver by travelling along the surface of the earth and bend around the surface of the earth and are formed in **VLF, LF, and MF**.

- ☞ NDB follows surface wave propagation.
- ☞ Frequency increases ionospheric attenuation increases
- ☞ Diffraction: Depends on frequency. Lower frequency = higher diffraction



FACTORS AFFECTING SURFACE WAVE RANGE

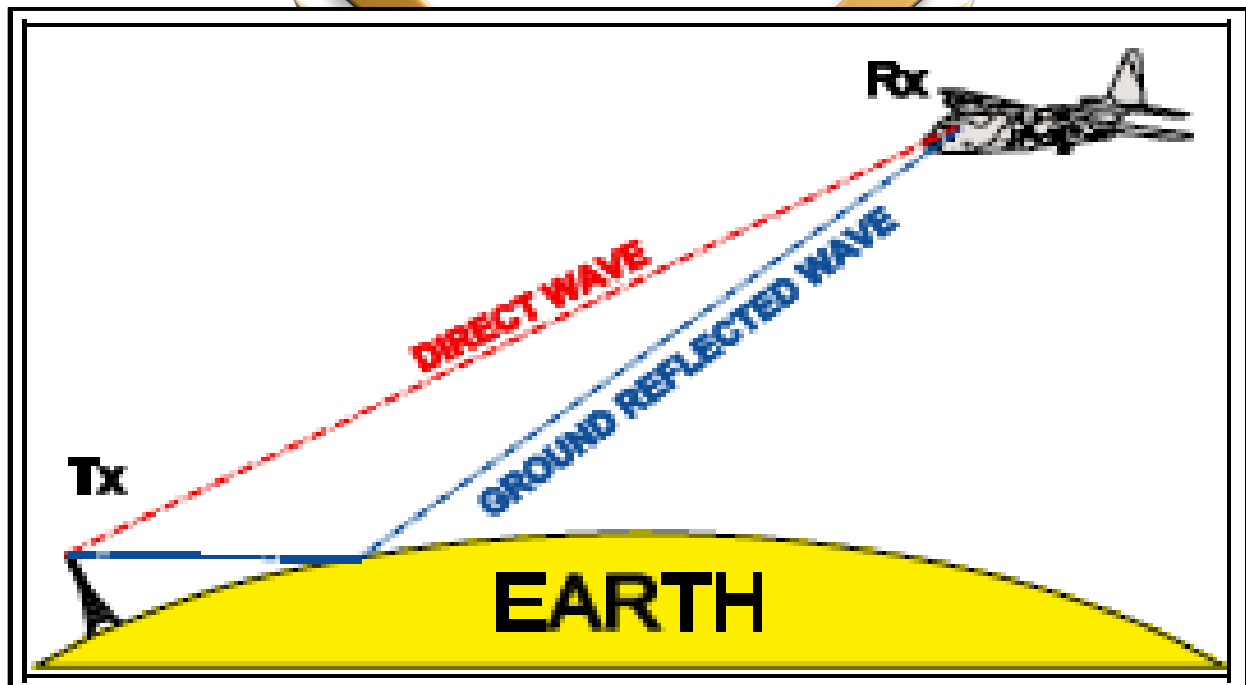
- ☞ Frequency decreases range increases
- ☞ Type of surface: - Range over sea is more than range over land
- ☞ Power: - To double the range power has to increase by four times.

$$\text{RANGE OVER LAND IN NM} = 2\sqrt{\text{Power}}$$

$$\text{RANGE OVER SEA IN NM} = 3\sqrt{\text{Power}}$$

SPACE WAVES – (Line of sight propagation) VHF and above

SPACE WAVE = DIRECT WAVE + GROUND REFLECTED



$$\text{RANGE} = 1.25(\sqrt{HT_x} + \sqrt{HR_x})$$

DIRECT WAVES- VHF Band and higher frequencies – the wave reaching a receiver in straight line (of sight) are called direct waves.

REFLECTED WAVES- Waves reflected by an object.

Ground waves = Space waves + Surface waves

or

Direct waves + Ground reflected waves + Surface waves

IONOSPHERIC PROPAGATION:-

SKY WAVES –

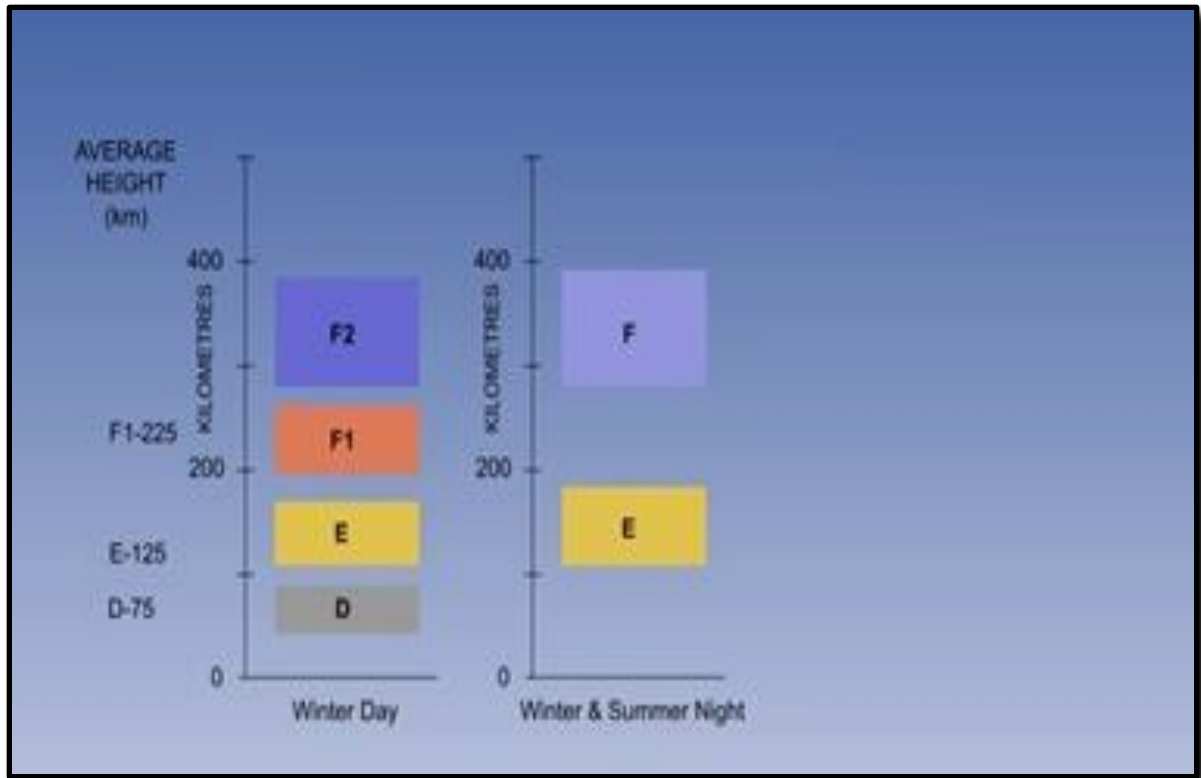
- ✓ The waves refracted by the ionosphere are called sky waves.
- ✓ Used in long range (HF).
- ✓ Frequency decreases ionospheric attenuation increases.

The Ionosphere causes electrons to be emitted. These free electrons from the ionosphere because the absorption of radiation is uneven; the ionosphere exists in 3 layers.

D Layer: 50 – 100 kms (avg 75 kms)

E Layer: 100 – 150 kms (avg 125 kms) also called **Kennelly Heaviside layer**.

F Layer: 150 – 350kms (avg 225kms) also called **Appleton layer**. This often exists as F1 and F2 Layer. Have max variations.



VARIATION DENSITY (OF ELECTRON) DEPENDS UPON

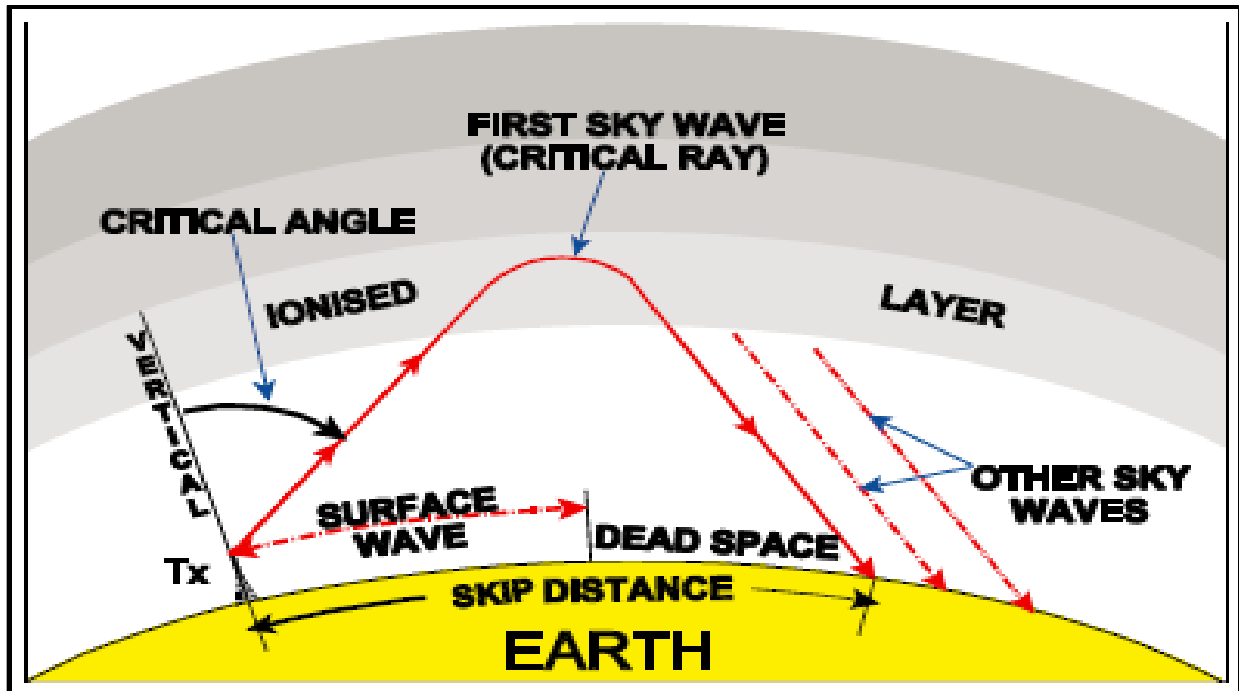
A. Diurnal activity: daytime more ionic density.

B. At night D Layer Vanishes. E layer intensity reduces and it rises. F1 and F2 Merge to form one.

C. Seasonal activity: max activity when sun closest.

D. Freq In use: Lower the freq, greater the attenuation (note opposite of Surface waves)

	CRITICAL ANGLE	SKIP DISTANCE	DEAD SPACE
↑ In Frequency	Must be increased	↑ as wave has to travel deeper	Will ↑ from both ends
↑ In power	Remains same	Remains same	Will ↓ from Tx End



Sky Waves: Conditions for refraction

Critical angle: a signal entering ionosphere will undergo refraction and may bend enough to return back to earth. Angle that this wave makes with the normal at the transmission point is called critical angle. And the returned wave is called critical ray or first sky return.

Freq in use: high freq requires higher electron density to refract it. D layer refracts low freq around 500 KHz. E layer upto 2 mhz. F layers from 2-30 mhz. above 30 MHz (VHF) no refraction.

Skip distance: distance between transmitter and the point on the surface where the first sky wave returns.

Dead space: the area between the limit of surface wave and point of reception of first sky wave.

Multi-hop refraction: if returning signal is strong, it will be reflected back to ionosphere again. This may happen several times.

Fading (fluctuation): two signals from same transmitter arriving at same receiver following different path may be out of phase causing fading.

Attenuation:

- ✓ Is caused as wave comes in contact with surface at induce currents, losing energy and slowing down at the bottom. This causes the waves form to tilt forward.

Attenuation depends upon:

- A. Type of surface
- B. Freq in use
- C. Polarization of radio wave: vertically polarized waves have min attenuation

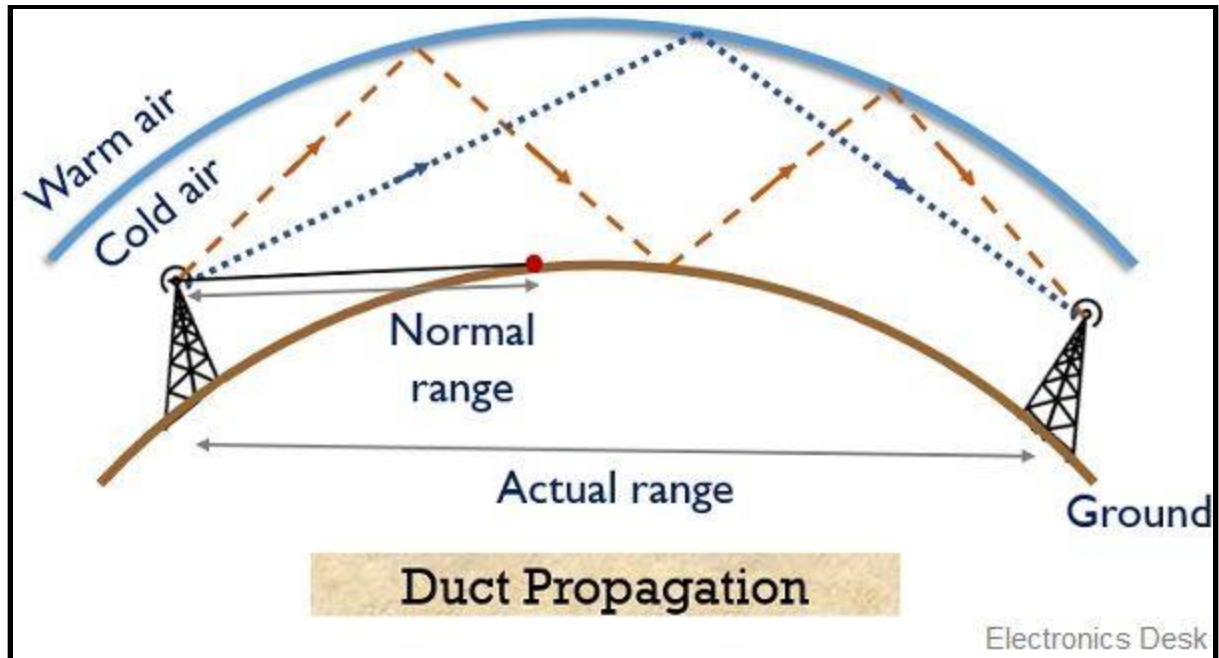
VHF RANGE FORMULA

Calculated by formula:



DUCT PROPAGATION OR SUPER REFRACTION

- ✓ When inversion layer is present. This effects relative permittivity of the layer. This gives greater refraction and causes surface wave ranges to exceed tangential horizon. Meteorological condition required:
 - A. Warm dry air blowing over cool sea.
 - B. Subsidence.
 - C. Pronounced radiation cooling.



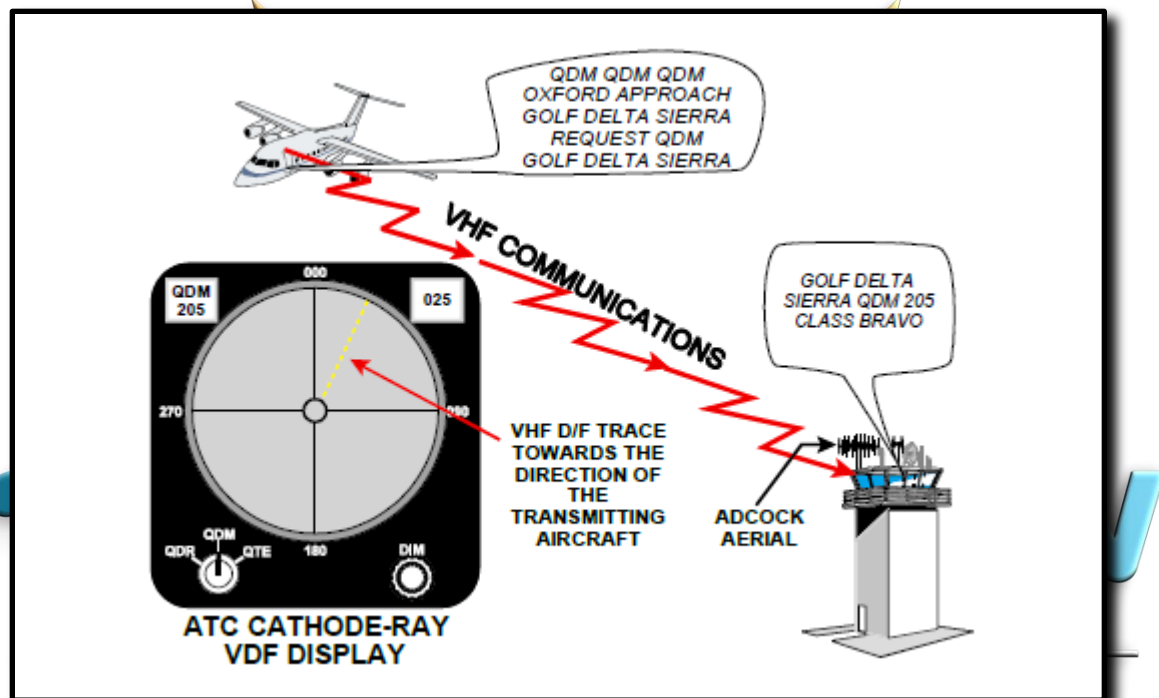
VHF DIRECTION FINDING (VDF)

Principle– Adcock aerial

Frequency – 118-137 MHz

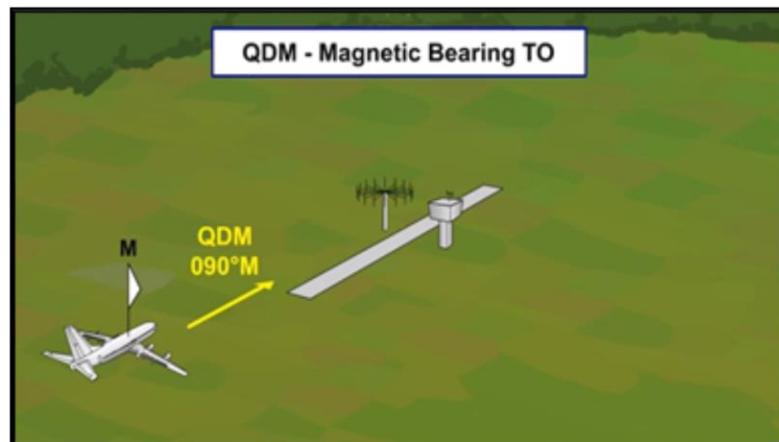
Band- VHF

Emission Code- A3E

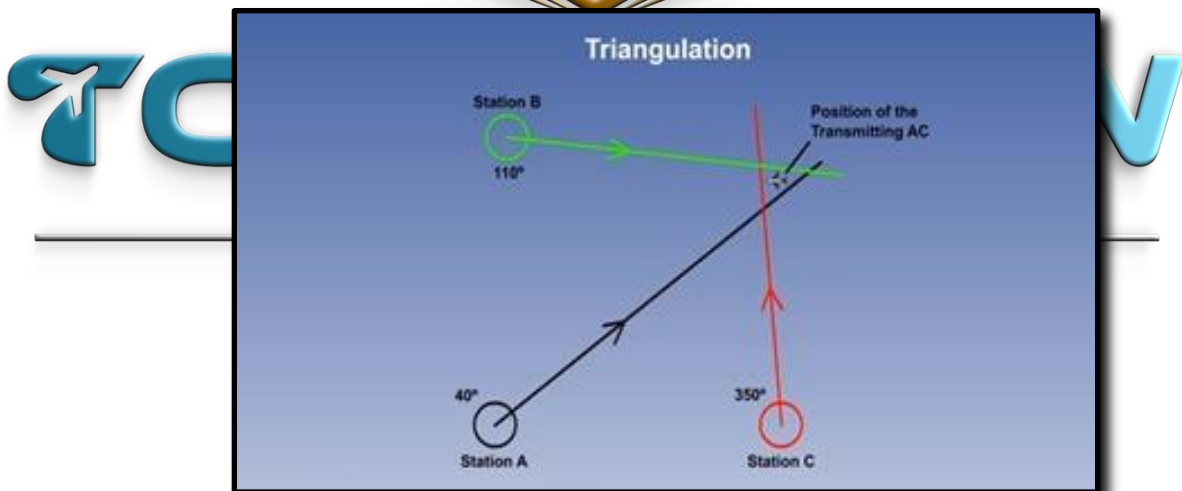


It is a ground station equipped to take bearing of an aircraft. When an aircraft transmits. It can give true or magnetic headings or bearings.

Q CODES



- ✓ Only VHT RT is required in cockpit for VDF.
- ✓ Emergency frequency 121.5 is use to provide position fixing by means of triangulation.



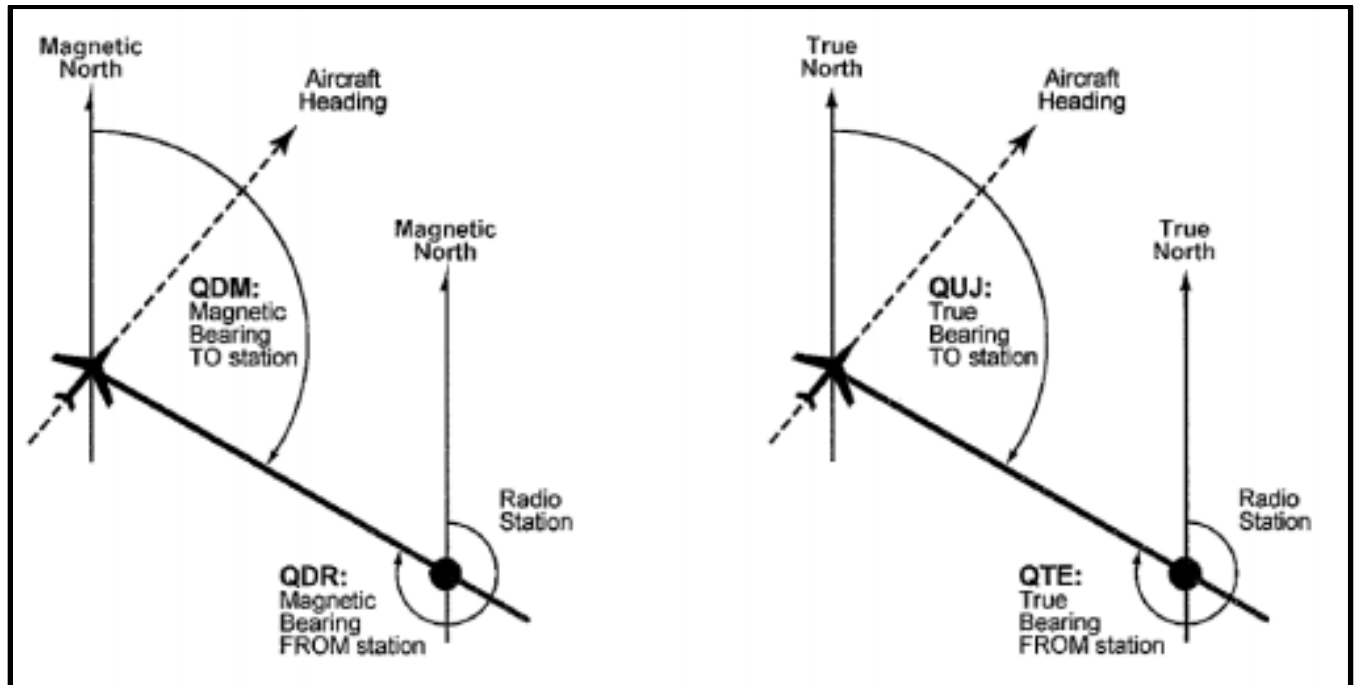
Bearing:

QDM- Magnetic Heading TO station in zero wind.

QDR- Magnetic Bearing FROM the station.

QTE- True Bearing FROM the station.

QUJ- True Bearing TO the station.



Uses: check track, position line, homing, Let-downs.

Class: A= $\pm 2^\circ$, B= $\pm 5^\circ$, C= $\pm 10^\circ$, D= $> 10^\circ$

Principle: Ground equipment- Adcock aerial, CRT.

Range: Line of sight, power of transmitter, intervening High ground, atmospheric conditions (ducting).

Accuracy: propagation error, site error, aircraft attitude, Overhead station, fading due to multi-path Signals.

Emergency service: position fixing by auto- triangulation.

Let- down service:

- ✓ QDM procedure- pilot interpreted.
- ✓ QGH procedure – controller interpreted.

COMMUNICATIONS

CHOICE OF FREQ FOR LONG RANGE COMM:

- ✓ From the given frequency band only HF and VHF can be considered. Of this VHF being LOS, HF is a better option.
- ✓ If frequencies bands are lower the HF there are problems of **aerial** and high **static noise**.

HF COMMUNICATION:

- ✓ **CRITICAL FREQUENCY:** the critical frequency is that frequency at which just starts to escape vertically to the ionosphere.

NIGHT COMM USING HF:

- ✓ At night the following happens to the sky wave:
 - A. Ionosphere goes up increasing refraction ht. so skip distance increases.
 - B. Due to reconstruction of electrons, the electron density reduces. So same wave has to travel deeper (higher) to be refracted enough to return.
- ✓ For these reasons the night frequency is lowered (generally to half of day), as this adjust skip distance because: Lower frequency refract from lower layers and requires a smaller critical angle.

OTHER POINTS ABOUT HF COMM.:

- A. frequency band 2-22 MHz (practically up to 18 MHz)
- B. Transmission is AM
- C. SSB may be used to economic in power and channel space

FACTORS AFFECTING HF RANGE:

- A. Transmission power (More Power more range)
- B. Time of day (During Night range would be less)
- C. Season of the year (Less range would be in Winters)
- D. Geographic location (More Range would be near Equator)
- E. Frequency in use (Lower the frequency greater would be the range)

CHOICE OF FREQ FOR SHORT RANGE COMMUNICATION-

(VHF as it is static free.)

- A. This provides range of about 80 nm at 5000 ft and 200 nm at 20,000 ft
- B. Frequency band is from 117.975 MHz to 137Mhz
- C. Transmission is AM
- D. Type emission is A3E

FACTORS AFFECTING VHF RANGE:

- 
- 
- 1. Height of transmitter
 - 2. Height of receiver
 - 3. Transmission power
 - 4. Obstacle near the transmission site affecting propagation
 - 5. Fading caused as aircraft may receive direct and ground reflected waves.

SELCAL (SELECTIVE CALLING SYSTEM):

- ✓ Relieves pilot of the tiresome task of maintaining a continuous listening, watching. When the ground station wants to contact an aircraft maintaining SELCAL, it transmits two coded tone pulses.
- ✓ This activates a cockpit call system by flashing a lamp or ringing a bell.

ACARS (AIRCRAFT COMMUNICATION ADDRESSING AND REPORTING SYSTEM):

- ✓ It is a common data link between the airline ground operations base and the aircraft.

SATCOM:

- ✓ Satellite communication through INMARSAT(international maritime satellite organization). These are a group of geostationary satellites provided by international maritime satellite organization. Provides comm. Between 80° N and 80° S. transmission is L band.



NDB (Non-Directional Beacon), ADF (Automatic Direction finder)

PRINCIPLE- Bearing by loop theory

FREQUENCY-

- ✓ 190-450 KHz (**NDB**)
- ✓ 190-1750 KHz (**ADF**)

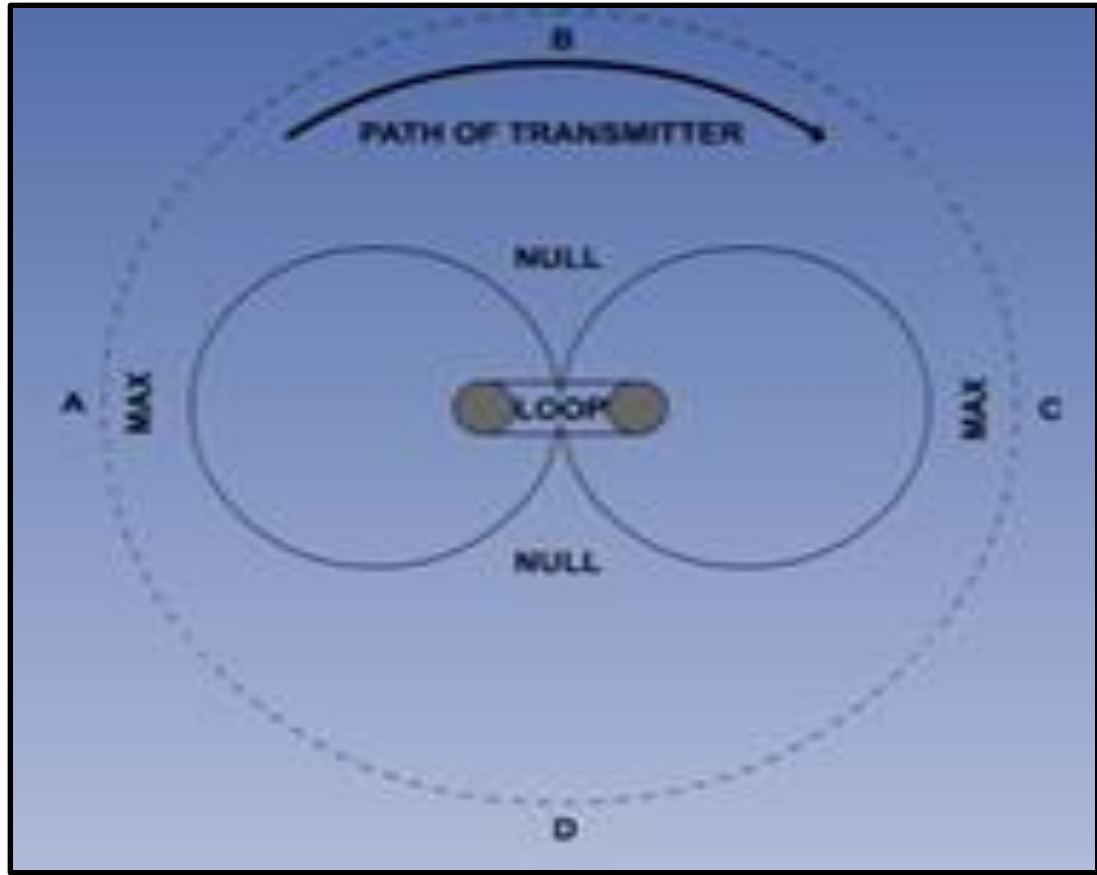
EMISSION CODE- N0N A1A, N0NA2A

BAND- Upper LF & Lower MF

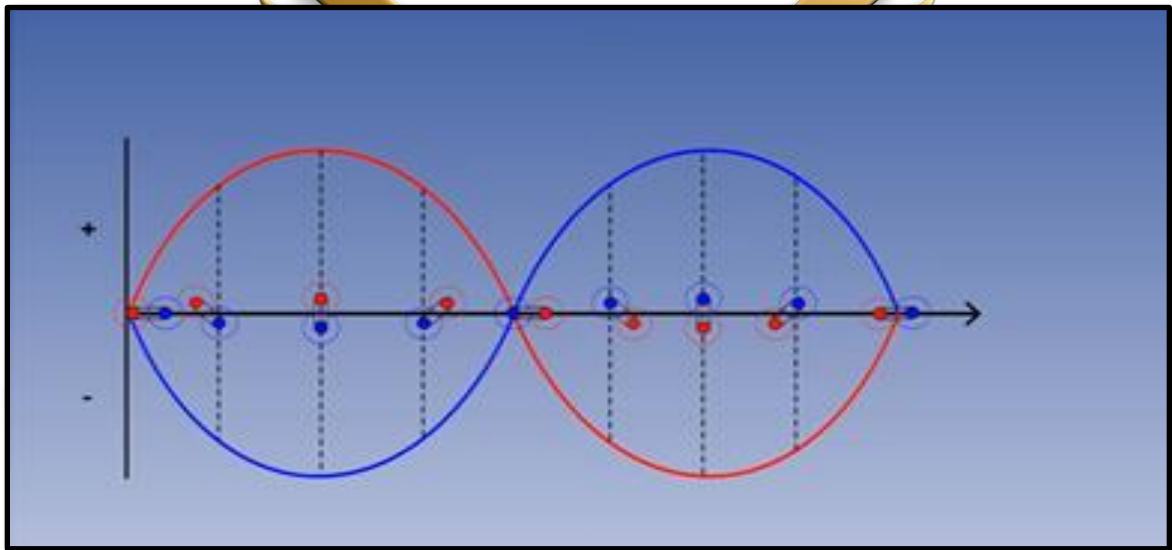
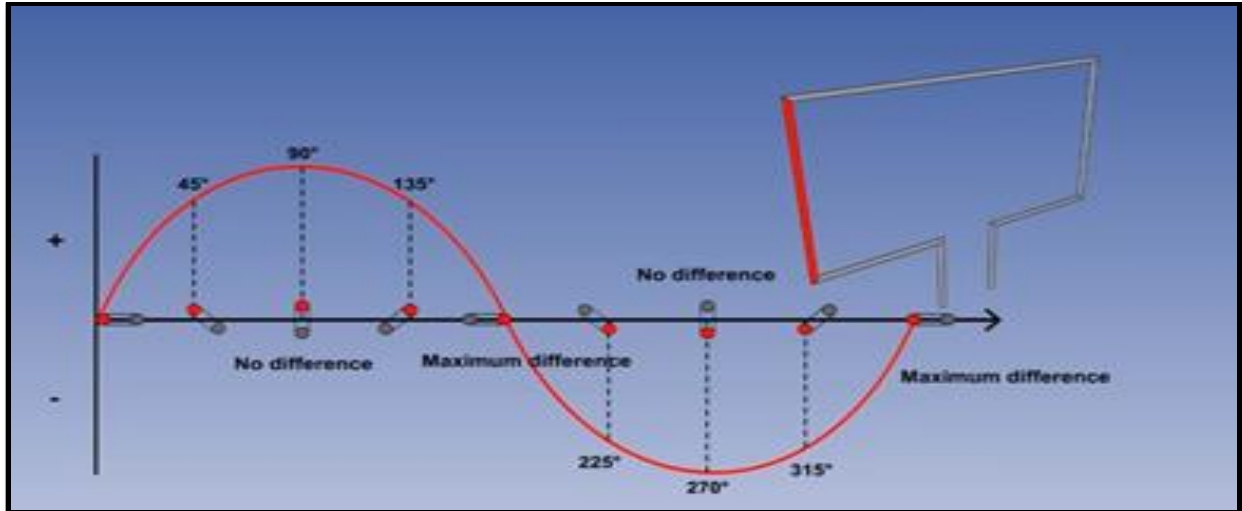
- ✓ NDB is an Omni directional beacon on ground transmitting vertically polarized signals
- ✓ NDB follows surface wave propagation.
- ✓ NDB – transmitter on Ground
- ✓ Accuracy: +/-2 deg up to 70 nm and +/-5 deg beyond that

WORKING OF ADF

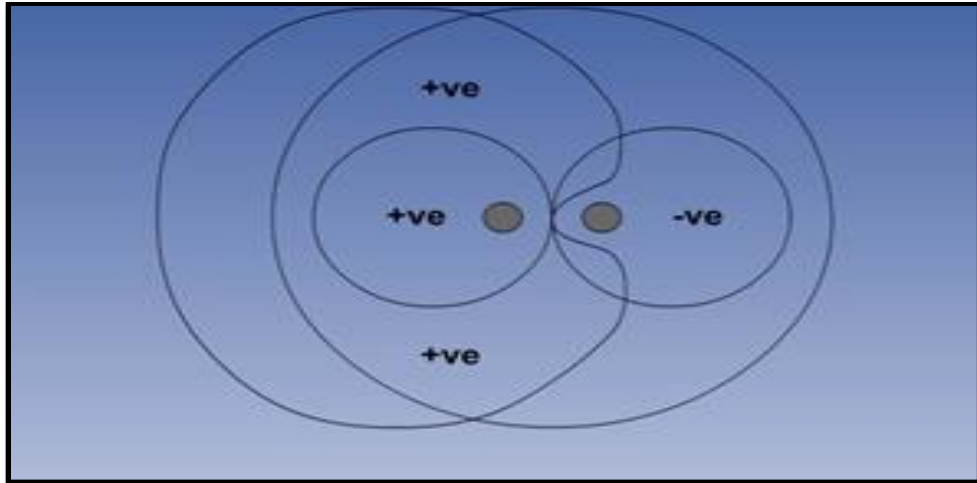
- ✓ ADF antenna has two aerials
 1. LOOP ANTENNA
 2. SENSE ANTENNA
- ✓ Two minima's and two maxima are causing 180-degree ambiguity
- ✓ 180-degree ambiguity is solved by a SENSE AREIAL, polar diagram of which is a circle.



- ✓ When plane of loop is parallel to direction of incoming signal max phase difference and max current.
- ✓ When at right angle both members will receive incoming signal together at the same phase so the current flow is nil.
- ✓ Polar diagram of signal strength produced by loop is a Figure of eight



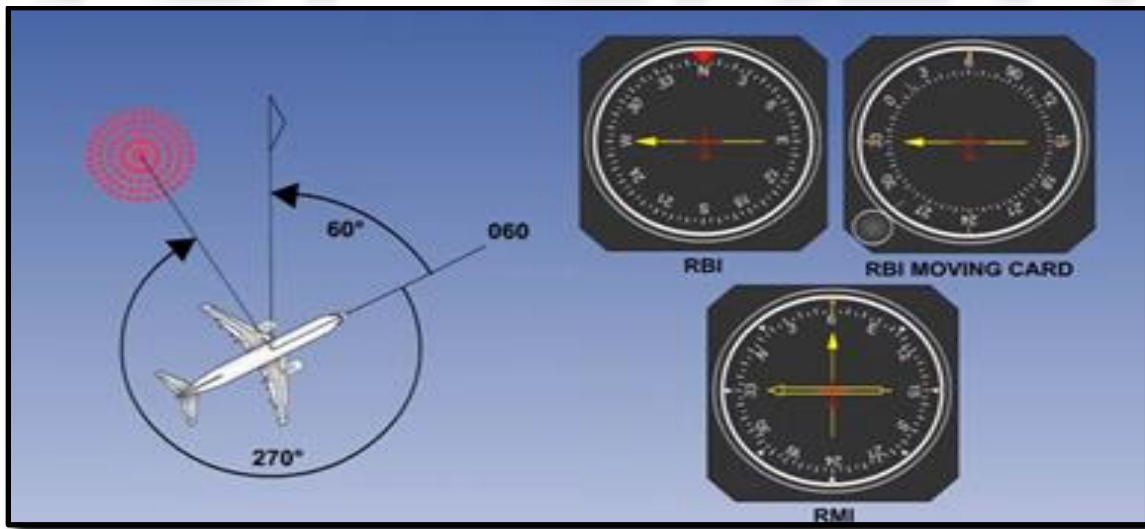
- ✓ The combined polar diagram at ADF antenna is a **CARDIOID** (It has a well-defined null)

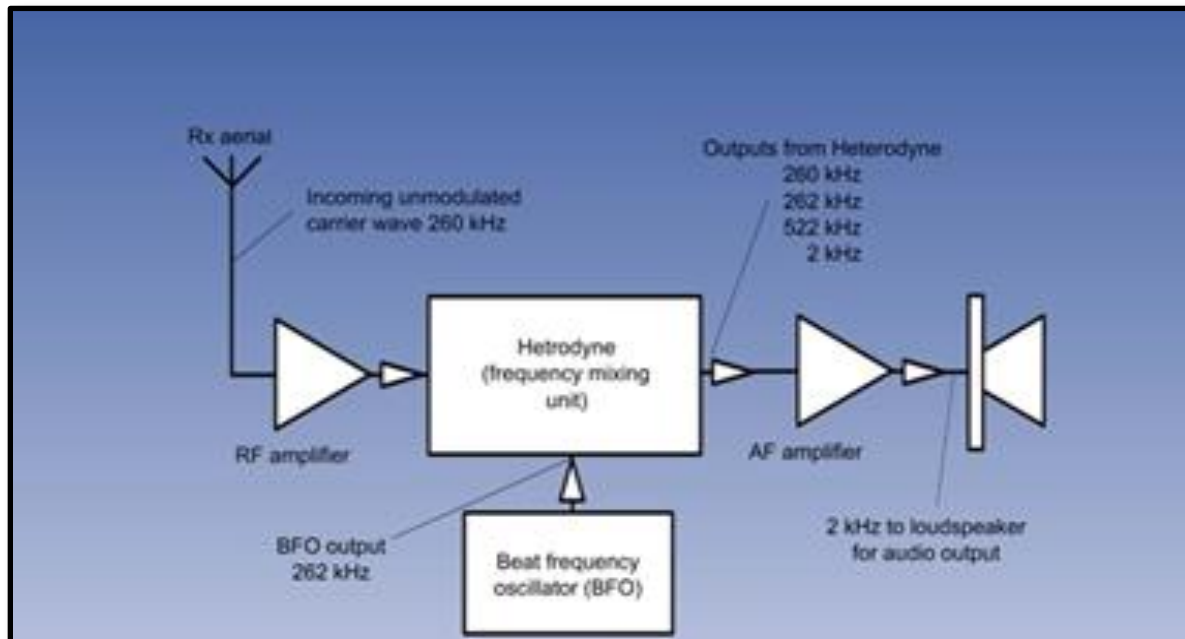


- ✓ POLAR DIAGRAM shows the position of points around antenna where the strength of the signal has reduced to a certain level, often half its original strength.
- ✓ Vertically polarized radio wave.

ADF is airborne equipment. That can tune in to signals from NDB and measure direction of the incoming signal.

ADF measure direction in relation to the nose of the aircraft. Hence, these are **RELATIVE BEARINGS**.



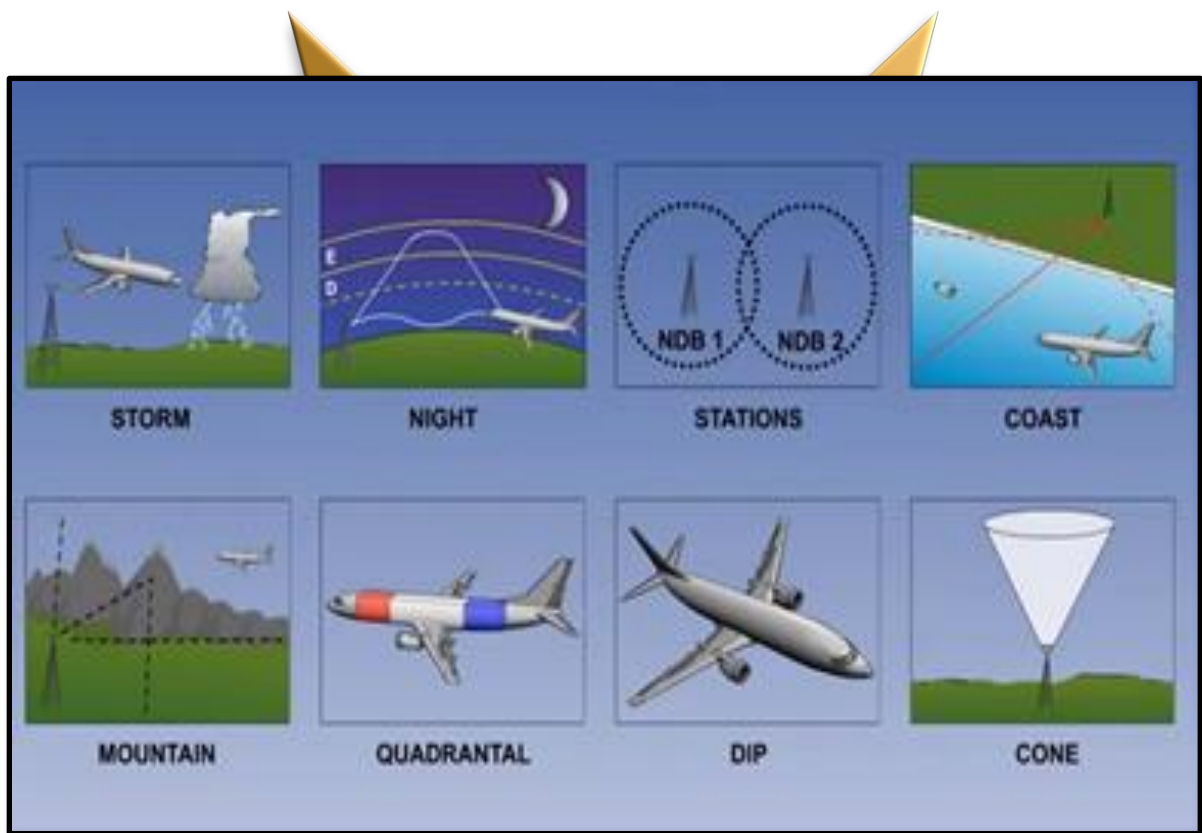
BFO (Beat Frequency Oscillator):

- ✓ It makes the NDB identification old Morse code audible to human ears.
- ✓ The signal is NON, so that carrier is un-modulated and hence carries no intelligence. The frequency is so high that it is beyond the human hearing range.
- ✓ With use of BFO signal is made audible. This is done by the use of an oscillator that produces a frequency slightly different from the transmitted signal.
- ✓ The difference of these two is fed into the earphone and this is an audible freq. for e.g. if the incoming signal is at 500 kHz, the BFO will oscillate at 499 kHz. On subtraction we get 1 kHz which is an audible freq.

BFO is used as follow:

TUNING	IDENT	MONITORING
N0N A1AON	ON	ON
N0N A2AON	OFF	OFF

FACTORS AFFECTING ACCURACY OF ADF:

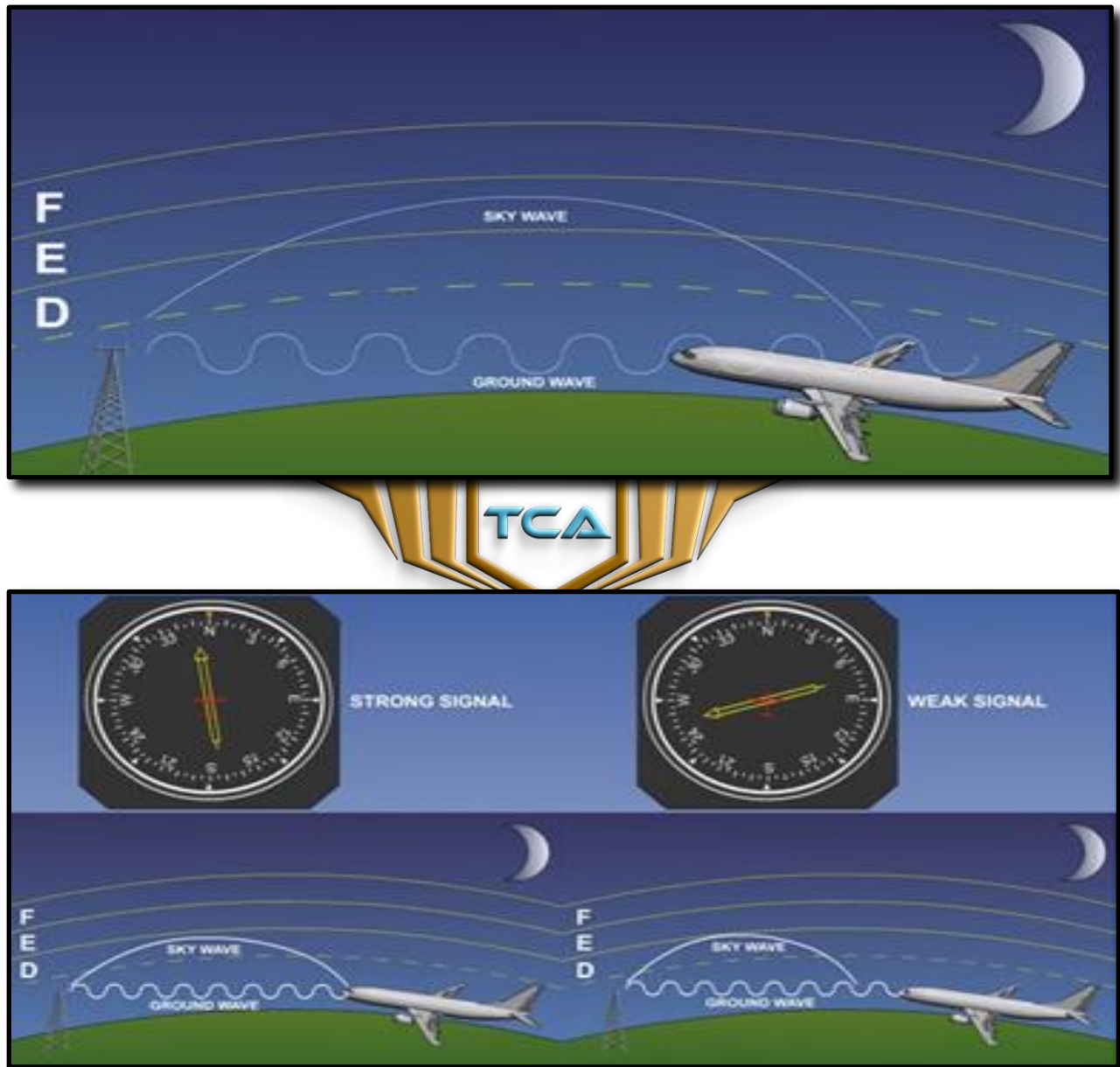


THUNDERSTORM:

- ✓ Due to thunderstorms causes ADF needle to point towards the thunderstorm(because of source of electrical conductivity).

NIGHT EFFECT:

- ✓ Due to presence of sky waves during night. Sky waves interfere with the loop aerial and cause it to hunt as the sky wave is constantly changing. Causes errors up to 30 deg.



Max during dusk and dawn, it can be felt at distances greater than 70 nm from the beacon.

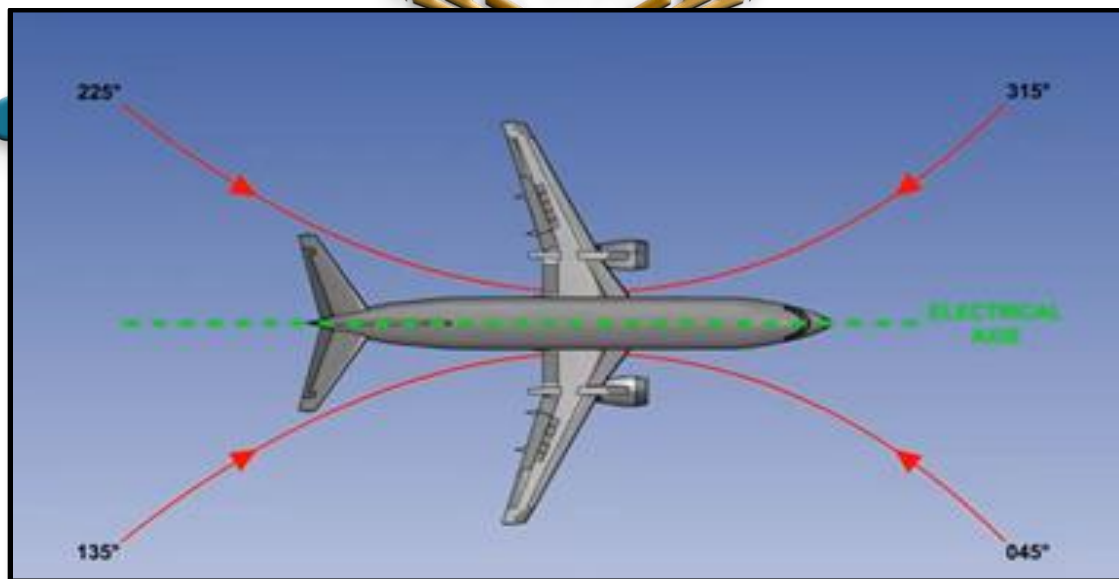
It can be reduced by

- ✓ Using beacon closer to a/c.
- ✓ Using beacon with lower frequency will get more surface range.
- ✓ Using beacon with greater power output.

COASTAL REFRACTION:

- ✓ Due to different radio energy absorption properties of water. If radio wave does not leave coast at 90 deg. It bends towards medium of high density i.e., land.
- ✓ From Land to sea signal bends away from the Normal and accelerate.
- ✓ We can reduce coastal refraction by:
 - ✎ Flying high.
 - ✎ Taking bearing when 90 degrees to coast.
 - ✎ By using the NDB as close to the coast as possible instead of in Land NDB.

QUADRANTAL ERROR



- ✓ Due to A/C metallic structure wiring, avionics etc. reflects and re-radiate signal. Electromagnet field is induced around an a/c which causes NDB signal to get refracted before being received.
- ✓ ADF gives maximum error when signal coming at 45 degree to fuselage and minimum when signal in line with fore and aft axis or abeam. Turning aircraft will suffer more from this error.

Maximum QE=A/C HDG+/-45° & A/C HDG+/-135°

Problem- A/C HDG is 185 find out Quadrantal Error?

INTERFERENCE ERROR:

- ✓ Interference can be caused by 2 nearby NDB's working on same frequency. ADF needle will lock in between as a resultant.

DOC- Designated operational Coverage

- ✓ It ensures that there will not be any interference from other nearby NDB.
- ✓ DOC is valid during day ONLY & expressed in NMs ONLY.
- ✓ It reduces range of NDB and specified in AIP.

Mountain Effect, Angle of bank error (DIP Error), Loop misalignment Error, Lack of failure warning system.

USES OF ADF:

- ✓ As an En-route aid by taking fixes.
- ✓ For flying airways accurately.
- ✓ For positioning at an enroute point.
- ✓ For let downs.

HOMING ON TO A STATION:

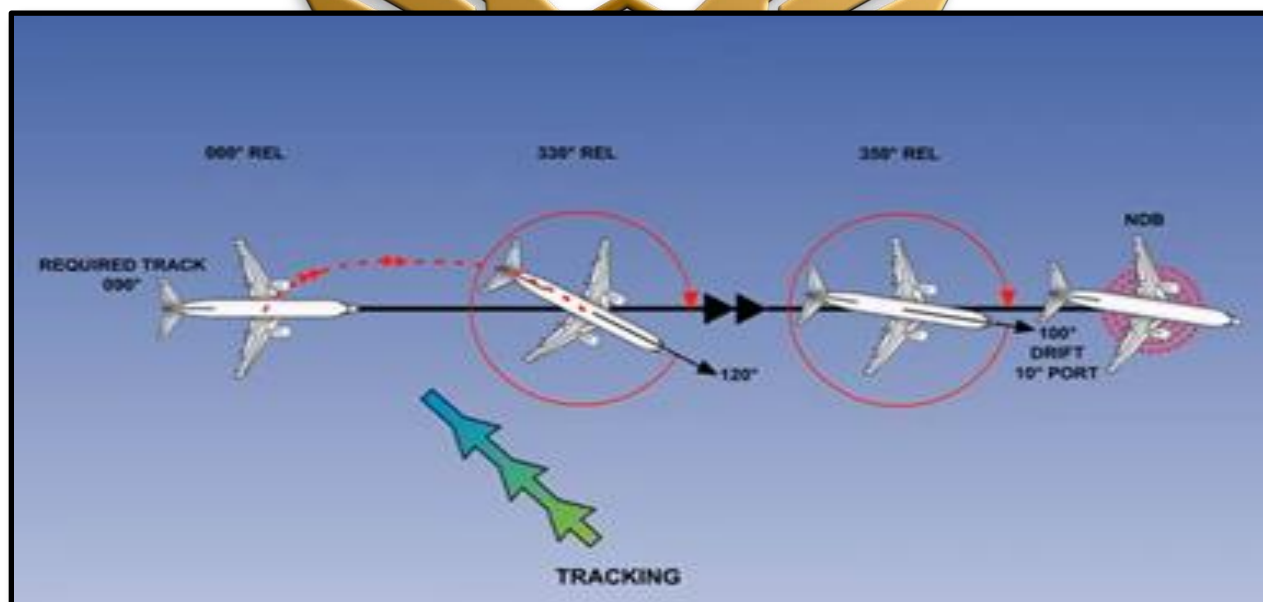
The thumb rule is:

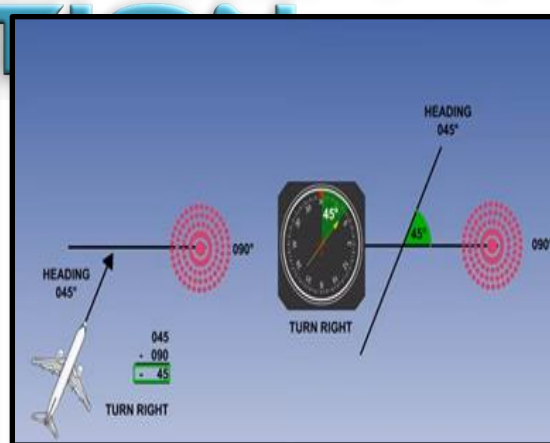
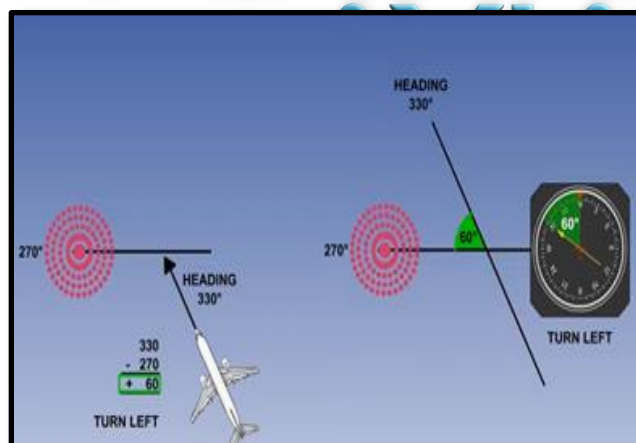
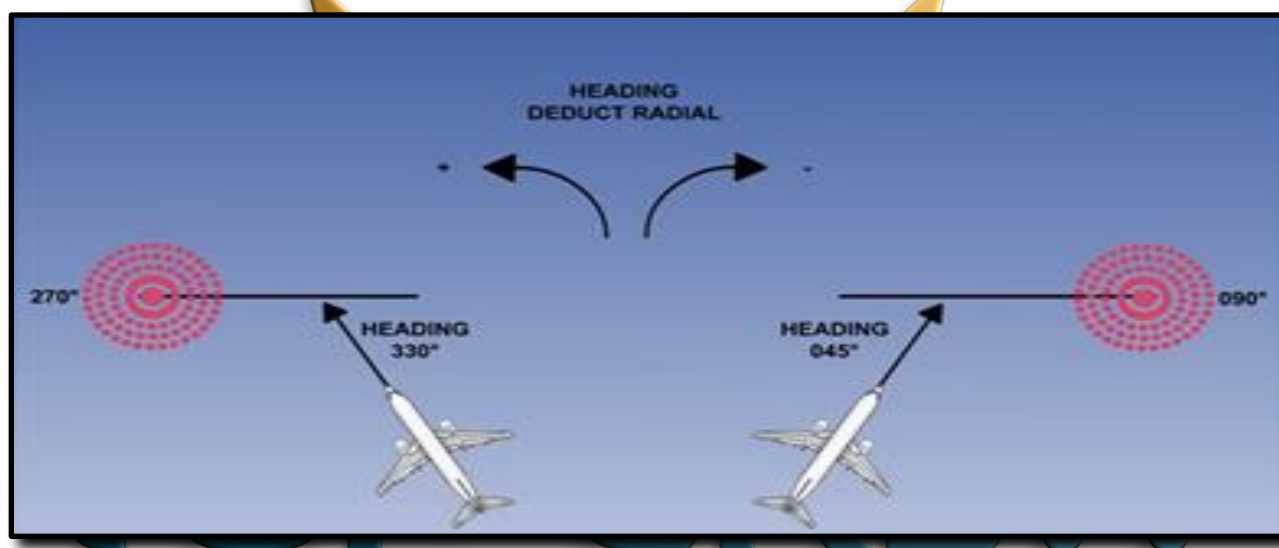
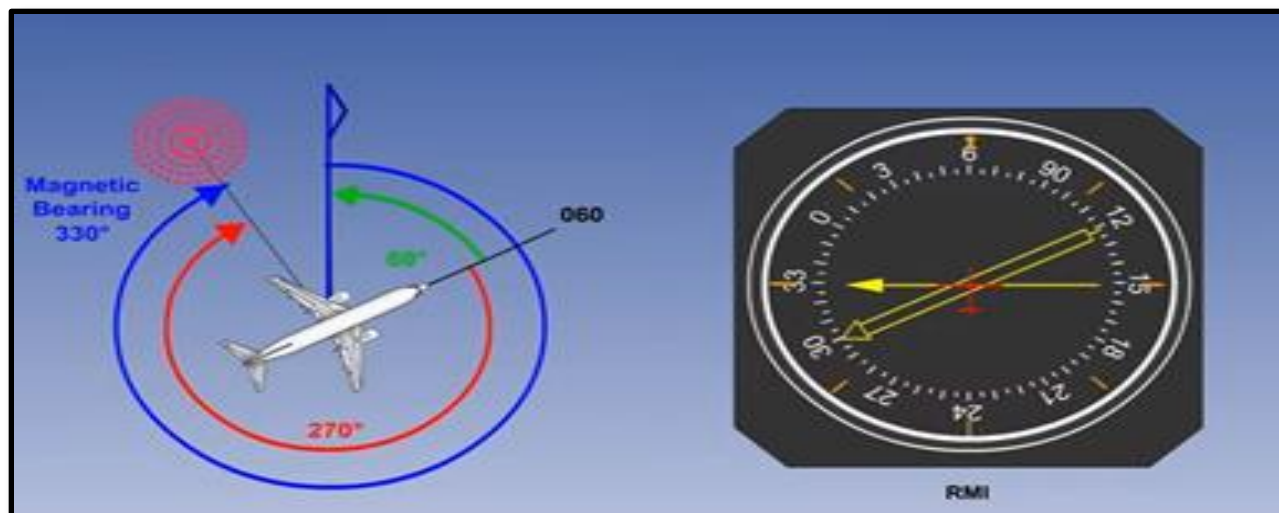
Winds from starboard	ADF < 360°
Winds from port	ADF > 360°

TRACKING OUT:

- ✓ In nil winds ADF is 180.
- ✓ In winds:

Port drift ADF	180-port drift
Starboard drift ADF	180 + starboard drift.





FACTORS AFFECTING RANGE OF NDB:

- ✓ **Transmission power:** to double the range power has to increase by four times.
- ✓ **Night effect:** Range is reduced to 70 NMs.
- ✓ **DOC:** reduces range and valid during day ONLY. Protection ranges are not valid for night due to night effect.
- ✓ **Frequency:** lower the frequency lesser the attenuation and greater the ground wave range.
- ✓ **Type of Surface:** longer range over sea is more than range over land.
- ✓ **Type of emission:** **NON A1A signals have larger range** than NON A2A signals.

TYPES OF NDB

1. **Locator NDB:** (low powered NDBs) usually co located with Marker Beacons (ILS) and placed at the outer and middle marker.
✎ Range- 10 to 25 NM.
2. **Homing And Holding NDB:** used for long-range bearing facility along airways.
✎ Range- up to 50 NM's
3. **En-route NDB:** Range more than 50 NM's



SUMMARY

NDB:

NDB is an omnidirectional beacon on ground transmitting vertically polarized signals in MF/LF band.

- ✓ Ground transmitter in LF or MF band (190 – 1750 kHz)
- ✓ Type of NDB: Locator (L) - airfield let- down (25 nm)
- ✓ En route - Nav-aid (50 nm or more)
- ✓ Range (nm):
 - A. $3 \times \sqrt{P}$ (W) over water
 - B. $2 \times \sqrt{P}$ (W) over land

ADF:

ADF is an airborne equipment that can tune in to signals from NDB and measure direction of the incoming signals

ADF measure direction in relation to the nose of aircraft

Airborne equipment – aerials, receiver, control unit, Indicator (RBI/ RMI)

RANGE 200NM by day and 70nm by night.

Principle of operation : (relative) bearing by switched cardioids

Frequencies: 190 – 1750 KHz (LF &MF)

Emission characteristics:

NONA1A – BFO ON for tuning Identification and monitoring

NONA2A – BFO ON for tuning, OFF Otherwise

Presentation: RBI or RMI

Uses of NDB: Homing, holding, approach, En-route nav-aid

ERRORS:

FACTORS AFFECTING ACCURACY	FACTORS AFFECTING RANGE
PRECIPITATION STATIC	NONA1A HAS MORE RANGE THAN NONA2A
THUNDERSTORM (BIGGEST ERROR)	MORE RANGE OVER SEA
NIGHT EFFECT	LOW FREQUENCY MORE RANGE
STATION INTERFERENCE	TRANSMITTER POWER
MOUNTAIN EFFECT	$RANGE = \sqrt{POWER \text{ IN WATTS }}$
COASTAL REFRACTION	TO DOUBLE THE RANGE THE POWER HAS TO BE INCREASED 4 TIMES
QUADRENTAL ERROR	
DIP ERROR	
NO FAILURE WARNING DEVICE	

VOR- (VHF OMNI DIRECTIONAL RADIO RANGE)

- ☞ **Freq – 108-118 MHz**
- ☞ **Band 108 -112MHz** shared with ILS so use only **even first decimals**. (40 channels)
- ☞ **Band 112 -117.975MHz** all frequencies (120 channels)
- ☞ **Band- VHF**
- ☞ **Emission Code – A9W**
- ☞ **Principle- Bearing by phase comparison**
- ☞ **Polarization – Vertical**



- ✓ Standard short range navigation aid. Can be used by day or night. When paired with DME gives instantaneous range and bearing. Operates in frequency range of 108 – 117.95 MHz

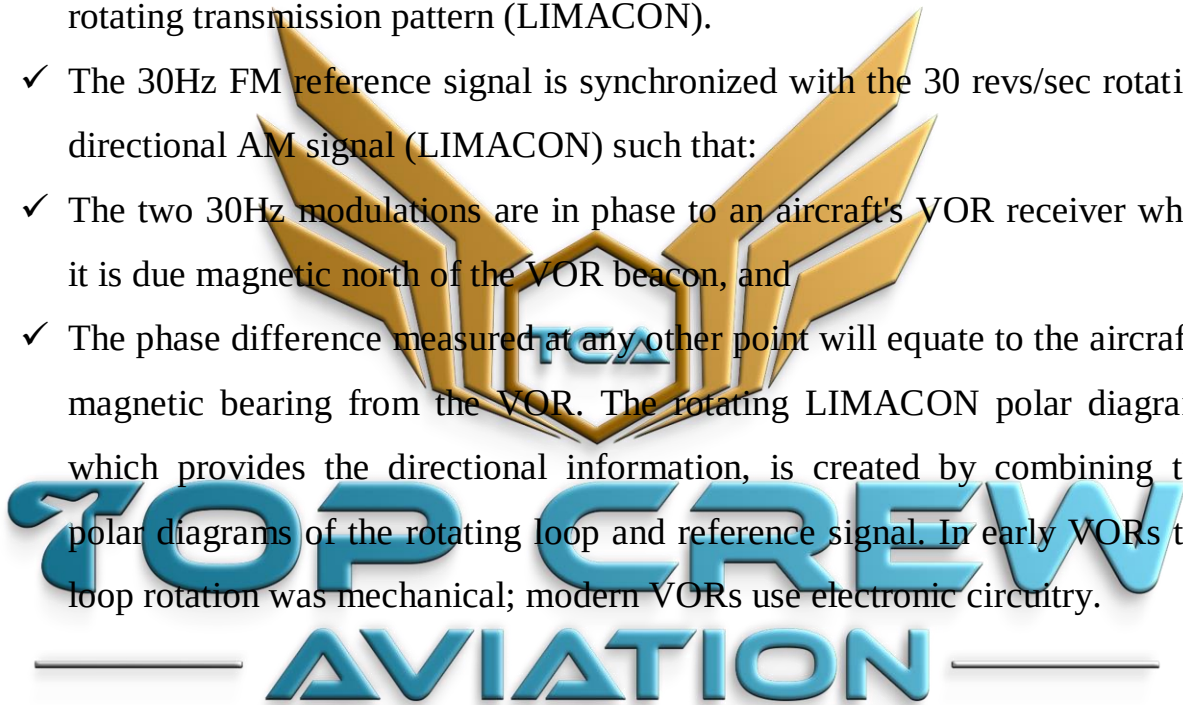
RADIAL:

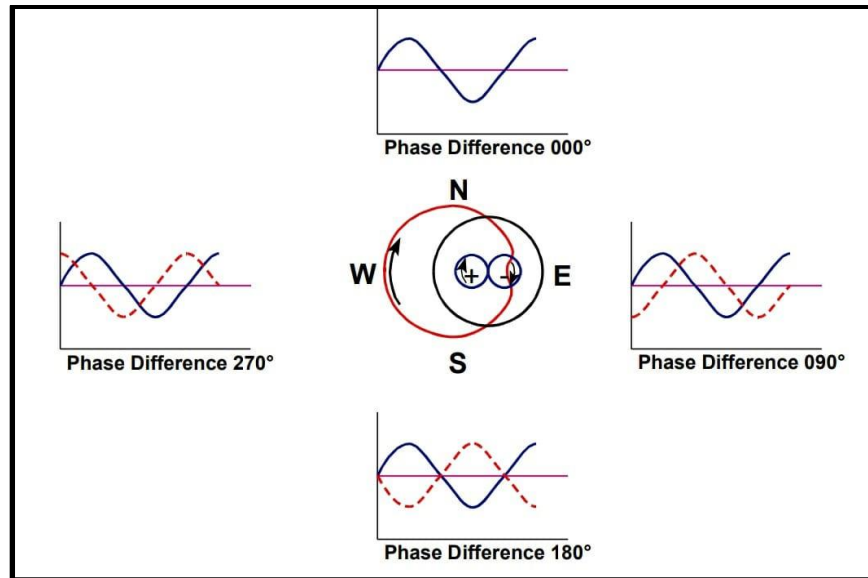
- ✓ Magnetic bearing of aircraft from the station.
- ✓ Tail always indicates radial.
- ✓ Radial does not change with the heading of aircraft.

WORKING:

- ☞ The ground beacon has 2 aerials. One fixed and one rotating.
- ☞ Fixed Antenna: 30Hz FM reference signal.

- ☞ Variable Antenna: 30Hz AM variable signal.
- ☞ VOR bearing is obtained by phase comparison.
- ☞ An aircraft's VOR receiver measures the phase difference (angular difference) between two signals from the VOR transmitter.
- ✓ 30Hz frequency modulated Omni-directional, reference signal which produces constant phase regardless of a receiver's bearing from the VOR, and
- ✓ 30Hz amplitude modulated variable phase (directional) signal created by the rotating transmission pattern (LIMACON).
- ✓ The 30Hz FM reference signal is synchronized with the 30 revs/sec rotating directional AM signal (LIMACON) such that:
- ✓ The two 30Hz modulations are in phase to an aircraft's VOR receiver when it is due magnetic north of the VOR beacon, and
- ✓ The phase difference measured at any other point will equate to the aircraft's magnetic bearing from the VOR. The rotating LIMACON polar diagram, which provides the directional information, is created by combining the polar diagrams of the rotating loop and reference signal. In early VORs the loop rotation was mechanical; modern VORs use electronic circuitry.

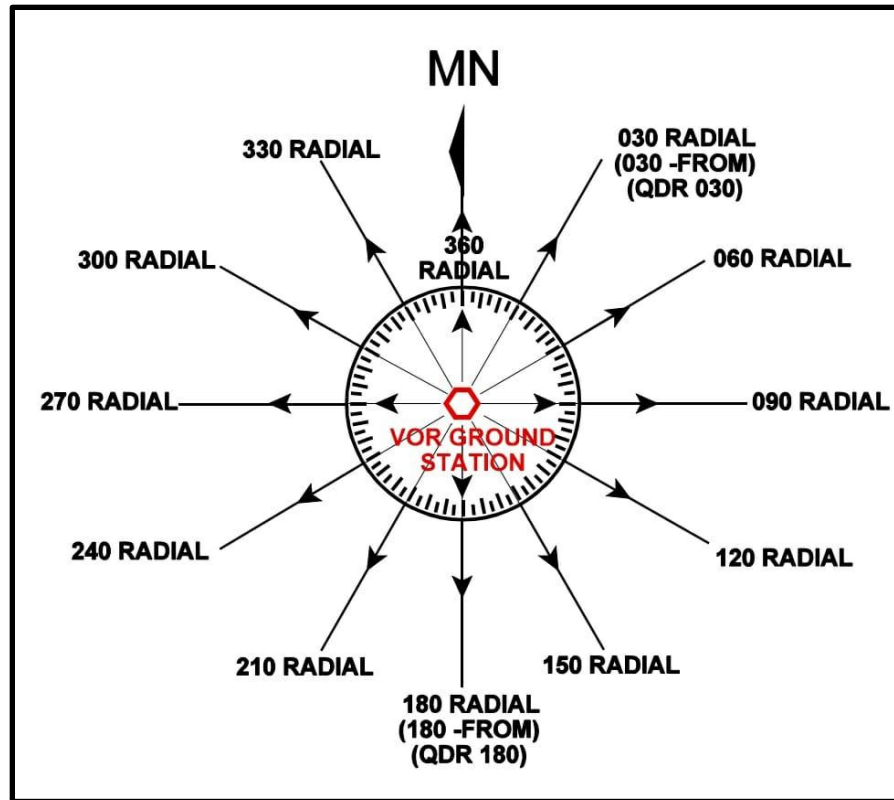




Above Figure shows one revolution of a limaçon with phase differences corresponding to four cardinal points. The blue sine wave is the reference signal. Hence, for example:

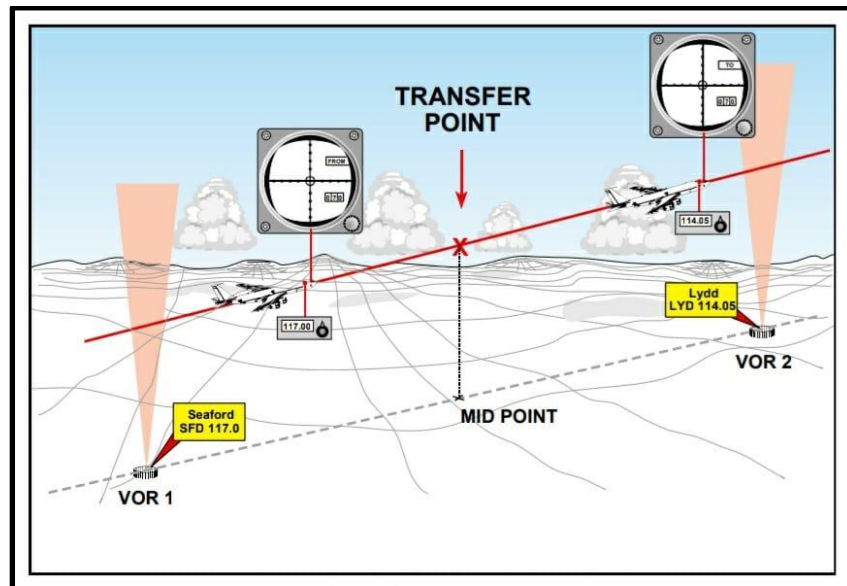
- ✓ A phase diff. of 227° measured at the aircraft = 227° Radial.
- ✓ A phase diff. of 314° measured at the aircraft = 314° Radial.
- ✓ Thus a VOR beacon transmits 360 radials continuously.
- ✓ The bearing information is supplied even during the identification period.

Radial (QDR) is a magnetic bearing FROM a VOR beacon.



- ✓ Phase of received signal will vary with the position of aircraft (receiver)
- ✓ Both signals are in the same phase when receiver is in magnetic north from transmitter
- ✓ Phase difference between two gives magnetic bearing of aircraft from station (QDR)
- ✓ Polar diagram is the resultant signal called **LIMACON**
- ✓ The net effect of these two polar diagrams i.e., circle and rotating 8 is almost like a cardioids but without a null. It is called a LIMACON.
- ✓ When the signal variable signal is phase compared with the reference signal, there would be a phase difference. This is so arranged that the phase diff is 0 at the magnetic north of station and 90 deg at 090, 180 deg and so on.

The phase difference measured at any other point will equate to the aircraft's magnetic bearing from the VOR



STATION IDENTIFICATION-

- ✓ Three Morse code letters transmitted every 10 seconds.
- ✓ DME transmits every 30 Sec at a higher pitch.

DIFFERENCE BETWEEN LIMACON AND CARDIOD:

CARDIOD	LIMACON
Fixed	Rotating Clockwise
Well defined Null	ILL-defined
Made at Rx (ADF Antenna)	Made at Rx (VOR Station)

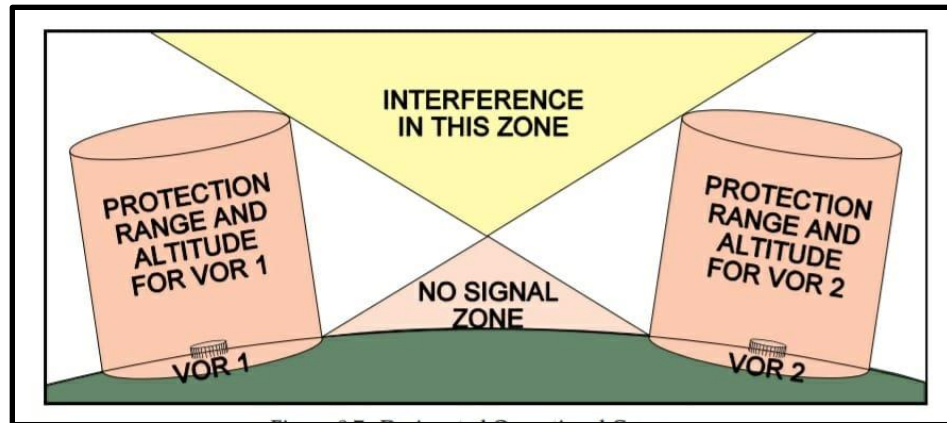
- ✓ Maximum deviation of CDI is 10 degree.
- ✓ **Warning flag:** Usually red flag will appear if any of the following is detected.
 1. No power or low power to the aircraft equipment.
 2. Failure of the aircraft's equipment.
 3. Failure of the ground station equipment.
 4. Failure of the indicator.
 5. Weak signals from either the reference or variable signal.
 6. During tuning.

MONITORING

- ✓ Every VOR station has a monitoring unit located near the transmitter. It will switch off the transmitter if:
 - ✎ Bearing information is wrong by more than 1 degree.
 - ✎ Reduced signal strength (15% below normal)
 - ✎ Failure of the monitor itself.

DESIGNATED OPERATIONAL COVERAGE (DOC)

- ✓ Range within which signal strength is guaranteed during day and night.
- ✓ Do not use VOR outside DOC.
- ✓ DOC is listed in AIP.
- ✓ **ACCURACY:** +/-5 degree on 95% of occasions.

**FORMULA:**

$$\text{Range (NMS)} = 1.25 (\sqrt{H1} + \sqrt{H2})$$

Where H1 and H2 are Tx and Rx height in feet above mean sea level and F is aircraft flight level.

✓ VOR DESIGNATED OPERATIONAL COVERAGE (DOC) in practice a beacon is protected as far as is deemed necessary. A DOC of 50/25000 published in AIPs means that an aircraft should not experience co-frequency interference within 50 NMS of VOR up to a height of 25,000ft.

USES OF VOR:

✓ HOMING, HOLDING, AIRWAYS FIX AND ENROUTE NAVIGATION.

AIRBORNE EQUIPMENT

- ✓ **OMNI BEARING SELECTOR**- Radial or its reciprocal is set on OBS.
- ✓ **TO/FROM INDICATOR**- indications depends upon the tracks selected on OBS.

TO/FROM INDICATOR:

- ✓ When the required magnetic track has been selected, the TO/FROM indicator will indicate TO if selected radial is close to ac position and FROM if selected radial is close to reciprocal of ac position.



IF – HDG and OBS in Conjunction – No reverse sensing

OMNI BEARING SELECTOR:

- ✓ OBS used by pilot to select magnetic track he wishes to fly either TO or FROM a VOR station.

- ✓ This can be used also to obtain a bearing, in which case the pilot centralizes the L/R needle by use of the OBS control and then reads of the radial.

NOTE: Full scale deflection is 10 deg, with 4 or 5 dots each indicating 2.5 or 2 deg each.

COURSE DEVIATION INDICATOR:

- ✓ Shows the pilot where his selected track lies, either to his left or his right.
- ✓ TAIL OF NEEDLE = VOR RADIAL
- ✓ HEAD OF NEEDLE = MAGNETIC TRACK TO VOR

TYPES OF VOR'S:

BVOR :(BROADCAST VOR)

- ✓ Which gives weather and airfield information between beacon identification.

DVOR: (DOPPLER VOR)

- ✓ It eliminates site error.

TVOR: (TERMINAL VOR)

- ✓ Used for terminal areas only, it has only low power; and is used at major airfields.

VOTs: (TEST VOR)

- ✓ This is found at certain airfields and broadcasts a fixed Omni- directional signal for a 360° test radial. This is not for navigation use but is used to test an

aircraft's equipment accuracy before IFR flight. More than $\pm 4^\circ$ indicates that equipment needs servicing.

- ✓ It is used at a designated place only, continues tone or series of dots it identified as VOT.

OBS	TO/FROM	CDI
360 $\pm$ 4°	FROM	CENTERED
180 $\pm$ 4°	TO	CENTERED

VOR TAC:

- ✓ Co- located VOR (Bearing) and TACAN (DME) beacons.
- ✓ TACAN (Tactical Air Navigation) gives only distance to civilians.



DOPPLER VOR:

	CVOR	DVOR
FIXED ANTENNA	30Hz FM	30Hz AM
VARIABLE ANTENNA	30Hz AM	30Hz FM
POLAR DIAGRAM	LIMACON- CLOCKWISE	LIMACON- ANTI-CLOCKWISE

- ✓ VHF beams used to suffer a lot due to site errors. DVOR aimed at reducing these.
- ✓ There is no rotating aerial.
- ✓ It is 50 aerials encircling a central aerial at a diameter of 44 ft.

- ✓ The modulation is reversed.
- ✓ The central aerial transmits the reference signal which is 30 Hz AM.
- ✓ The 50 fixed aerial in combination transmit variable signal which is 30 Hz FM.
- ✓ The combined transmission of all 51 aerial produces a rotating limaçon. Freq is same and the airborne EQPT is the same so there is no difference to the pilot.



FACTORS AFFECTING ACCURACY:

1. **Beacon alignment** – 000 radial is not aligned with magnetic North.
2. **Site Error** – uneven terrain or physical obstacles in the vicinity of a VOR transmitter affect its accuracy.

3. **Propagation error** – after the signals leave the transmission site, they can still be reflected by the terrain over which they pass.
4. **Airborne equipment error** – manufacturing inaccuracies
5. **Pilotage error** – difficulty in holding the aircraft on the desired radial by pilot.

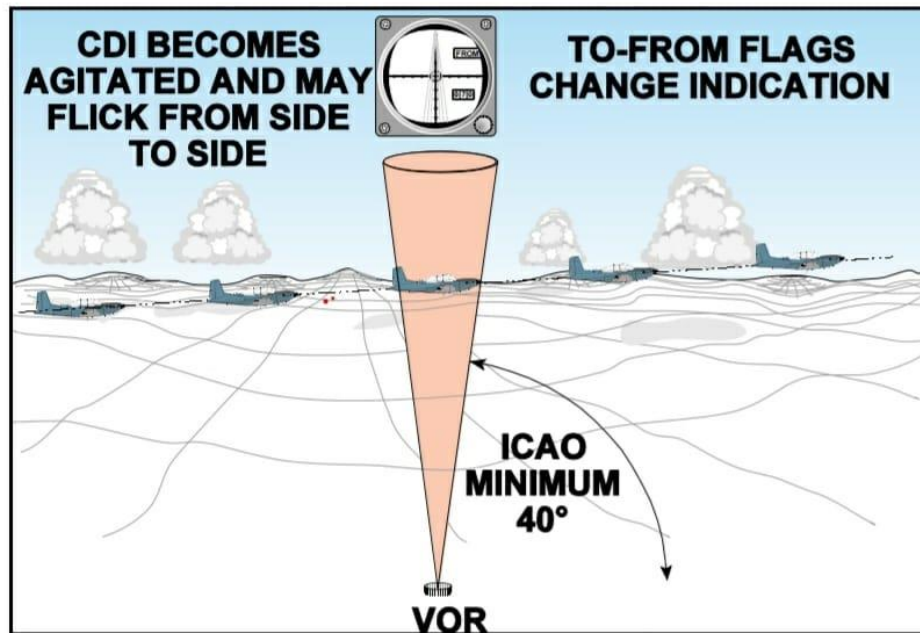
NOTE: Errors A is due to GROUND PROPAGATION and B due to AIRBORNE EQUIPMENT. Together they are called AGGREGATE ERRORS. Can be found out using Pythagoras theorem. Say if the propagation errors are 4 deg and airborne equipment errors are 3 deg, then aggregate error is 5 deg.

VOR FACTORS AFFECTING OPERATIONAL RANGE AND BEARING ERRORS

- ✓ The higher the transmitter power the greater the range.
- ✓ Maximum theoretical reception range (nm) = $1.25 \times \{\sqrt{h_1 (ft)} + \sqrt{h_2 (ft)}\}$,
- ✓ VOR transmissions are affected by terrain. - SCALLOPING

FACTORS AFFECTING VOR RANGE:

- ✓ Transmission power
- ✓ Transmitter and ac ht.
- ✓ Protection range and altitude/DOC application in range and altitude for all 24 hrs
- ✓ Nature of terrain

CONE OF CONFUSION:

Overhead a VOR beacon the VOR transmits vertically polarized signals inclines at 60-80 deg to the horizontal plane. There is no transmission vertically overhead in the shape of a cone. But some weak spill over signals may be there. Indicators when a/c in cone of confusion is:

- ✓ Warning flag will come and go off
- ✓ CDI (L/R) needle will oscillate left and right.
- ✓ TO/FROM will flicker between to and from

ADVANTAGE OF VOR OVER ADF:

- ✓ VOR is free from static and does not suffer from night effect.
- ✓ Cockpit indications are easier to see and interpret
- ✓ No night effect
- ✓ Coverage ranges can be accurately forecast
- ✓ VOR is frequency paired with DME

- ✓ NO warning device in ADF to tell pilot if unreliable or failed.
- ✓ Smaller aerals
- ✓ VOR can be used for AREA NAVIGATION- RNAV and can be coupled with autopilot



VOR SUMMARY

CHARACTERISTICS: Magnetic bearings, valid day and night

FREQUENCY: 108 to 117.95 MHz; 160 channels

USES: Airways; Airfield let-downs; Holding points; En-route navigation

PRINCIPLE OF OPERATIONS: Phase comparison of two 30 Hz signals

IDENTIFICATION: 3 letter aural Morse or Voice every 10 s, continuous tone for VOT (also ATIS using AM on voice)

MONITORING: Automatic site monitor $\pm 1^\circ$ Ident suppressed when standby transmitter initially switched on

TYPES:

1. **CVOR** - reference signal is FM; variable phase signal is AM - Limacon polar diagram rotating clockwise
2. **DVOR** - more accurate than CVOR due to less site error - reference signal is AM; variable phase signal is FM - simulated anticlockwise rotation of aerial
3. **TVOR** - low power Tx at airfields
4. **VOT** - Test VOR giving 180 radial - aircraft should have $< \pm 4^\circ$ error

OPERATIONAL RANGE: Transmitter power Line of sight DOC valid day and night

ACCURACY AFFECTED BY:

1. Site error (less with DVOR)
2. Propagation error

3. Scalloping (bending due to reflections from terrain)
4. Airborne equipment error (+/- 3°)

CONE OF CONFUSION: OFF flag may appear; TO / FROM display and bearings fluctuate

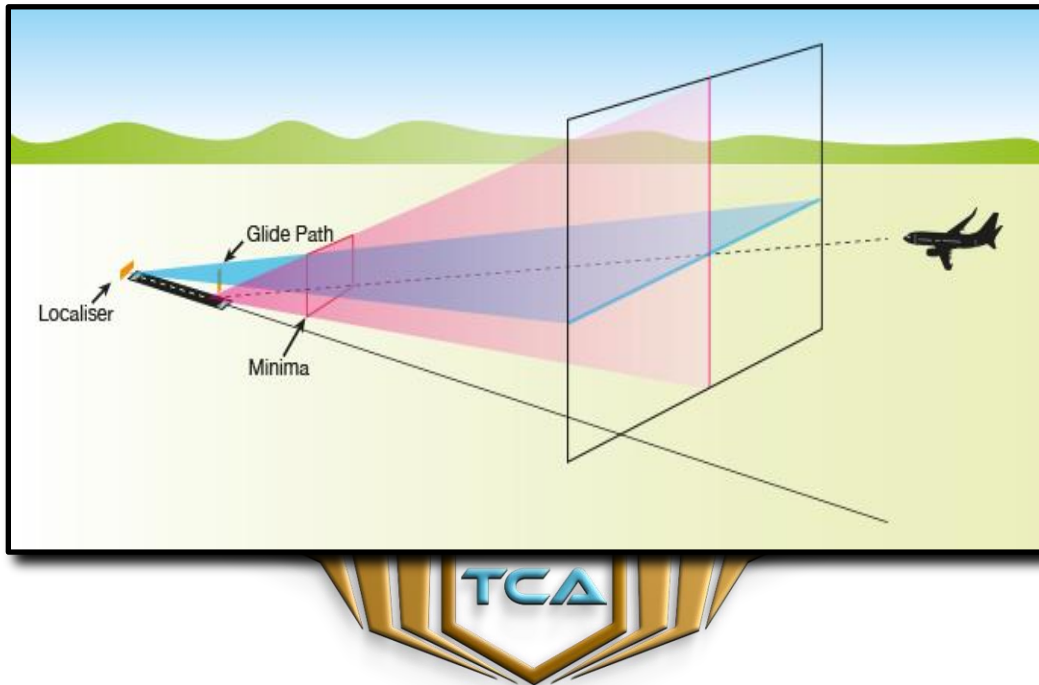
AIRBORNE EQUIP: Aerial, Receiver, Display (CDI / RMI) CDI: 2° per dot; max 10°; relationship between indication and aircraft position

RMI: arrowhead gives QDM; tail gives QDR;

USE MAGNETIC VARIATION: At station, In Flight procedures, Radial interceptions; Track keeping; Station passage.



INSTRUMENT LANDING SYSTEM (ILS)



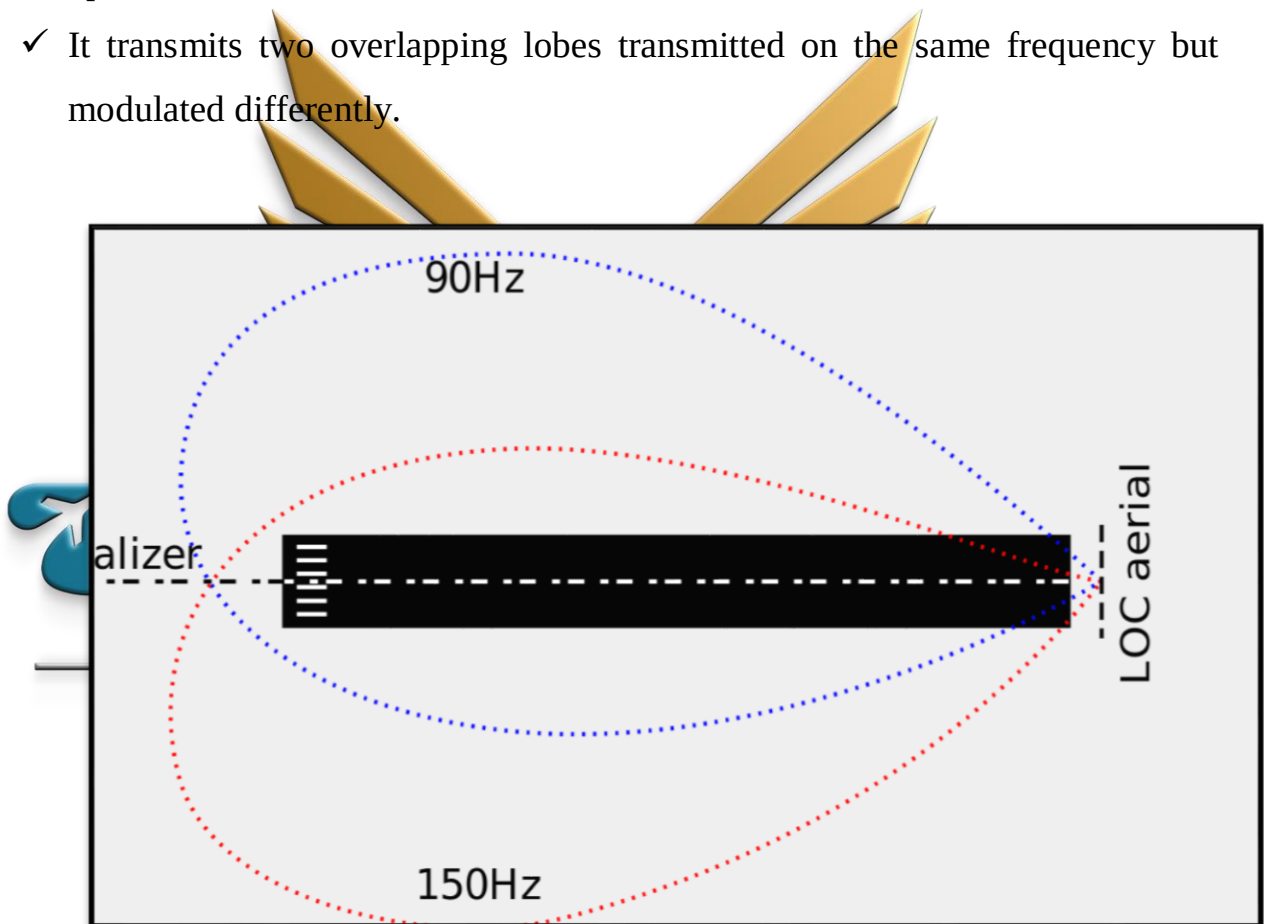
ILS provides azimuth and elevation guidance for a precision approach; it is pilot interpreted R/W approach aid. No assistance from ground control is required. The R/W served by ILS or PAR is called a precision approach runway.

It has following components:

1. Localizer
2. Glide slope
3. Marker beacon
4. Approach lighting system

LOCALIZER TX:

- ☞ **Freq** – 108-112 MHz (Odd Decimal)
- ☞ **Band**- VHF
- ☞ **Emission Code** – A8W
- ☞ **Principle**- Difference in depth modulation
- ☞ **Frequency Channel** – 40 Paired
- ✓ Localizer is located in line with the R/W centerline approx. 300m from upwind end of R/W.
- ✓ It transmits two overlapping lobes transmitted on the same frequency but modulated differently.

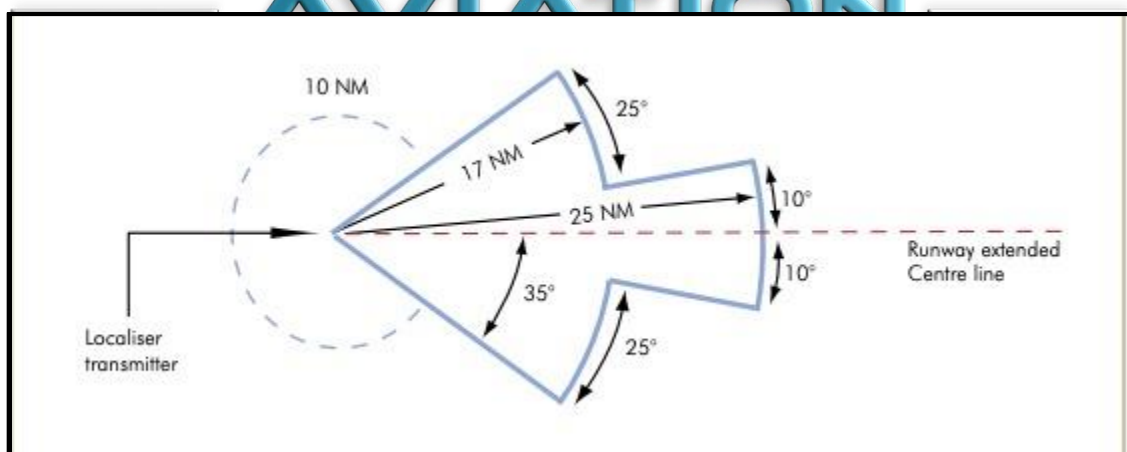


WORKING:

- ✓ The localizer Antenna produces two overlapping lobes along with the runway approach; the lobes are transmitted on VHF frequency.
- ✓ Right hand lobe as seen by a pilot making an approach is modulated at 150 Hz and is called blue sector.
- ✓ Left hand lobe is modulated at 90 HZ and is called the yellow sector.
- ✓ A/C left of the centerline will get more of 90 HZ signal as compared to 150 HZ signal. This will result in vertical needle indicating a right turn.
- ✓ The DDM (difference in depth modulation) will be 0 along the extended runway centerline.

NOTE –

1. DME is paired with the localizer reads 0 at the touchdown point.
2. ILS cockpit indicator tells us where to go?
3. Localizer full scale deflection is 2.5° , so each dot would be 0.5°

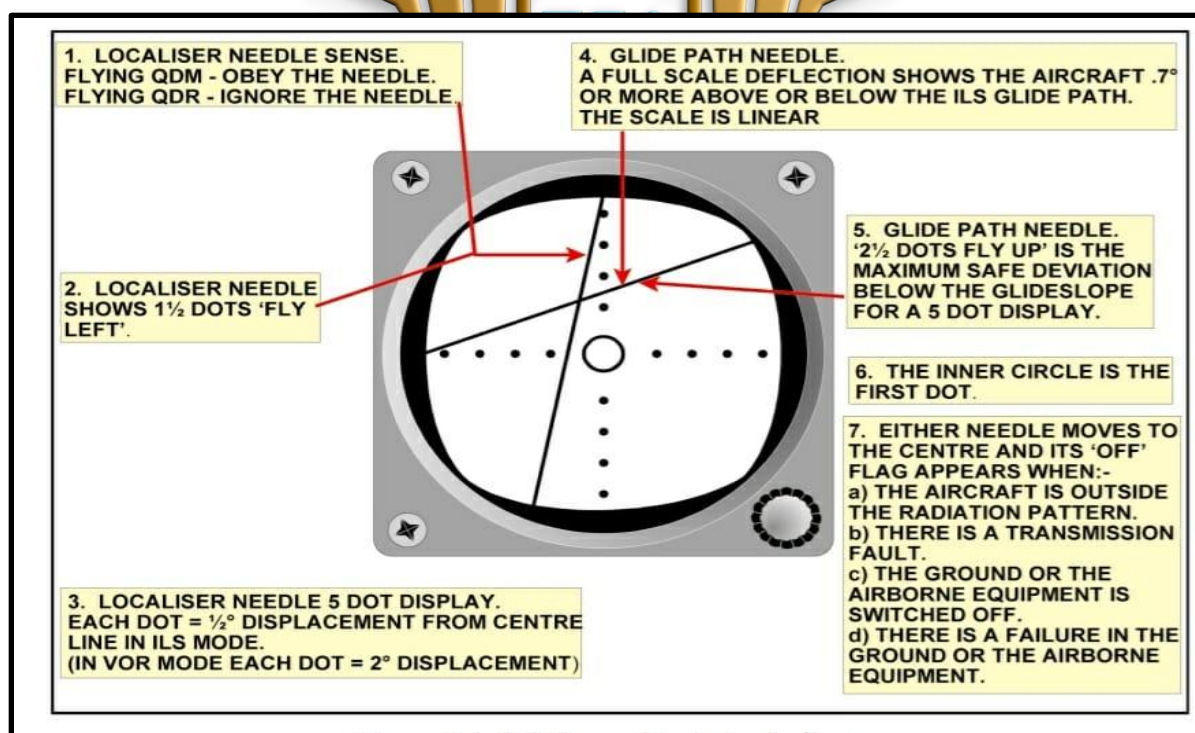
LOCALISER COVERAGE:

10° L/R of C/L Range	25 NM
35° L/R of C/L Range	17 NM
35- 10° = 25° L/R of C/L Range	17 NM

NOTE: The coverage may be reduced to 18° and 10 NMs, if there is additional nav facility available.

LOCALISER INDICATION:

- ✓ The deviation scale gives indications up to 2.5 deg on either side of the centerline. So, on a 5-dot indicator each dot is 0.5 deg and on a 4-dot indicator it is 0.625 deg.



***Warning flag on cockpit CDI display:**

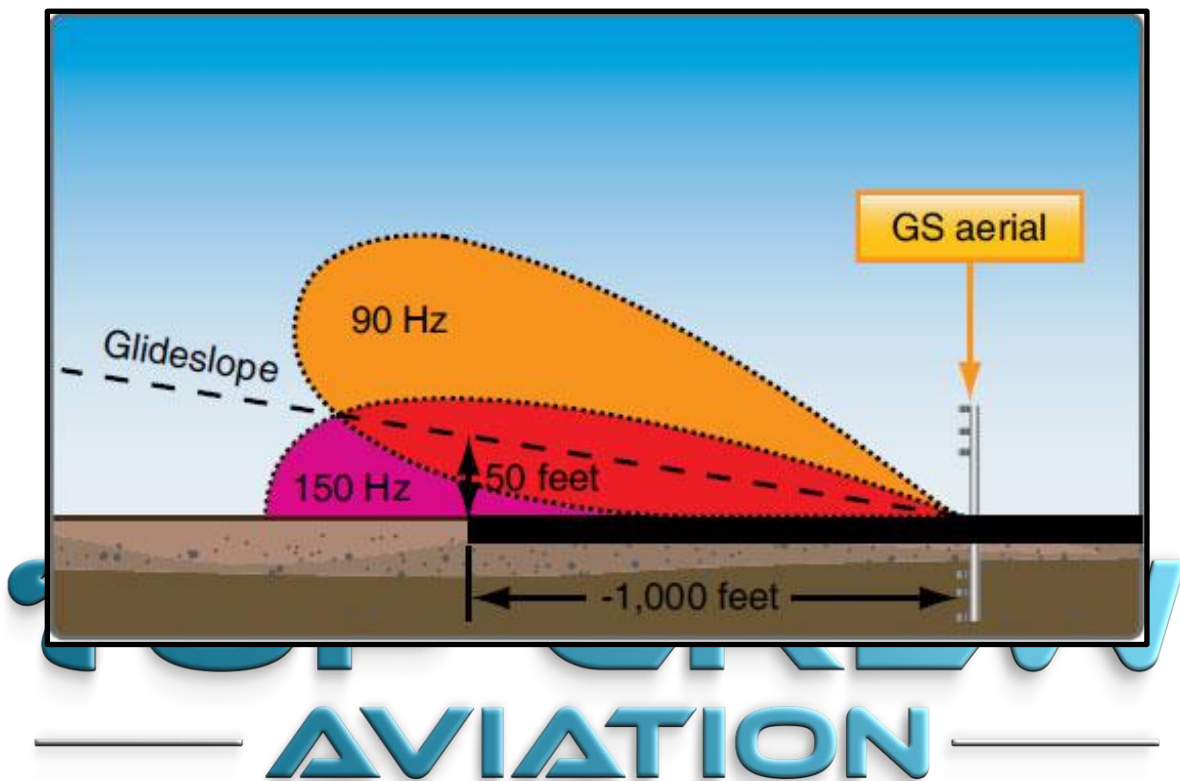
The warning flag may appear in following cases:

1. In case no frequency selected.
2. In case of absence of both modulation signals.
3. In case of percentage of both modulated signal is 0.



GLIDESLOPE Tx:

- ☞ **Freq** – 329-335 MHz
- ☞ **Band**- UHF
- ☞ **Principle**- Difference in depth modulation
- ☞ **Frequency Channel** – 40 Paired
- ☞ **Indication**- Elevation guidance



- ✓ Glide slope is fixed approx. 150m from centerline and 300m upwind one side of the R/W.
- ✓ The transmission is the same as localizer except that the 2 beams are in vertical plane with upper lobe having 90 Hz modulation and lower lobe 150 Hz modulation.
- ✓ DDM is zero at the selected glide slope (generally 3 deg)

- ✓ When it is above the glide slope, more of 90Hz modulation signal is received and glide slope needle in cockpit deflect down suggesting crew to go down and vice-versa, when more of 150 Hz signal received.

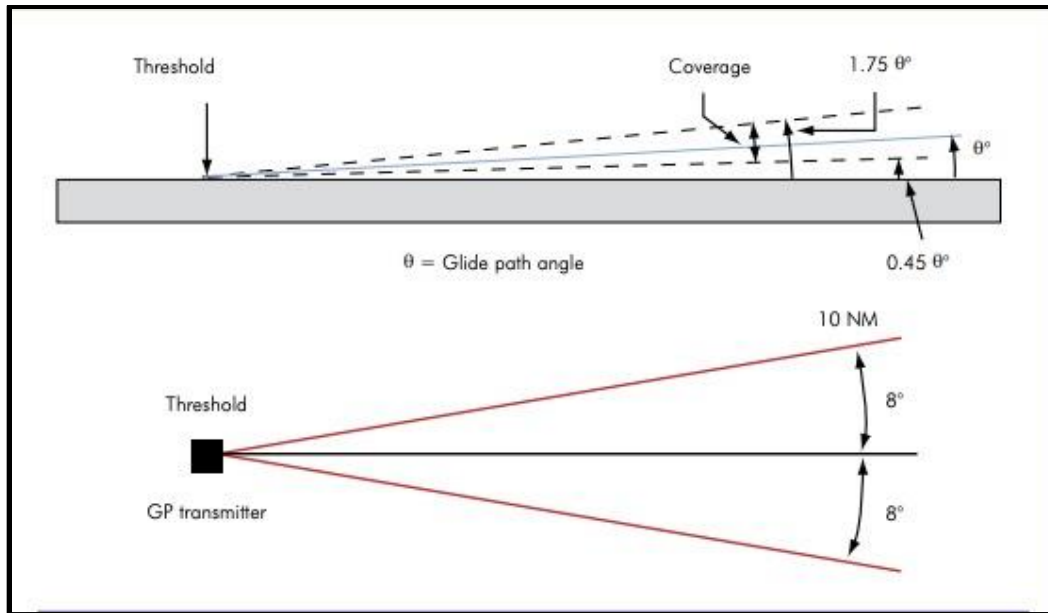
FALSE GLIDEPATH:

There could be more than one locus of points having DDM zero other than the one at the correct glide path. This is because due to metallic structures at the radiated lobe of glide path is repeated several times above the true glide path. These produce several other equipment signals. However, these are not dangerous to the pilot due to:

1. The first false glide path does not occur below 6 deg.
2. They are above and not below.
3. Normally in an ILS approach you first intercept localizer and then meet the glide path from below. So you are likely to miss the false glide paths.
4. Signals would be weak as they are outside the rated coverage.
5. **SIGNAL REFLECTION (BEAM BENDING)** signal will diffract (bend) around obstructions and even ac and vehicles.

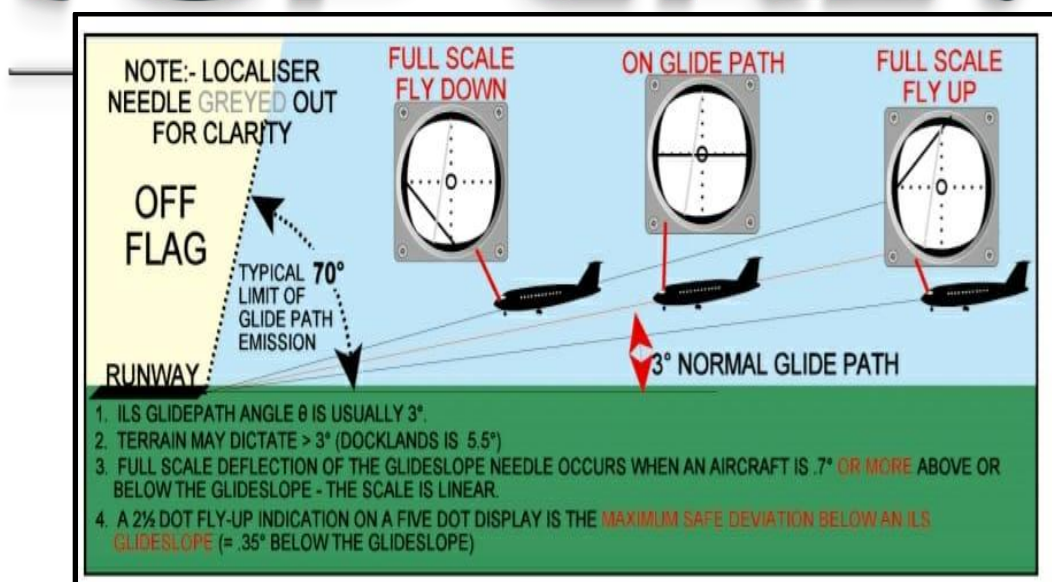
GLIDEPATH COVERAGE:

The glide path coverage is from 0.45 to 1.75° of the selected glide path angle. So for a 3 deg glide path it will be from 1.35 deg to 5.25 deg. In azimuth this coverage is upto 8 deg on either side or upto distance of 10 nm.

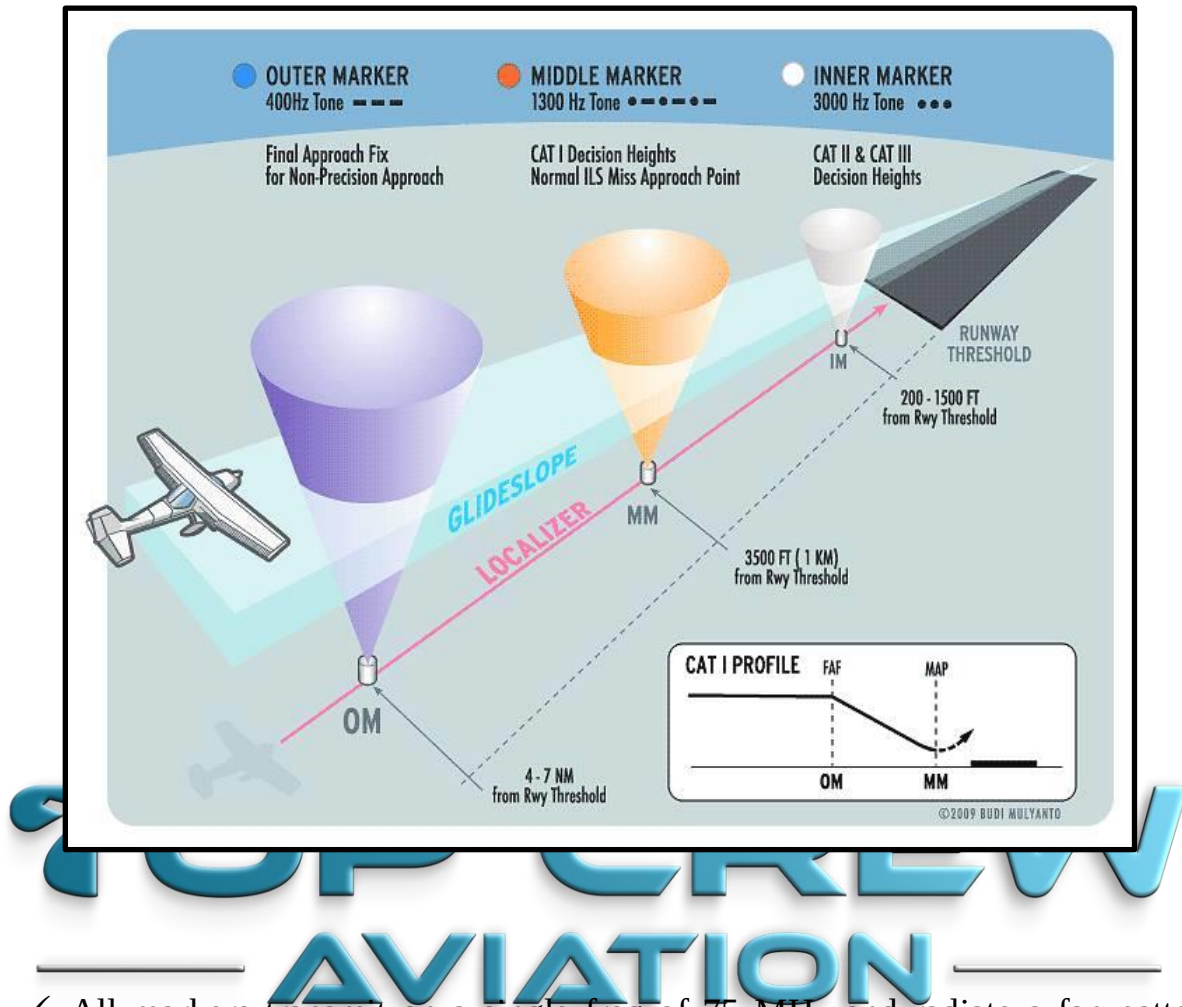


GLIDEPATH INDICATIONS:

The deviation scale gives indications up to 0.7 deg above or below the glide path. So on a 5 dot indicator each dot is 0.14 deg. The MAX DEFLECTION below glide path is 0.35 deg or 2 dots in a 4 dot indicator or 2.5 dots in a 5 dot indication. On seeing this indication an immediate climb is to be carried out.



MARKER BEACONS: 2 or sometimes 3 marker beacons are installed along the extended centerline to give range indication on approach. This allows pilot to check his height as he passes over each marker.



- ✓ All markers transmit on a single freq of 75 MHz and radiate a fan pattern upwards to a ht of approx.3000 ft.

	cockpit light	Ident	Modulating Frequency	Pitch	touch down Range
OM	blue	2 dashes/sec	400Hz	low	6.5 – 11.1 km (3.5 – 6 NM)
MM	AMBER	Alternate dots and dashes 3/sec	1300Hz	Medium	1050 m ± 150 m (3500' ± 500')
IM	WHITE	6 dots/sec	3000Hz	high	75 - 450 m (250' – 1500')

- ✓ A DME that is frequency paired with an **ILS** supplement or replaces the range information from markers/locators.
- ✓ The DME range is zero referenced to the ILS **runway** threshold.
- ✓ The DME is protected only within the **ILS localizer service** area up to **25,000 feet**.

TYPE OF EMISSION: A8W for localizer, glide path and marker.

IDENT: As localizer and glide path are freq paired they both need not identify themselves separately. So only localizer transmits indent. Its carrier is AM by horizontally polarized 1020 Hz. It is in the form of Morse code.

ILS REFERENCE DATUM: defined as the point of specific height (usually 50 ft) located vertically above the intersection of the R/W centerline and the ILS

landing threshold through which the downward extended path portion of the ILS glide path extends.

FORMULA FOR ROD AND OTHER CALCULATIONS:

$$\text{ROD} = \text{GP ANGLE} \times \text{GS (KTS)} \times 100/60$$

$$\text{HT} = \text{Angle} \times \text{Range} \times 101$$

LIMITATION OF ILS:

- ✓ Bends in localizer and GP beams.
- ✓ High cost of installation.
- ✓ Permanent fixture and cannot be moved from runway to runway.
- ✓ To minimize interference the landing rate has to be kept low and vehicles are not permit near the eqpt.

The back-course ILS has the following disadvantages:

- ✓ The glide path indications are incorrect (they would, if guide the aircraft to the wrong end of the runway).
- ✓ The CDI needle (localizer) is sense reversed. (Flying to R/W)
- ✓ There are no range-check markers

MONITORING ILS TRANSMISSION:

- ✓ Monitoring equipment act in case of Localizer shift of more than 35 ft.

- ✓ Glide slope angle change of more than 0.75 deg of selected glide path angle.
E.g. for 3 deg is $3 \times 0.075 = 0.225$ deg
- ✓ Reduction in power output of any of the T× by 50%.
- ✓ It will cause any of the following of happen:
 - A. Stop all transmissions
 - B. Remove indent
 - C. May permit operation to a lower category.

Categories of ILS

Category	Decision Height	Visibility/RVR not less than
CAT 1	Not Lower than 200FT (60M)	800M VIS/ 500M RVR
CAT 2	Lower than 200FT (60M – 100M)	350M RVR (300M)
CAT 3A	Lower than 100FT (50FT)	200M RVR
CAT 3B	Lower than 50FT (15FT)	200-50M RVR 75-200M
CAT 3C	0	0

TOP CREW
— AVIATION —

SUMMARY

Components and frequencies:

Localizer VHF – 108 to 111.975 MHz (40 channels). Aerial at Upwind end

Glide path UHF – frequency paired. Aerial abeam touchdown

Markers VHF – 75 MHz Fan shaped vertical radiation OM, MM and IM

Back beam - From localizer. Non-precision approaches

Locator- low power NDB at OM DME freq paired.

Possibly in place of marker - Zero- Referenced to threshold

Ident: 2 or 3 letters, 7 groups/min, suppressed when ILS U/S. continuous tone during maintenance

Markers:



OM	blue	2das/s	400Hz	6.5 – 11.1 km
MM	amber	3 dot-das/s	1300 Hz	1050m
IM	white	6 dots/s	3000 Hz	75-450m

Ground monitoring: localizer within 35' (Cat I) at ref datum. GP

Within 0.075 θ . Power within 50%

Otherwise: cease radiation, remove ident or Lower category

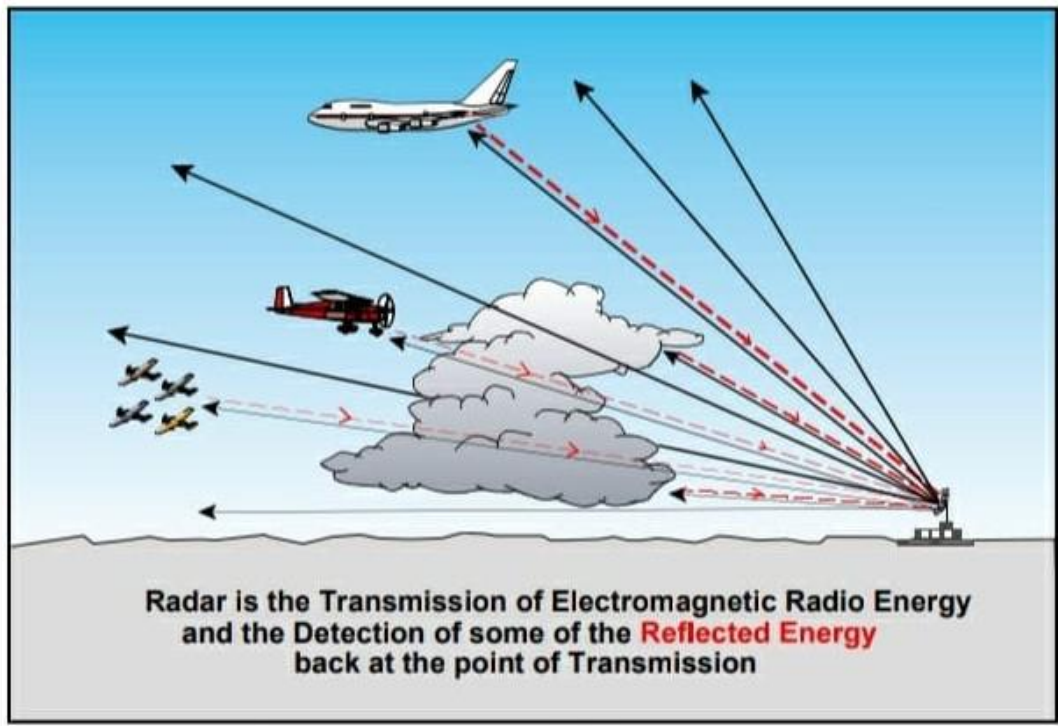
ILS Coverage: LIZ: 25 nm $\pm$ 10°, 17nm $\pm$ 35°, (plus 10nm $>$ 35°

GP: 10nm $\pm$ 8°, 0.45 to 1.75 θ .

Error: Beam bends, scalloping, beam noise; restricted vehicle movement during low vis ops; Check failure flags; monitor ident



RADAR (RADIO DETECTION AND RANGING)



TOP CREW AVIATION

TYPES OF PULSE RADARS

- ✓ **Primary radar** uses pulses of radio energy reflected from a target i.e. it uses one frequency throughout.
- ✓ **Secondary radar** transmits pulse on one frequency, but receives on a different frequency. i.e. the object transmits its own energy.
- ✓ It is a system utilizing an interrogator and transponder; the transponder can be located in the aircraft or on the ground.

USES OF RADAR APPLICATION

- ✓ ATC uses radar to monitor aircraft in relation to each other whilst they are flying on airways, separate traffic and to vector the aircraft if necessary.
- ✓ Provide radar talk-down to a given runway: surveillance radar approach monitor aircraft on ILS let-down, or during airfield instrument or a military precision approach radar (PAR)
- ✓ **Secondary** surveillance radar provides ATC with information regarding an aircraft's call sign, altitude, speed, track history, destination and type of emergency when appropriate.
- ✓ **Distance measuring equipment (DME) - SECONDARY RADAR** provides a pilot with very accurate slant ranges from a ground-based receiver/transmitter known as a transponder.
- ✓ **Doppler radar.** This is a self-contained airborne system, needing no ground-based equipment, which provides a pilot with a continuous indication of the aircraft's drift and groundspeed.
- ✓ **Airborne weather radar (AWR)** is used to:
 1. Depict the range and bearing of clouds.
 2. Indicate areas of the heaviest precipitation and associated turbulence.
 3. Calculate the height of cloud.
 4. Ground map.
- ✓ **RADIO ALTIMETER:** Is used to indicate height above ground level (AGL)
 - Primary radar

RADAR FREQUENCIES:

Radar systems are in the VHF and above frequency bands because:

- ✓ These frequencies are free from external noise/static and **ionosphere scatter** the shorter wavelengths produce narrow, efficient beams for **target discrimination** and **bearing measurement** efficient reflection from an object depends upon its size in relation to the wavelength; shorter wavelength **are reflected** more efficient.

PULSE TECHNIQUE:

- ✓ Primary, secondary and Doppler radar systems use the pulse technique which is the transmission of radio energy in very short bursts.
- ✓ Each burst of energy is in a pulse form of a pre-determined shape.
- ✓ The duration of the pulse is equal to the pulse length or width. Although a pulse is of short width (time) it can contain many cycles.

AIRPORT SURVEILLANCE DETECTION EQUIPMENT (ASDE)

- ✓ This is also known as **AIRFIELD SURFACE MOVEMENT INDICATOR (ASMI)** And is a very accurate radar display (in all weather and conditions of visibility) of the aerodrome infrastructure, taxi-ways, runways, aprons etc. vehicular traffic and aircraft that are stationary, taxiing, landing or taking- off.
- ✓ ASMI radar is designed to provide a detailed, bright and flicker-free display of all aircraft and vehicles on runways and taxi-ways so that Air Traffic Control Officers can be certain that runways are clear of traffic before

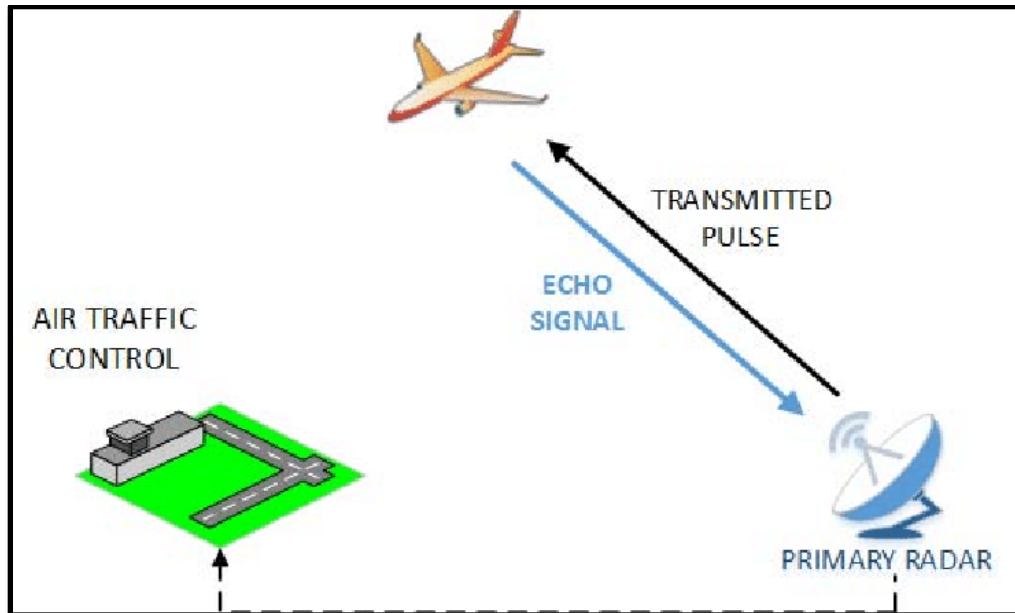
landing or take-offs, and to enable them to ensure the safe and orderly movement of traffic on taxi-ways.

AIRFIELD SURFACE MOVEMENT INDICATOR (ASMI)

1. A very narrow beam in the order of $.25^{\circ}$ to 1°
2. A scanner rotation rate of 60rpm.
3. A prf in the order of 4000 to 20000pps.
4. Pulse widths in the order of .03 microseconds.
5. Frequencies of 15 to 17 GHz (shf), 2 to 1.76cm wavelengths.
6. Ranges of 2.5 to 6nm in light precipitation.



PRIMARY RADAR



- ✓ **Primary radar** uses pulse of radio energy reflected from a target i.e. it uses one frequency throughout.

- ✓ **PRINCIPLE** – Echo & Search Light

Echo – Range

Search Light – Bearing of the object

Frequency band – VHF & Above

- ✓ In primary radar normally a parabolic is used which transmits omnidirectional signal in the form of pulse energy. The signal strikes the object and reflects back.

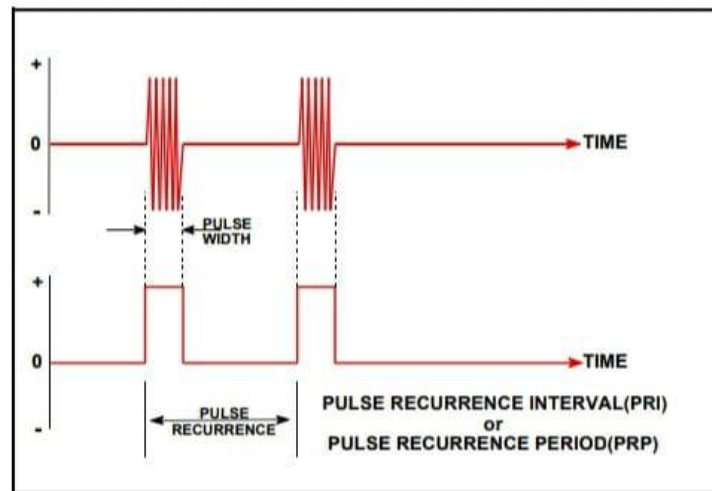
$$\text{MAXIMUM RANGE} = \frac{C}{2\text{PRF}}$$

$$\text{Range} = C \times \frac{\text{PRI}}{2}$$

$C = \text{Speed of light} = 3 \times 10^8 \text{M/S}$

PULSE WIDTH

- ✓ Time elapsed in transmission one pulse is called pulse width, it is expressed in micro/sec.



PRI (Pulse Recurrence Interval) / PRP/PRT

- ✓ Elapsed from starting of the pulse to starting of another pulse expressed in microseconds.

PULSE RECURRENCE INTERVAL (PRI)

- ✓ Is the time interval between two pulses. It is also known as the pulse recurrence period. (PRP)

PRF/PRR (Pulse Recurrence Frequency)

No of pulse per sec: It is also known as pulse recurring rate (PRR)

$$\text{PRF} = \frac{1}{\text{PRI}}$$

$$\text{PRF} = \frac{10^6}{\text{PRI}} \text{ Sec}$$

Note: - Maximum range of radar is determined by PRF.

MINIMUM RANGE RECEPTION ERROR:

- ✓ In primary radar the basic concept is when transmitter is transmitting the pulse during that period if reflecting signal is received the Rx will not receive the signal because the Tx will still be transmitting the tail of the pulse there for primary radar has minimum range reception error.

Note: - Minimum range of radar is determined by the pulse width.

$$\text{MINIMUM RANGE} = \text{PULSE LENGTH IN SECONDS} \times \frac{c}{2}$$

BEAM WIDTH

- ✓ Determines the angular resolution or angular discrimination between two aircraft.
- ✓ If the beam width is wide than two aircraft which are less laterally separated may be shown as one blip on PPI (Plan Position Indicator) screen.

Note: - Narrower the beam width better is the angular discrimination.

FACTORS EFFECTING PRIMARY RADAR RANGE

Power:

$$\text{Range} = \sqrt[4]{\text{Power}}$$

To double the range power must be increased by 16 times.

Similarly, to triple the range to triple the range power must be increased by 81 times.

1. Radar follows line of sight propagation therefore:
Range = $1.23 (\sqrt{HTx} + \sqrt{HRx})$
2. **PRF** – Increase in PRF reduces the range.
3. **Target Characteristics** – Small target provides poor reflection while larger tower gives better reflection.
4. **Rain Clutter** – The rain drops may attenuate the signal there for range reduces.
5. **Pulse width** – It determines the minimum range.

Note: Moving target indicator (MTI) ensures that radar only detect by using Doppler technique.

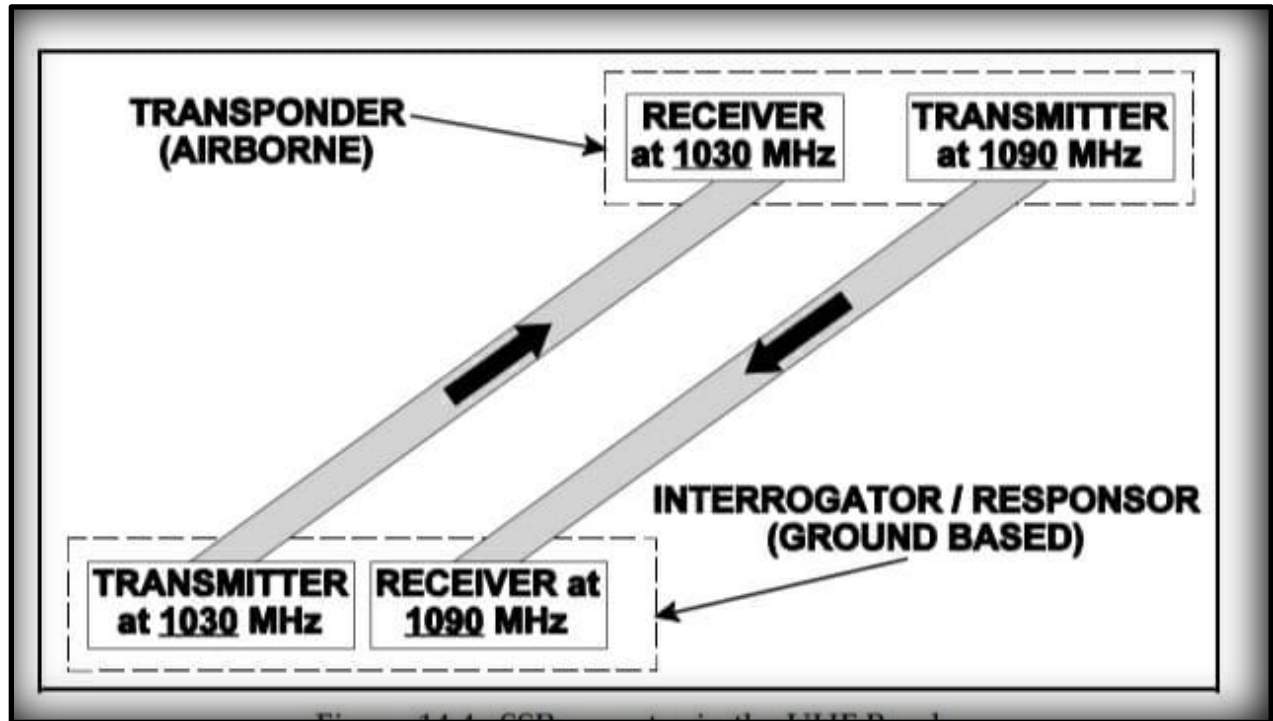
TYPE OF AERIAL OR ANTENNA

1. Parabolic
2. Phased array / Slotted / Fixed plate

Advantages of Slotted Aerial

1. Less side lobes (Side Lobber suppression)
2. Less power required
3. Less Energy Wasted
4. Better resolution

SSR (SECONDARY SURVEILLANCE RADAR)



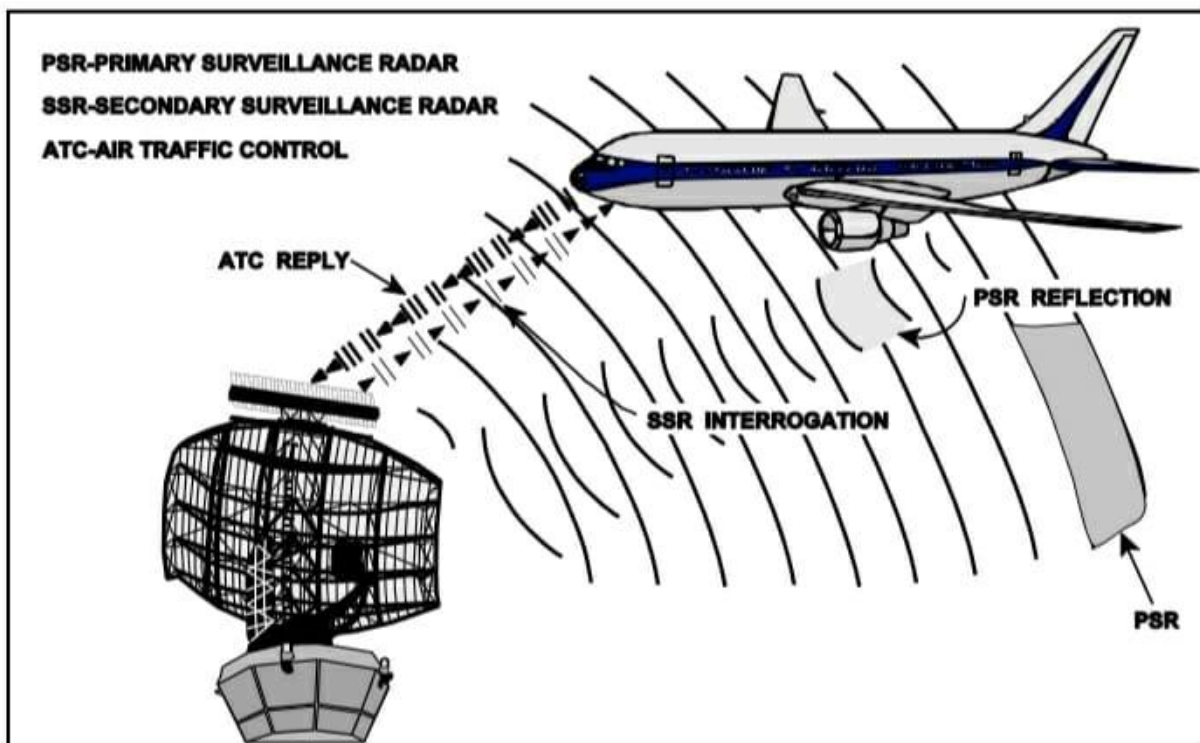
PRINCIPLE – Coded and Moded Pulses

FREQUENCY – 1030 & 1090 MHz

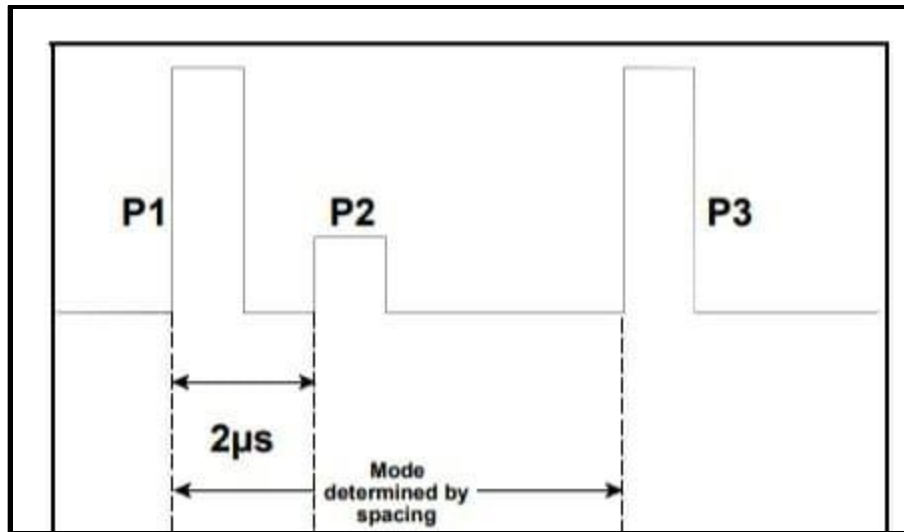
BAND – UHF

WORKING

- ✓ The ground station radar is the 'interrogator'. It transmits a rotating beam of pulse.
- ✓ This is picked up by the ac 'transponder'. Which decodes the signal and after a fixed delay replies on a different freq, the ac not only identifies itself but even gives its ht. on the ground, range and bearing are displayed on the screen and also shown are ac ht, destination or point of leaving airspace and call sign.

**CURRENT MODES AND CODES:**

- ✓ The signal from the interrogator is basically 2 pulses timed at specific intervals. The pulse intervals are called modes. There are 4 modes. Their time intervals are as follows:



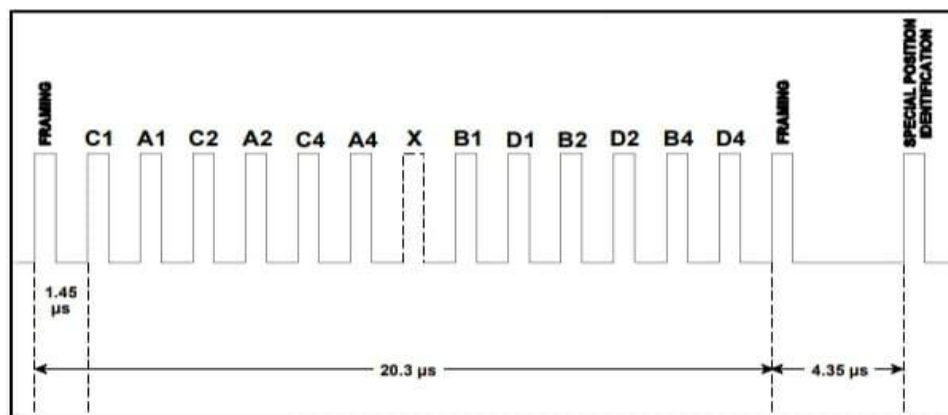
1. Mode A – 8 micro sec apart – Used for identification
2. Mode B – 17 micro sec apart – Used for identification STAND BY
3. Mode C – 21 micro sec apart – Provides identification and altitude
4. Mode D – 25 micro sec apart – Experimental

AIRCRAFT TRANSPONDER REPLY

- ✓ Once the pilot decides to respond the transponder replies with a code obtained by inclusion or omission of up to 12 pulses.
- ✓ This train of 12 pulses is contained between 2 framing pulses, 20.3 pulses is 4096. These are numbered from 0000 to 7777 using all numbers except 8 and 9.
- ✓ Each of these pulses in the transponder is 0.45 micro sec wide.

SPECIAL POSITION IDENTIFIER PULSE

- ✓ Whenever controller asks the crew for squawk ident and as the crew presses the squawk the aircraft transponder immediately transmits another pulse called the SPI special identification 2nd framing pulse and gets transmitted for 20 sec after pushing the button.
- ✓ After 4.2 Micro sec from the last framing pulse, which discreet aircraft position on ATC screen. (Radar Screen)

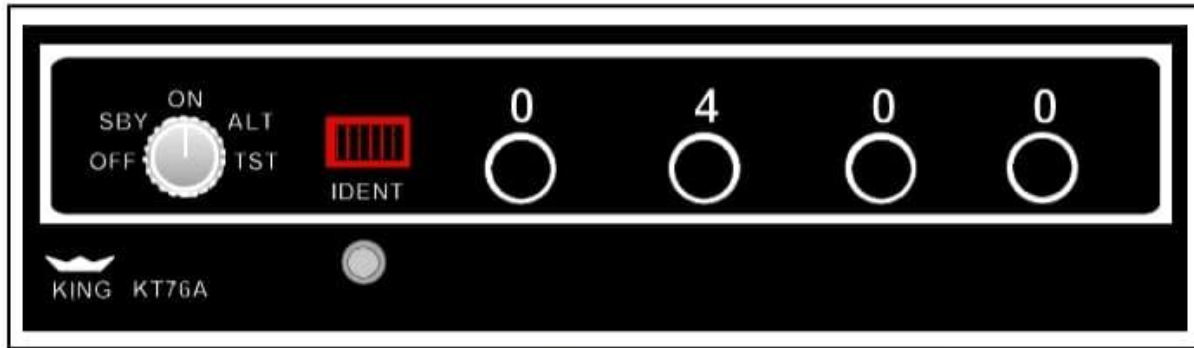


TOP CREW AVIATION

MODE C

- ✓ On getting mode C interrogations the transponder will produce one of the 4096 codes.
- ✓ This code is decided as per the altitude the ac is flying.
- ✓ This is referred to 1013.2 hpa irrespective of what the pilot selects on the altimeter.
- ✓ The EQPT can provide automatic altitude tele metering upto 128000 ft with a change of output every 50 ft. The controller automatically gets info about ac ht.

Note: If the altitude indicates by mode C transponder and the altitude given by the pressure altimeter differs by +/- 300', the crew is supposed to turn off mode C.



MODES:

- ✓ Mode S is intended to provide data-link of info between ac and ATC reducing the need of verbal communication.
- ✓ Unlike the earlier mode A and C, mode S can handle more than 5×10^{33} different uplink and downlink messages.
- ✓ The uplink message will be specifically addressed to a particular ac.
- ✓ This will permit communication with up to 16 million discrete ac.
- ✓ This will be spoken to separately.
- ✓ The encoding altimeters will have a capacity of 25 increments instead of 50 ft increments of mode C, enhance info in the form of ac bank angle, accelerations, HDG, groundspeed will be passed automatically to ATC.

Note: Mode S level 7.1 is the most recent SSR

SQUAWK CODES	MEANINGS
7700	Ac in state of distress (mayday)
7600	Radio failure
7500	Hijack (unlawful interference)
2000	Ac has no code allotted
7000	Ac has no code allotted
7004	The ac is carrying out aerobatic maneuver
0000	ac has transponder malfunction

ADVANTAGES OF SSR:

- ✓ Only one way travel so less power is required.
- ✓ No problem of reflections from GD.
- ✓ Information other than just the ac position can be exchanged.
- ✓ No maneuver is required by the ac to identify itself.

DISADVANTAGES OF SSR:

- ✓ Because all EQPT operate on the same frequencies, a transponder will reply to interrogations from neighboring GD stations. This is called **fruiting**. This can be avoided by allotting neighboring radars different PRFs and disregarding a sequence of returns at the wrong PRFs
- ✓ If 2 ac are in the same interrogating beam at the same time and close to each other so that their transponder signals may overlap at the Rx and causes interference called **GARBLING**. This occurs when ac are less than 1.7 nm apart.

- ✓ If an ac is being interrogated by more than one ground station at once (over interrogation). It may get saturated and not reply to one of them.

DIFFERENCE BETWEEN PRIMARY AND SECONDARY RADAR:

PRIMARY	SECONDARY
Frequency is constant throughout	Tx and Rx frequency is different
Requires no external help	Work with the help of GD beacon
Minimum range reception error	No such error
Uses single pulse	uses a pair of pulses
Requires high peak power	less peak power is required
Affected by weather	not affected by weather
Tx and Rx both are on ground	Tx and Rx both are at different position
depends on characteristics of reflection object	independent of such considerations



DISTANCE MEASURING EQUIPMENT (DME)

PRINCIPLE – Jittered PRF/ Random PRF/Unique twin pluses

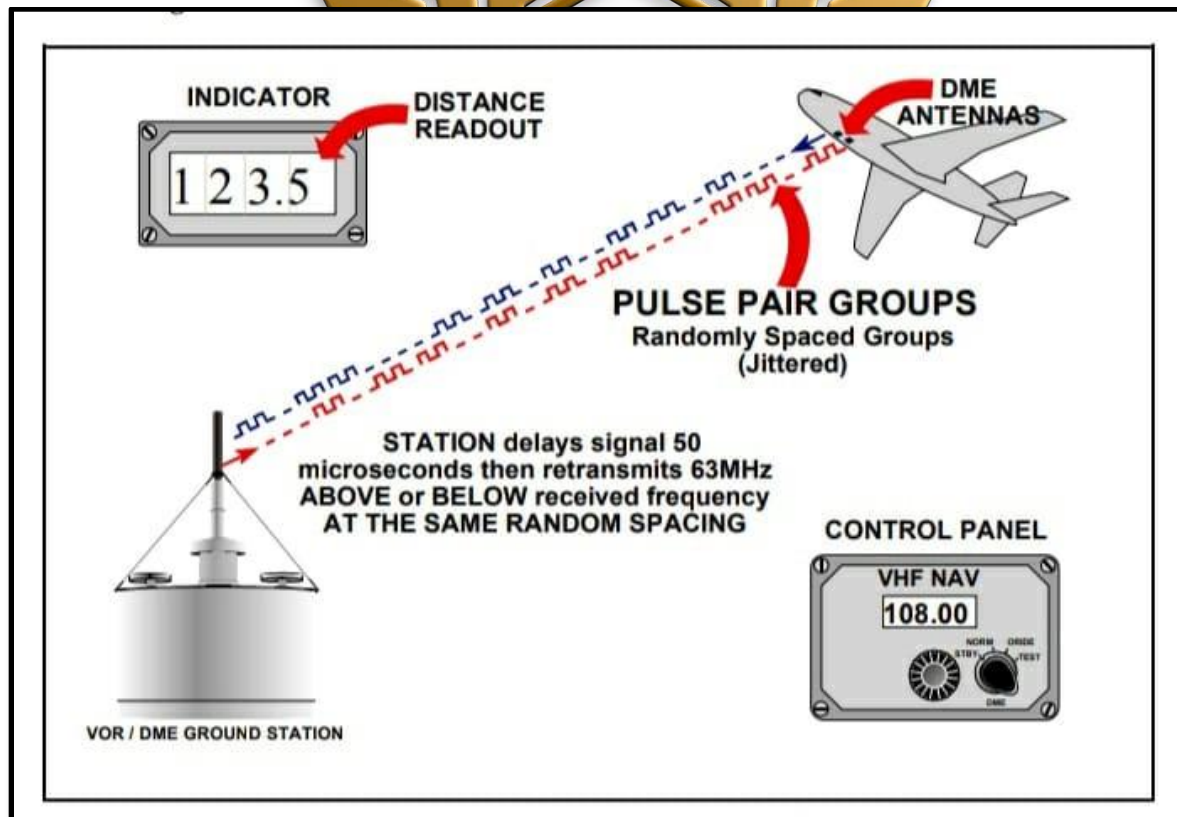
FREQUENCY – 962-1213 MHz

BAND – UHF

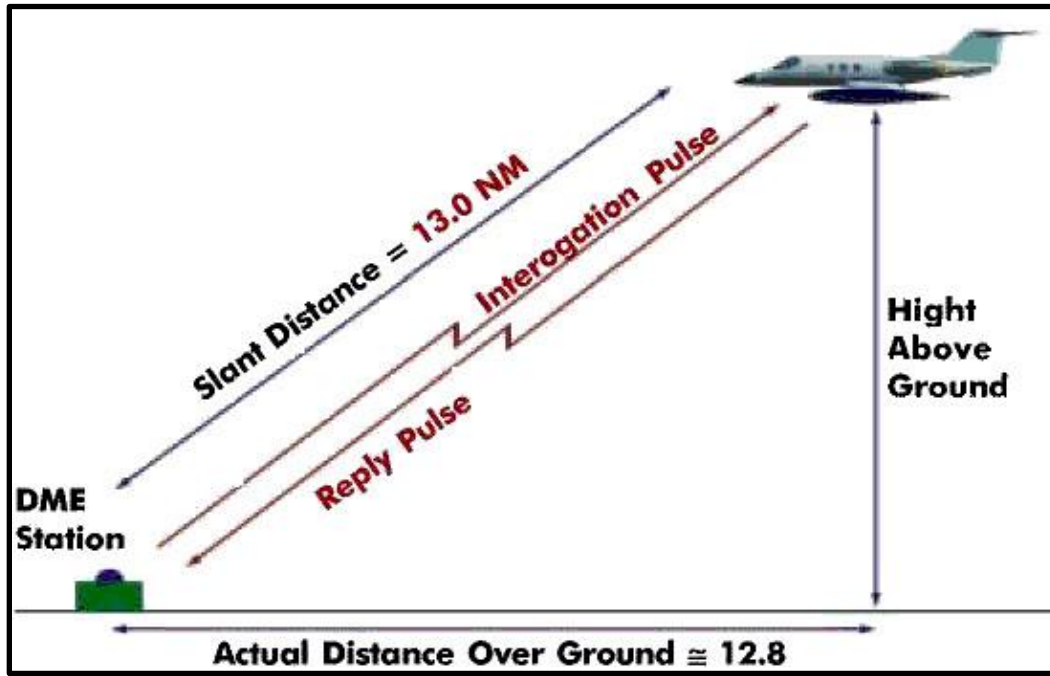
INDICTION – Slant range and ground speed

OPERATION BASE – SSR

EMMISSION CODE – P0N



- ✓ It is a secondary radar system which provides accurate SLANT RANGE of the a/c from the ground transmitter.
- ✓ The interrogator is on the a/c and the transponder is a ground.



WORKING

- ✓ An ac wishing to interrogate and find its range sends a series of pairs of pulses at random time intervals on the ground station freq. the ground station receives these pairs of pulses and after fixed delay of 50 microseconds sends a reply in the same pattern (PRF) that the pairs of pulses arrived.
- ✓ But the reply is on the freq. which is 63 MHz different (plus or minus) from the freq. transmitted by the ac.
- ✓ The ac equipment and when it gets a reply of a similar pattern but on a freq. different by 63 mhz. it recognizes these as replies to its transmission.

- ✓ The time delay between transmission and reception and adding the extra 50 microseconds gives the ac EQUIPMENT. The distance between the ac and the ground station.
- ✓ The distance between 2 pulses of a pair is 12 microseconds. The time interval between the pairs is varied at random (called transmission at random PRF)
- ✓ The ground equipment gives a reply on a freq different from ac freq by 63 MHz

This is as per the following:

	Ac transmits	ground station transmits
➤ Low band	1025-1087	962-1024 (at-63) called Y Channel
➤ High band	1088-1150	1150-1213 (at + 63) called X channel

So there are 63 X channels and 63 Y channels. A combination of ac and ground stn is called a channel.

MODES OF OPERATIONS

1. Search Mode – PRF 150-PPS upto 15000 Pulse
2. Stand By – PRF60-PPS
3. Lock Mode – PRF (27-30) PPS

JITTERED PRF

- ✓ The receiver on ground receives this pattern of random PRF sent by us and many other patterns sent by other ac.
- ✓ The transponder replies to all, but our ac equipment does not accept all these replies. It knows the pattern that it sent and opens the gate to accept only those replies which match our pattern.

SELF TRIGGERING:

In case the ac and ground equipment had been on the same freq. then there would be a problem.

- ✓ The transponder (on ground) transmission will be reflected by objects and obstacles in its vicinity.
- ✓ Some of these reflections would come back to the transponder. Transponder will think that these are from the ac will reply to these and out range of these reflecting objects.
- ✓ This process is called self-triggering and is avoided in secondary radar by transmission and reception being on different frequency

BEACON SATURATION:

- ✓ DME beacon gets saturated when more than 100 ac trigger it at the same time.
- ✓ The transponder transmits a total of 2700 PPS random whether or not is being triggered by ac or not.
- ✓ We know that ac transmit at 25-30 PPS. If we take an average of 27PPS, then total no. ac to which these 2700 PPS can send replies is $2700/27 = 100\text{ac}$.

- ✓ So if there are more no of ac, the equipment will reply to only the stronger 100 signal by reducing its gain.

The range of DME depends upon:

1. Ac ht
2. Transmitter ht
3. High ground in the direction

ACCURACY: ADVANTAGES OF DME:

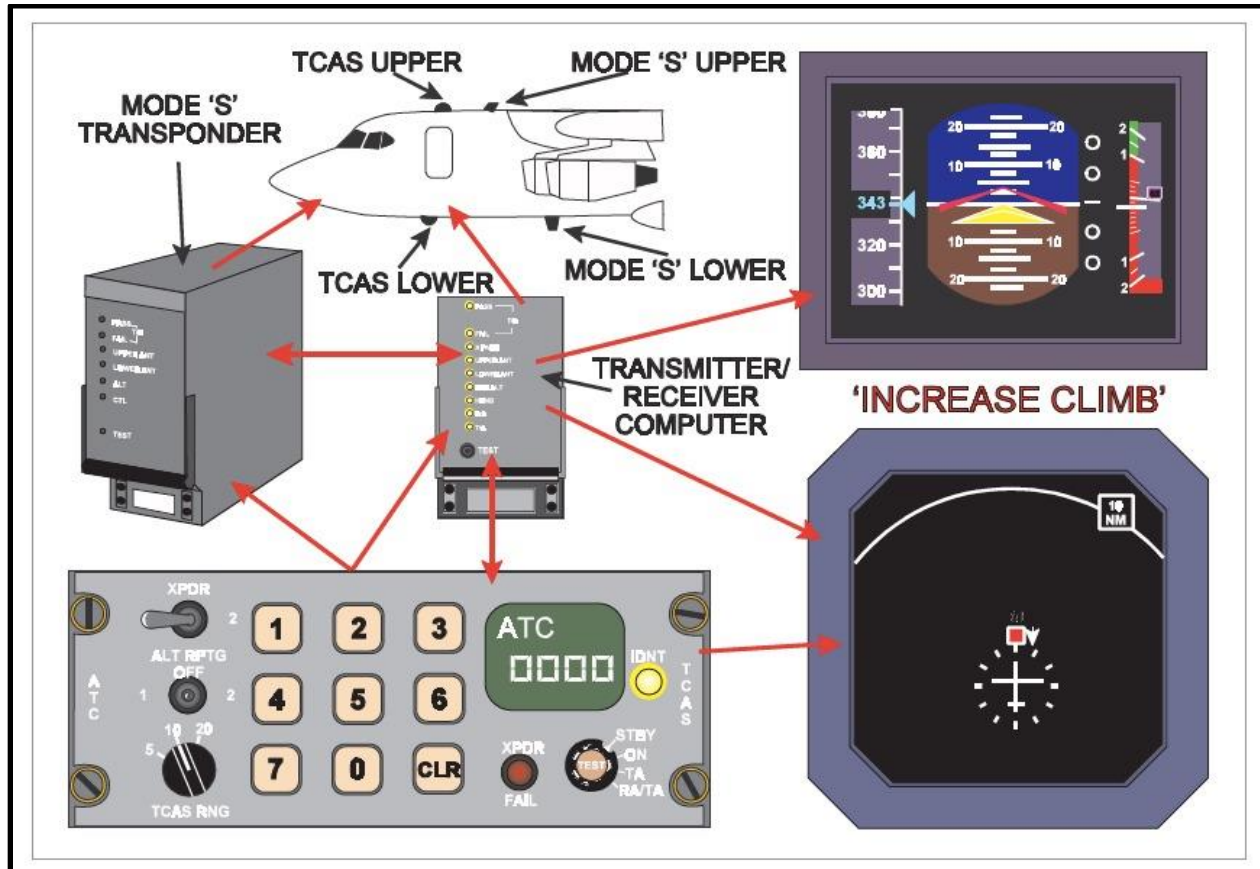
- ✓ No interference due weather as reflection from clouds are on same freq as Tx, so these will not be different by 63 MHz and so not accepted valid return
- ✓ Transmission power required is less as 2 ways travel not required.
- ✓ Self-triggering does not take place.
- ✓ **ACCURACY:** As per ICAO accuracy 95% of the times is ± 0.25 nm plus 1.25% of ac range measured.
- ✓ Accuracy of precession DME is 0.20 NM.
- ✓ **IDENTIFICATION:** 3 letter call sign transmit every 30 sec. If DME and VOR are co-located then their ident can be distinguished because DME has a higher tone.
- ✓ **RANGE:** almost 200 NMS. VHF FORMULAE
- ✓ **SLANT RANGE:** DME measures slant range which will always be more than actual range.

- ✓ **TACAN (TACTICAL AIR NAVIGATION):** military NAV aid uses a DME station to provide range. The facility cannot be used by civilian ac. But its DME station can be used by civilian ac
- ✓ **Conditions when VOR & DME can be frequency paired:-**

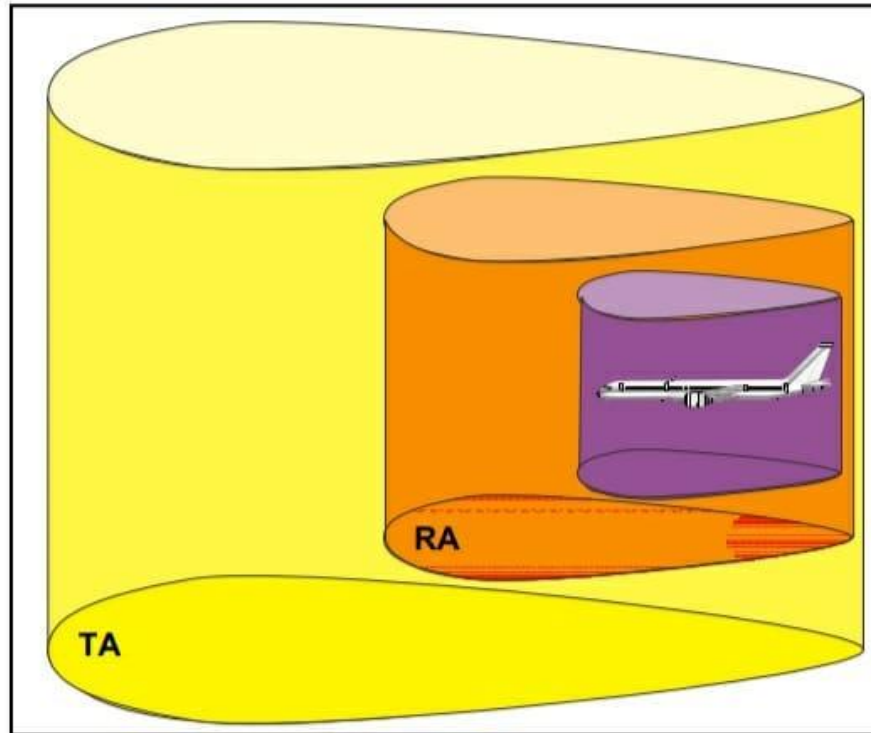
Relative PSN of both Tx	Frequency Paired
Both transmitter are within 30m/100 ft within the TMA	Paired
Both Tx are 600m/2000' outside TMA	Paired
Co-located	Paired
If both Tx are less then 6NMs apart & serving the same station	Paired, letter Z will appear in the end of ident



TRAFFIC ALERT & COLLISION AVOIDANCE SYSTEM (TCAS/ACAS)



AVIATION



PRINCIPLE

- ✓ Secondary radar
- ✓ Range 30 nm
- ✓ Frequency 1030 & 1090 MHz

TYPES OF TCAS

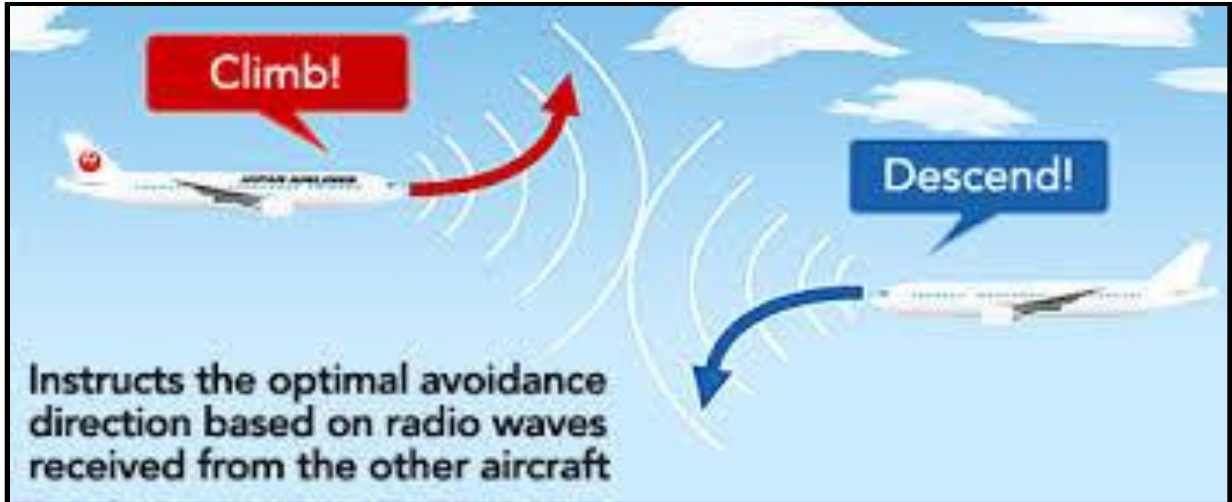
TCAS/ACAS-1

- ✓ ORAL WARNING – TRAFFIC ADVISORY- TRAFFIC TRAFFIC
- ✓ VISUAL WARNING – YELLOW CIRCLE 
- ✓ Designed to aid visual acquisition
- ✓ Gives traffic advisories (TAs). TAs are visual display of all responding air traffic

- ✓ Normally gives about 40 seconds warning of a possible collision
- ✓ Gives bearing and distances of other aircraft
- ✓ Pilot has to then look out in the direction and take avoiding action as deemed fit

TCAS/ACAS II

- ✓ **ORAL WARNING** - Resolution Advisory – **Climb, Increase Climb Descend, Increase Descend, Monitor Vertical Speed.** These are enough.
- ✓ **Visual Warning** - RED SQUARE 
- ✓ TCAS II provides not only traffic advisory but also resolution advisory in vertical plane only.
- ✓ More accurate
- ✓ When computer calculates that a conflict (crash) is possible, it generates resolution advisories (RAs) describing avoidance maneuvers only in climb and descent (voice messages)
- ✓ Can track up to 45 aircraft and can display up to 30
- ✓ Can coordinate RAs up to 3 conflicting aircraft
- ✓ Will not generate any RAs below 1000 feet above ground
- ✓ Nor will an aircraft be given RAs to descend below 1000 feet
- ✓ TCAS III, ACAS III future system will give horizontal RAs also



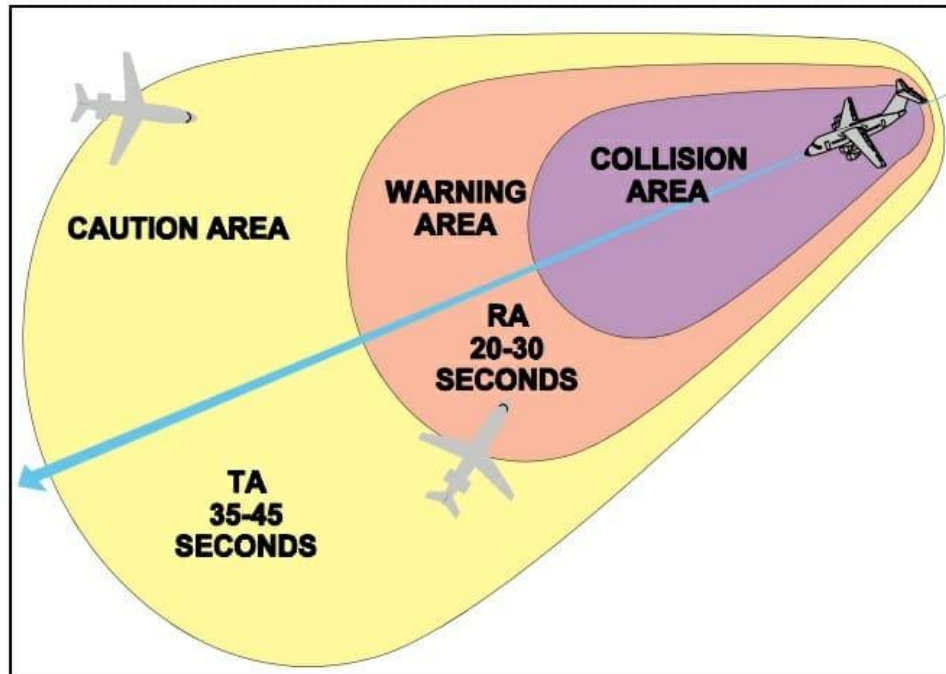
TYPE OF RA

1. Preventive RA – In this crew need not to take immediate action to prevent in pending collision.

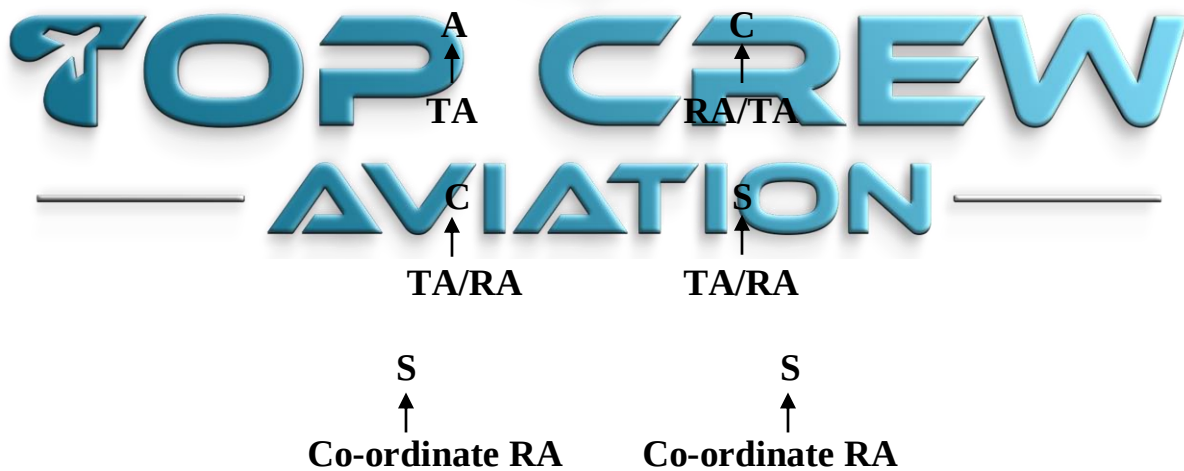
Example – Monitor vertical speed

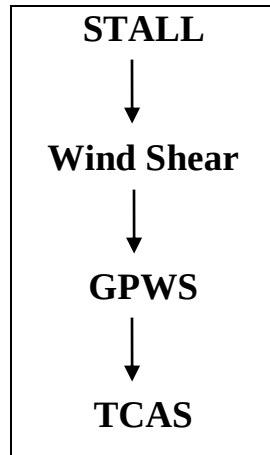
2. Corrective RA – Resolution advisory in which crew needs to take appropriate action immediately to avoid in pending collision





NOTE: TCAS can track 45 A/C at a time. It can display 30 of them and provide co-ordinate RA to 3.



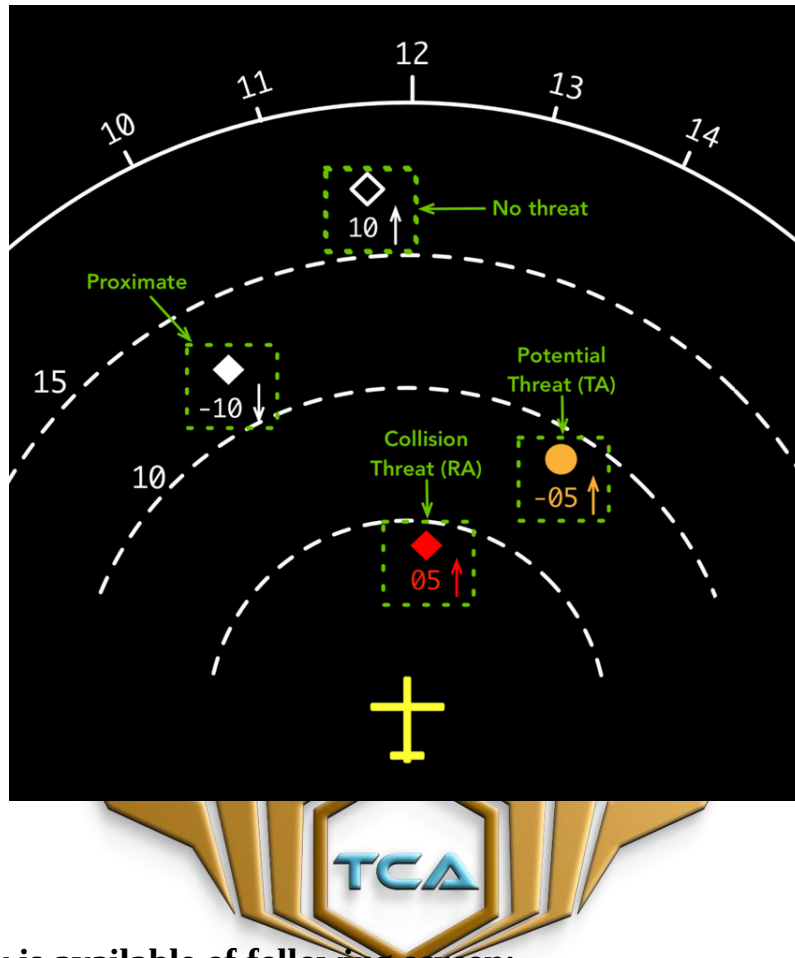
Prioritization of warning:**VISUAL INDICATION OF TCAS****OTHER TRAFFIC**

- ✓ Other Traffic appears as hollow cyan diamonds  which represent transponder equipped aircraft within range of the display and within +/- 2700 feet relative height.

PROXIMATE TRAFFIC

- ✓ Proximate Traffic appears as a solid cyan diamond  and represents transponder equipped aircraft within range of the display and within +/- 1200 feet relative height. TCAS does not consider this traffic a threat and displays it to improve crew situational awareness. Any traffic within 6NM $\pm$ 1200ft

**TA****RA**



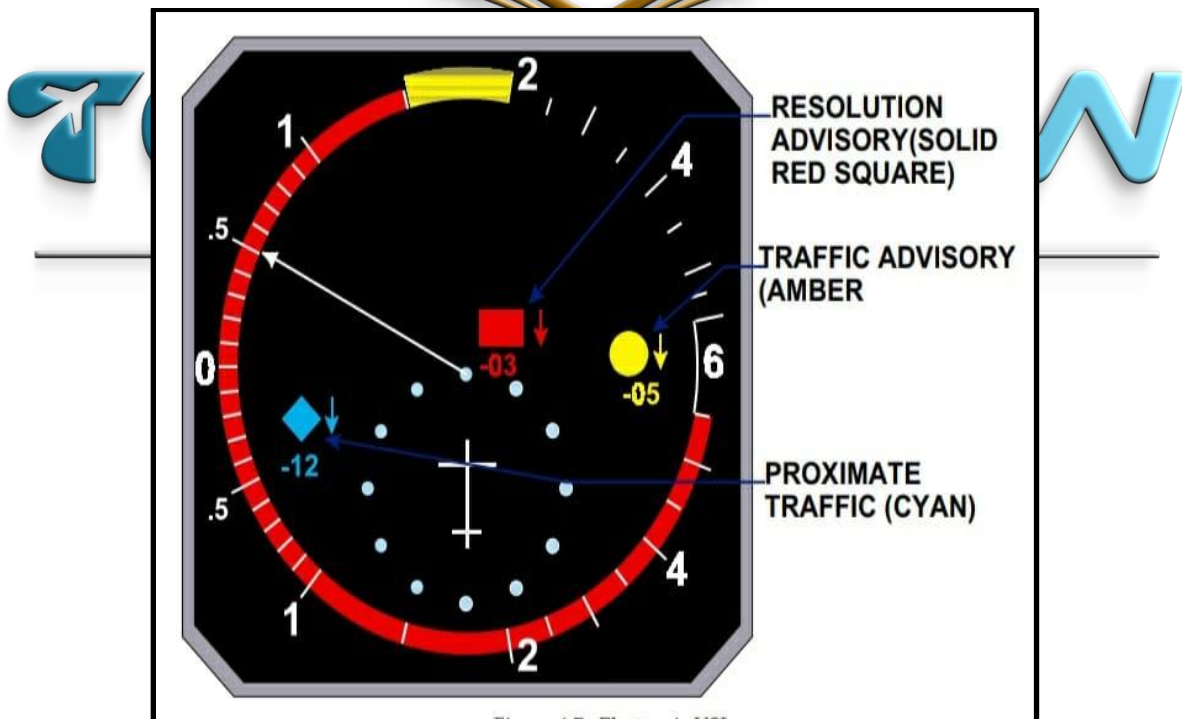
TCAS Display is available of following screen:

1. Electronic horizontal situation indicator/ND (Navigation display)

TOP CREW
— AVIATION —



2. Electronic VSI (RAVSI)



3. Dedicated TCAS display

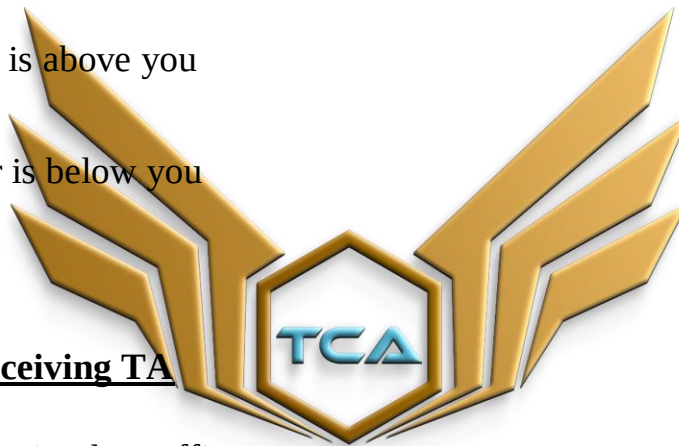
DATA TAG

↓ Arrow down – Descending more than 500 ft/pm

↑ Arrow UP - Climbing more than 500 ft/pm.

+ Means Intruder is above you

- Means Intruder is below you



Crew action of receiving TA

1. Visually Identify the traffic
2. Co-ordinate with ATC
3. Be prepared for subsequent RA

Note: Crew may know TA if after visual identification confirms that no flight path deviation is required.

Crew action on receiving RA

- ✓ Comply the RA command immediately disengage auto pilot & FD's. Once clear of the conflict inform ATC of deviation.

Note: In case of conflict B/W ATC instruction & TCAS command follow the TCAS commands.

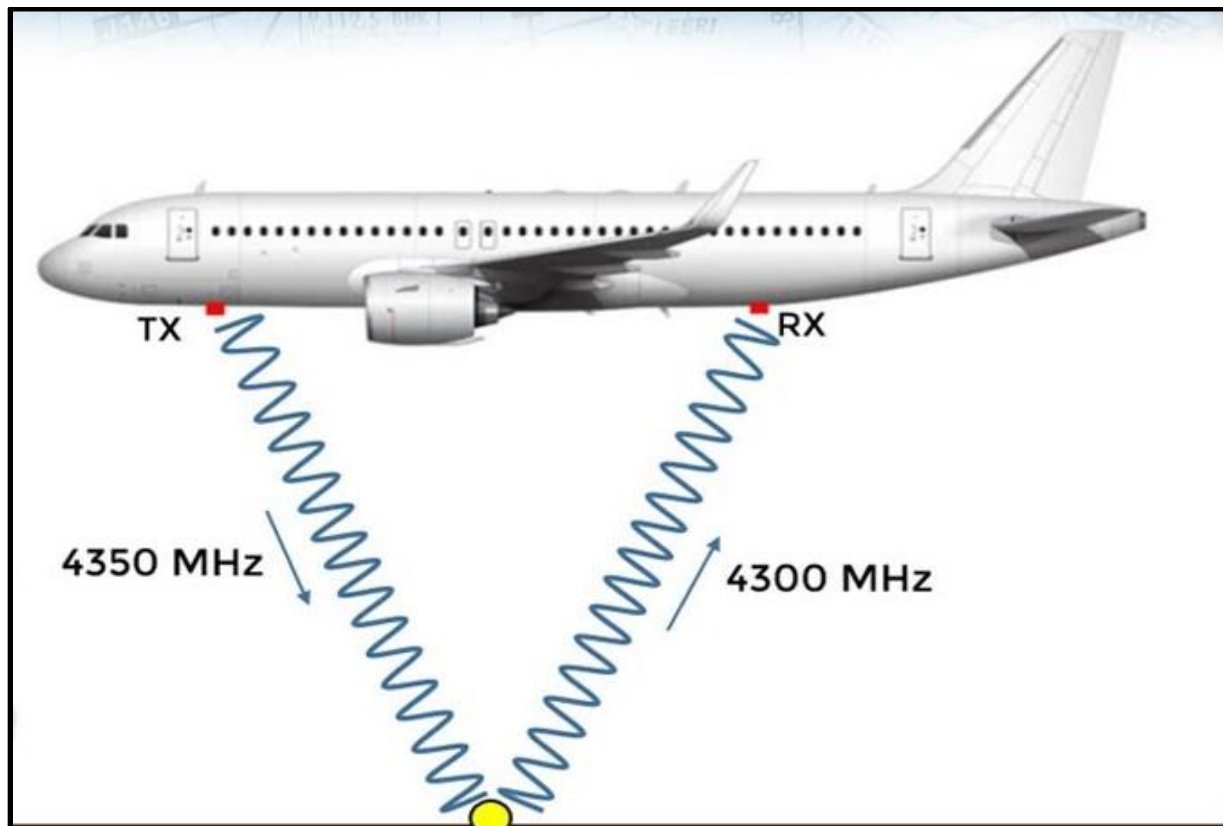
TCAS Operational Restrictions:

1. No increase descend RA below 1400 agl.
2. No descend command below 1000' agl.
3. No command below 400' agl.

DISREGARDING RA

- ✓ Manoeuvres should never be made in a direction opposite to that given in an RA; this is because the sense may have been determined following an exchange of data with the established threat. For this reason:
- ✓ RAs may be disregarded only when pilots visually identify the potentially conflicting traffic and decide no deviation from the current flight path is needed.
- ✓ If pilots receive simultaneously an instruction to manoeuvre from ATC and an RA, and both conflict, the advice given by TCAS should be followed

RADIO ALTIMETER



Principle – Frequency Modulated continuous wave

Frequency – (4200-4400) MHz, Mean frequency – 4300

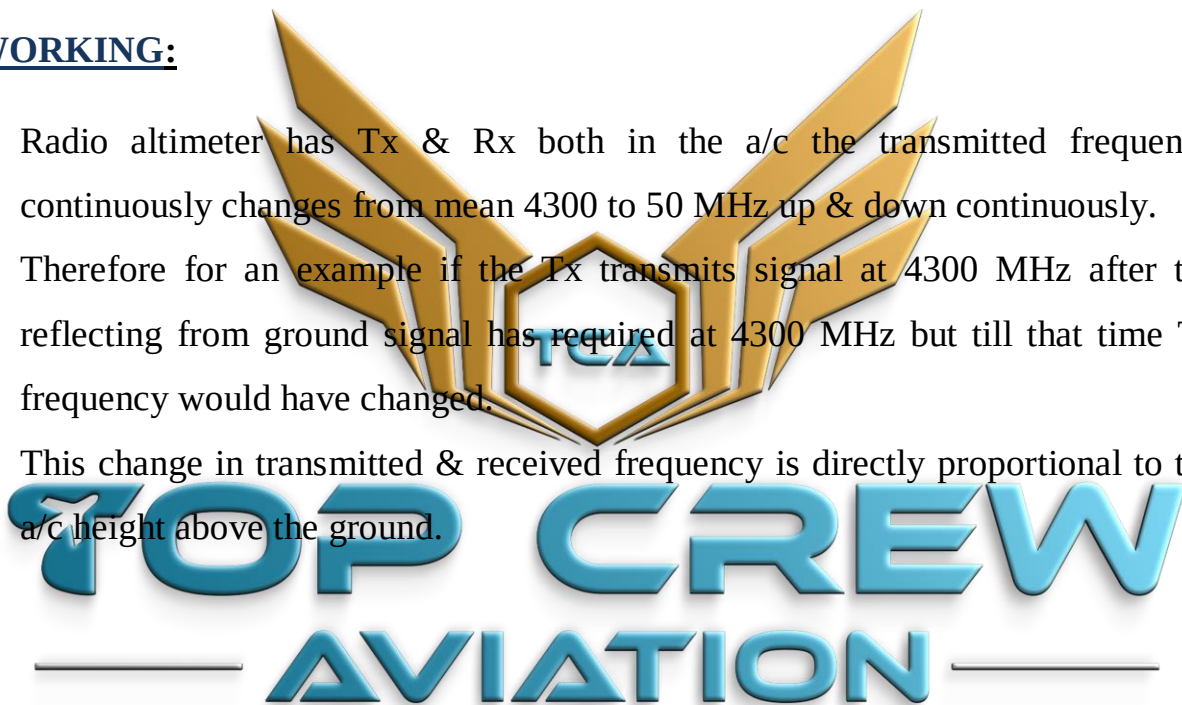
(High altitude radar altimeter from – 1600 MHz UHF)

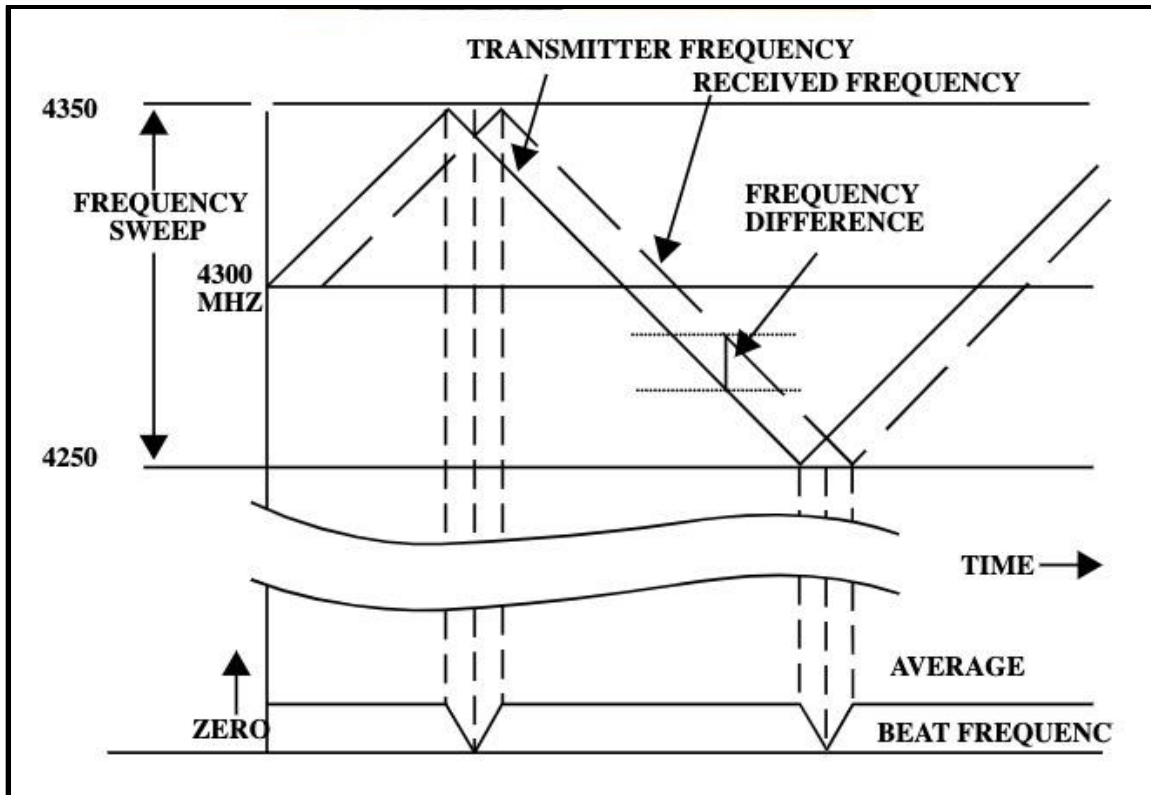
Indication – Height above the ground level in LDG configuration

Band– SHF

WORKING:

- ✓ Radio altimeter has Tx & Rx both in the a/c the transmitted frequency continuously changes from mean 4300 to 50 MHz up & down continuously.
- ✓ Therefore for an example if the Tx transmits signal at 4300 MHz after the reflecting from ground signal has required at 4300 MHz but till that time Tx frequency would have changed.
- ✓ This change in transmitted & received frequency is directly proportional to the a/c height above the ground.





TYPES OF DISPLAY:

1. Stand alone radio altimeter

This is a linear display in which there is a needle which indicates radio height and a mask. The needle of radio altimeter disappears in following cases:

- A. Operating above 2500'
- B. Transmitter fault
- C. Power failure

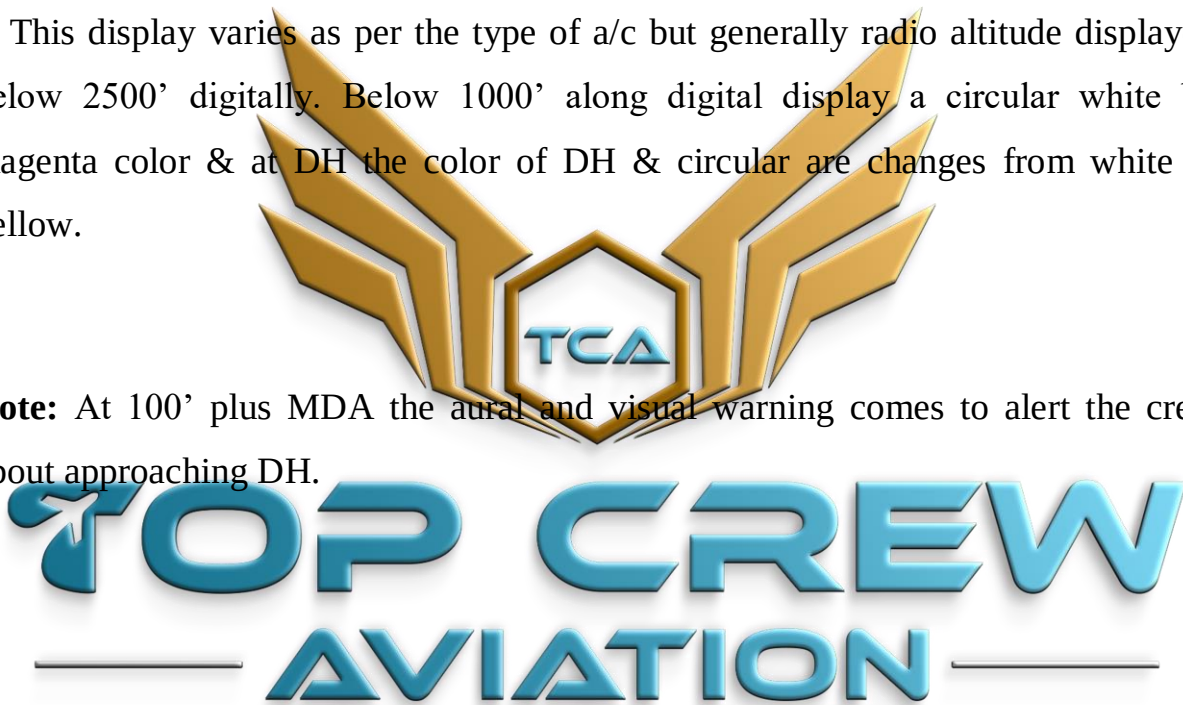
Note:

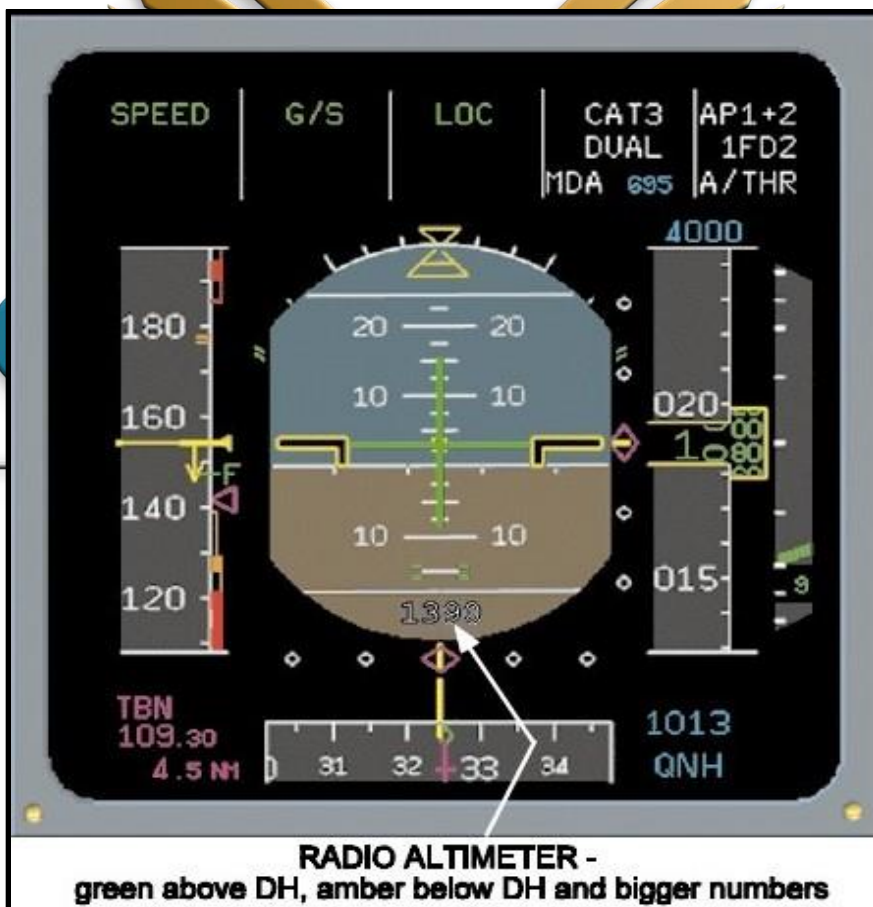
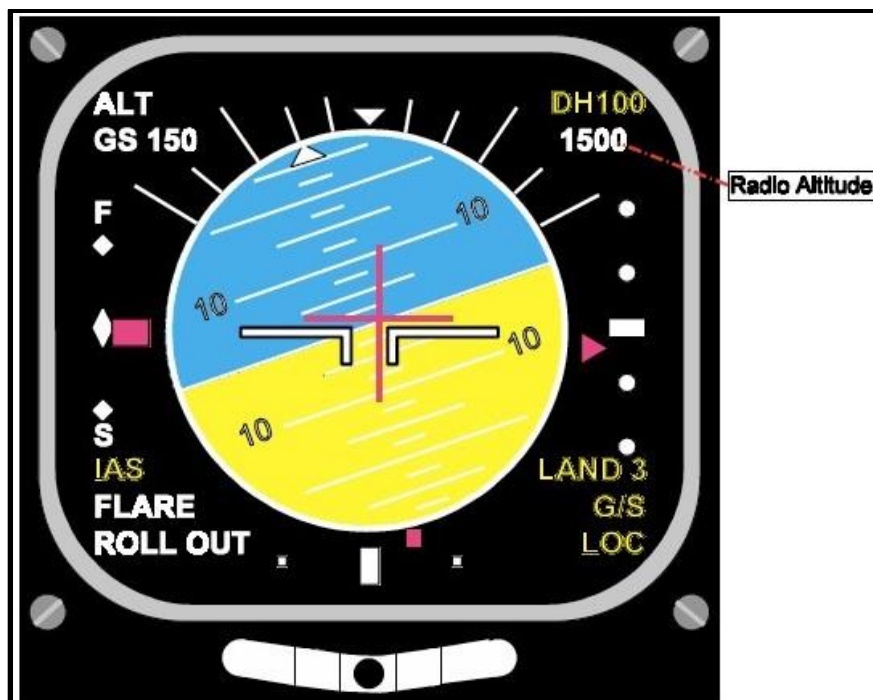
1. There is a warning flag which occurs when system is switched off or during a self test, system detect signal strength.
2. There is a DH selecting NOB where crew can select the decision height upon reaching a DH oral or visual alert is provided to the crew

2. EFIS integrated radio altimeter display

This display varies as per the type of a/c but generally radio altitude displayed below 2500' digitally. Below 1000' along digital display a circular white by magenta color & at DH the color of DH & circular are changes from white to yellow.

Note: At 100' plus MDA the aural and visual warning comes to alert the crew about approaching DH.





EQUIPMENT:

- ✓ As the equipment uses FMCW principle there is separate Tx and Rx with their respective aerials.

USES:

- ✓ Used while flying at low levels. Most equipment caters for ht up to 2000 to 2500 ft and is compatible with the GPWS. But the major disadvantage is that it gives the ht below and not ahead required for warning.

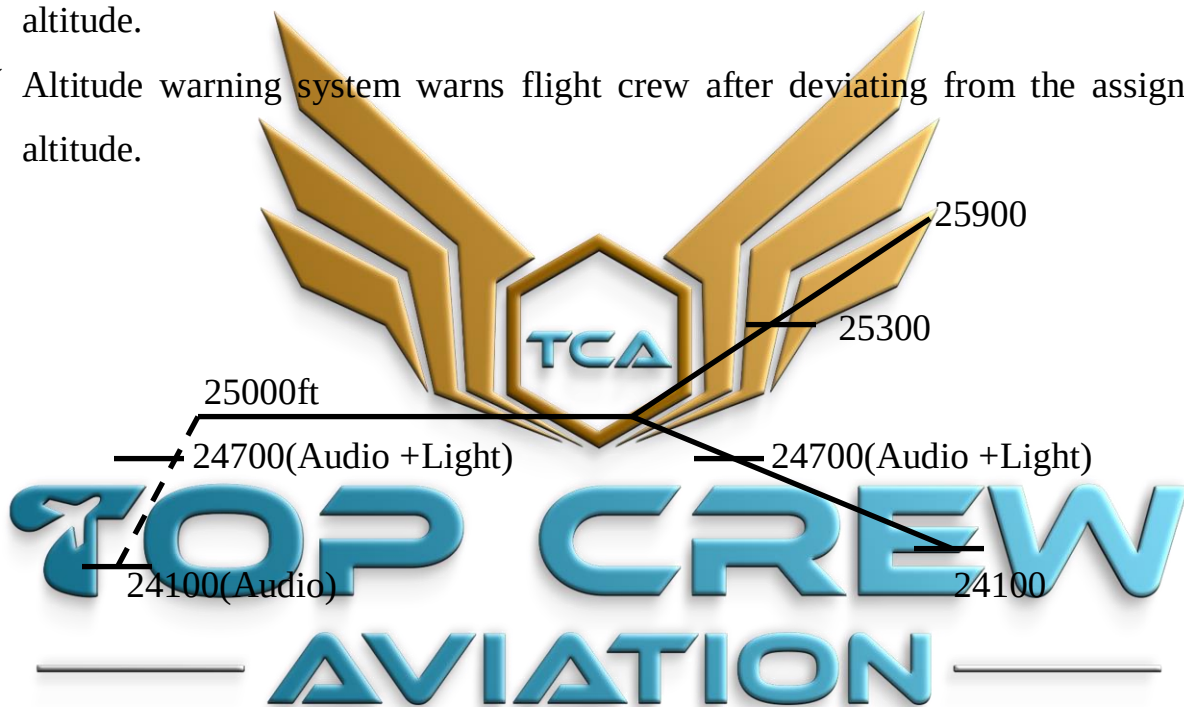
ACCURACY:

- ✓ The equipment measures in steps of 5 ft, so that is the min accuracy. Overall accuracy is 5 ft +/- 3% of the indicated ht. the equipment is mainly read to most accurate when coming for a landing so it is calibrated to read 0 with wheels down. This means that when ac is in level flt with wheels up, it will read slightly less than the actual ht.

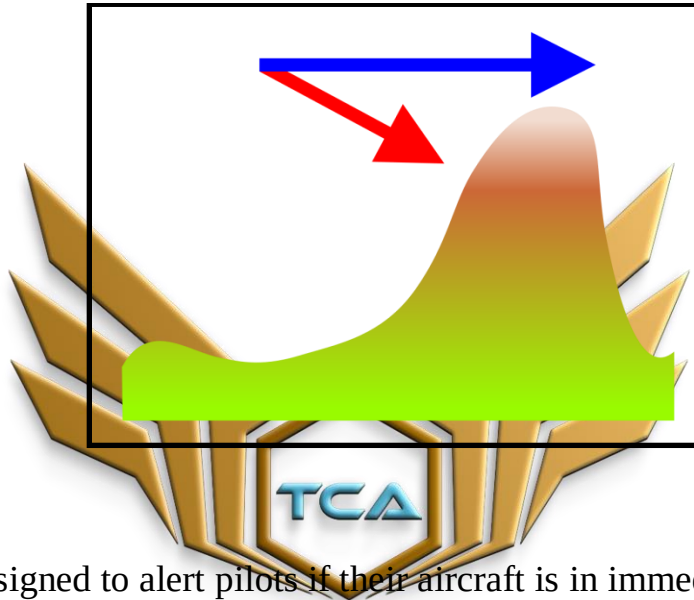
ALTITUDE WARNING

- ✓ An altitude alerting system must be capable of warning the pilot of approaching the selected altitude and also when deviating from the selected altitude. The alerts are visual and / or aural.
- ✓ It is similar to DH but is used in conjunction with pressure altimeter and takes its inputs from the air data computer.
- ✓ The altitude selected knob is at the bottom right of the equipment and the warning light on the top right.

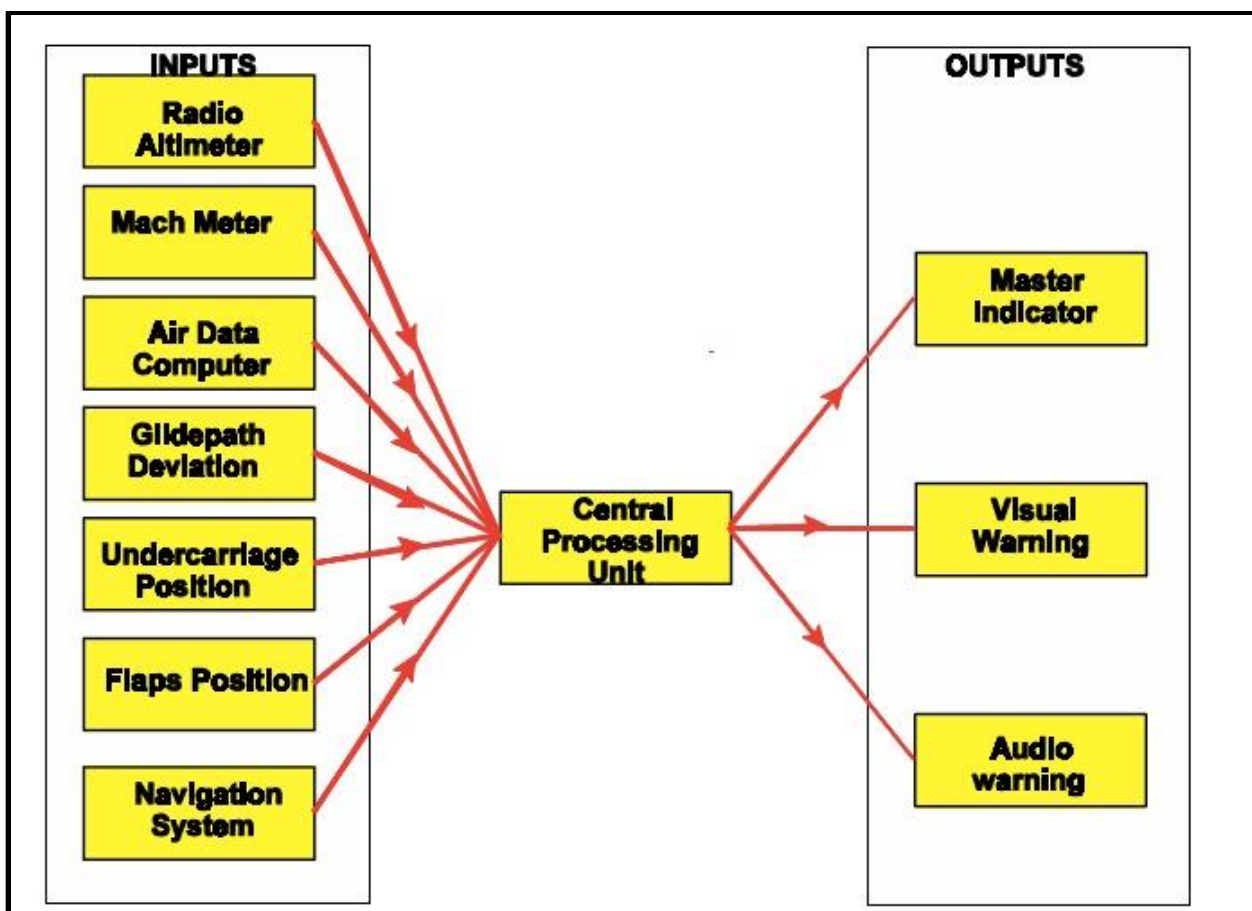
- ✓ As the ac approaches the selected altitude, 900 ft before reaches it gets an audio warning.
- ✓ When 300 ft from the selected altitude, the warning light goes out. Subsequently while flying at that altitude, if the ac climbs or descends by more than 300 ft, the light flashes and audio warning sounds.
- ✓ Once the ac deviates by more than 900 ft, all warning stop and the system resets itself. Such systems are mandatory for flying in RVSM airspace.
- ✓ Altitude alerting system warns flight crew before reaching at the selected altitude.
- ✓ Altitude warning system warns flight crew after deviating from the assigned altitude.



GPWS (GROUND PROXIMITY WARNING SYSTEM)



- ✓ Is a system designed to alert pilots if their aircraft is in immediate danger of flying into the ground or an obstacle.
- ✓ It gives audio and visual warnings. Inputs are taken from radalt VSI, guide path indicate, U/C position and flap position.
- ✓ These are fed to a CPU and relevant outputs given. The envelope of operation is from 50 ft to 2450 ft.



GPWS MODES

GPWS MODE		ADVANCED EQUIPMENT Alert(60 sec) Warning(30sec)	
1. Excessive descend rate	50-2450'	'sink rate' 'sink rate' Repeated each 1.5sec+Amberlight	'whoop whoop pull up'
2. excessive terrain closure rate	50-2450'	'terrain terrain' Repeated each	'whoop whoop pull up'

		1.5sec+Amberlight	
3. Negative ROC after take-off or go-around	50-700'	'don't sink'	
4. unsafe terrain clearance while not in the landing configuration	4A. proximity to terrain. Gear not locked down (from 1000') 4B. proximity to terrain- flaps not in a landing position (from 1000')	'too low gear' 'too low flaps'	'too low terrain' 'too low terrain'
5. descent below glide slope	From 1000' or less	'glide slope'	-
6. Decision Height	50-1000'	'minimums'	-

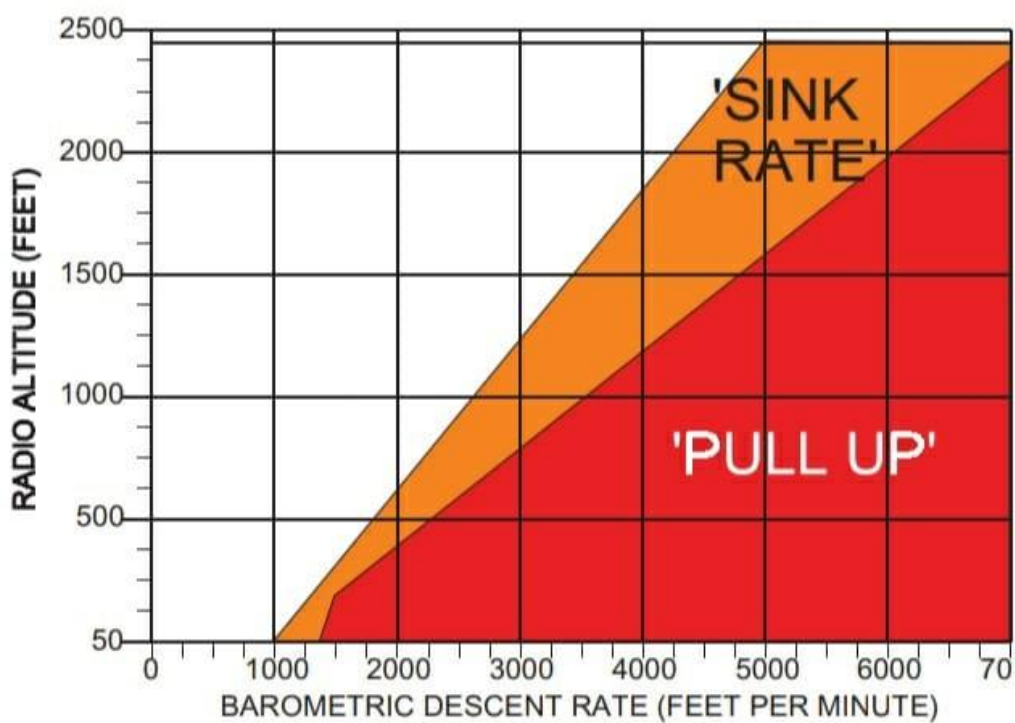
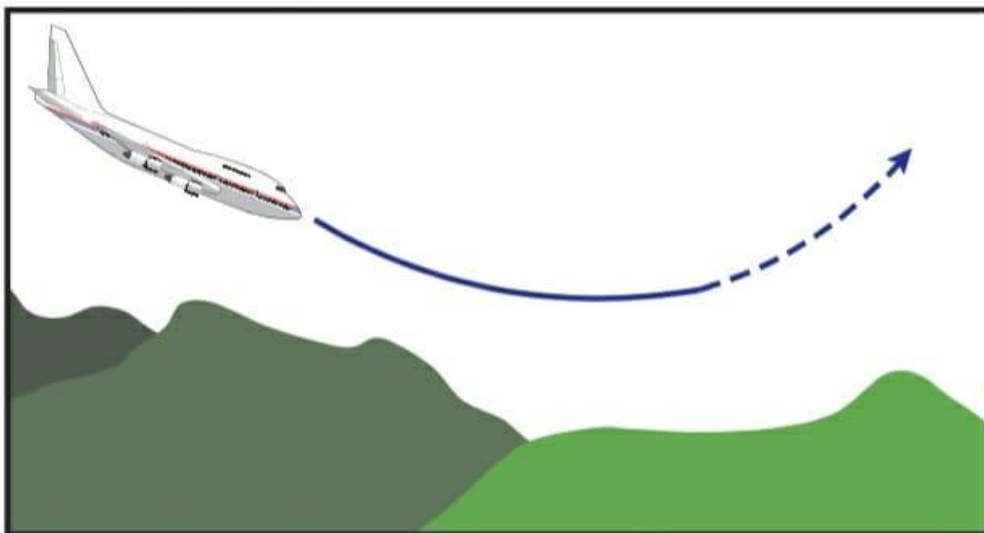
TOP CREW
— AVIATION —

MODE 1

AURAL ALERT - SINK RATE, SINK RATE

AURAL WARNING - 'WHOOOP WHOOP PULL UP'

VISUAL -

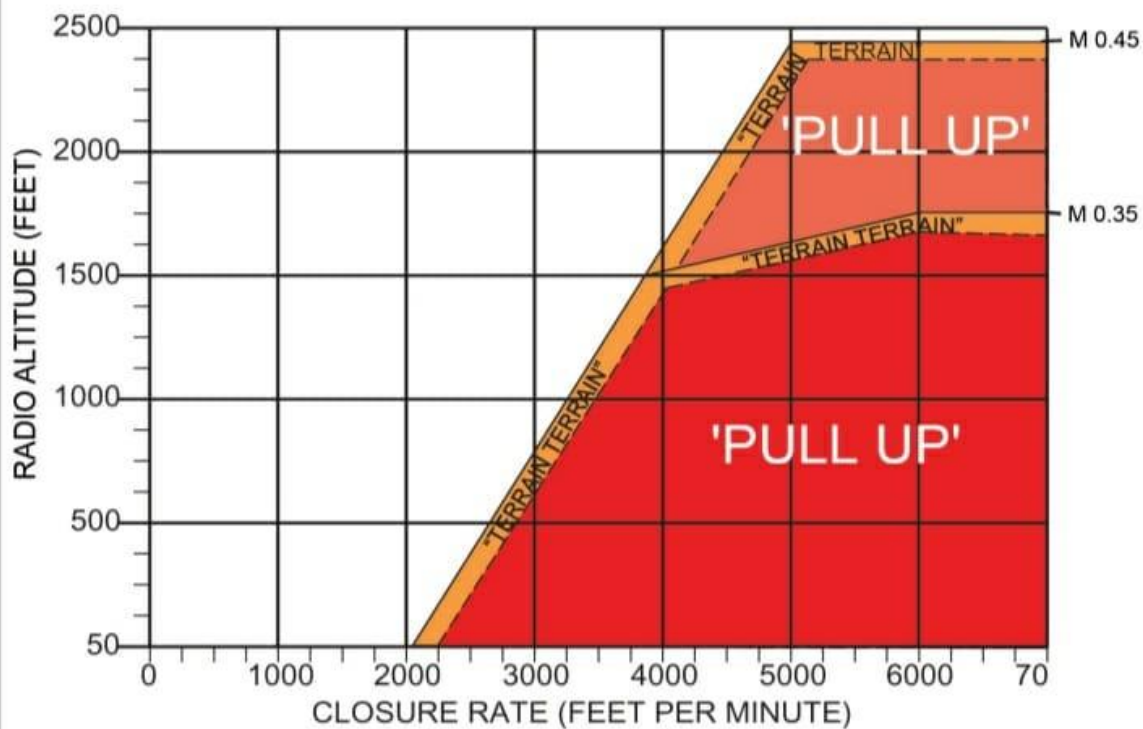
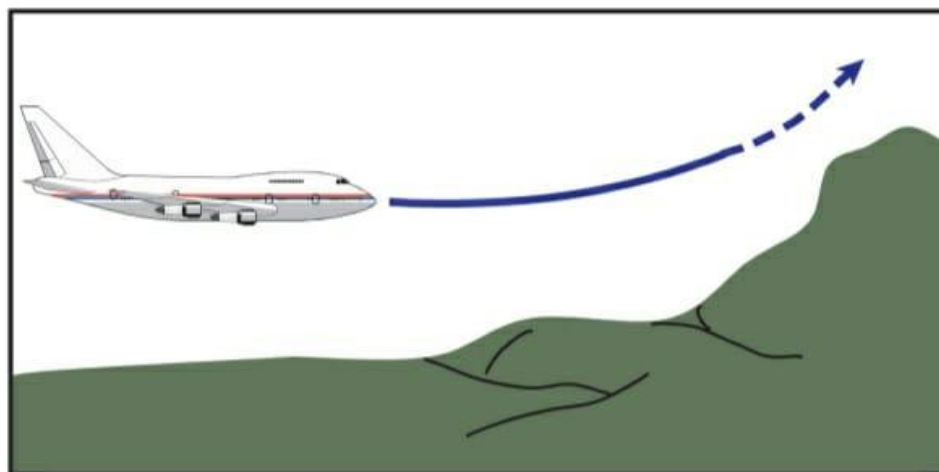
PULL UP

MODE 2

AURAL ALERT - 'TERRAIN, TERRAIN'

AURAL WARNING - 'WHOOOP WHOOP PULL UP'

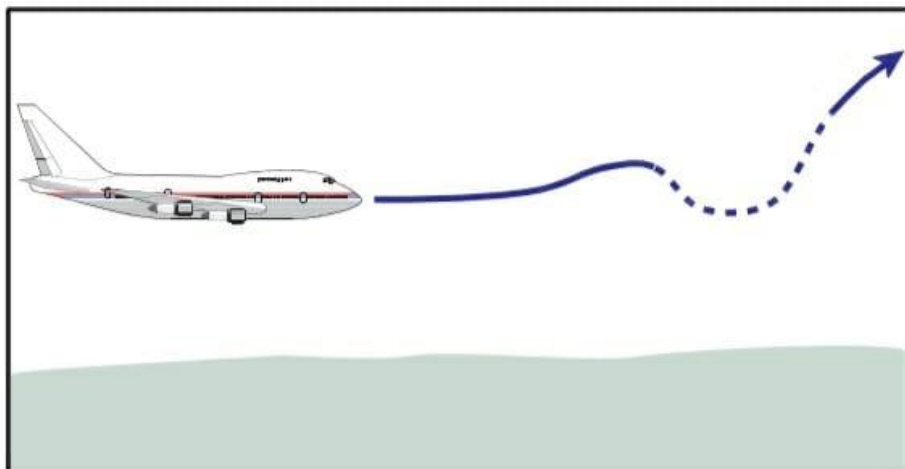
VISUAL -

PULL UP

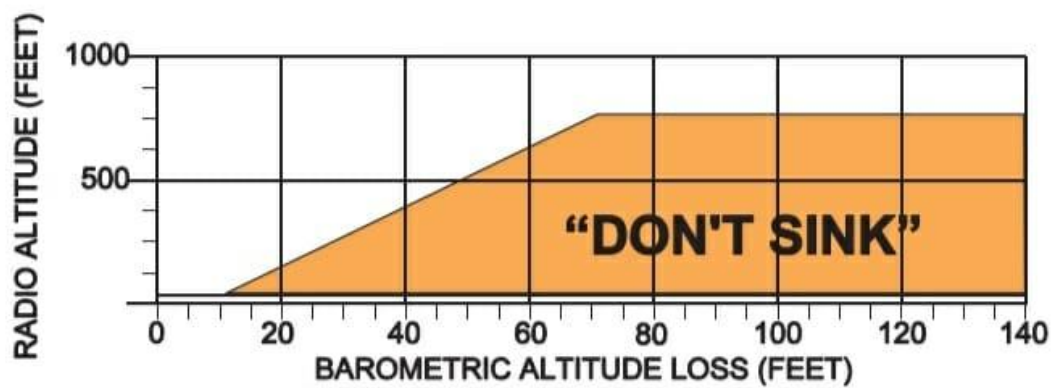
MODE 3

AURAL ALERT - "DON'T SINK"

VISUAL -

PULL UP

Note: Mode 3 arms when the aeroplane descends below 200 ft in the landing configuration

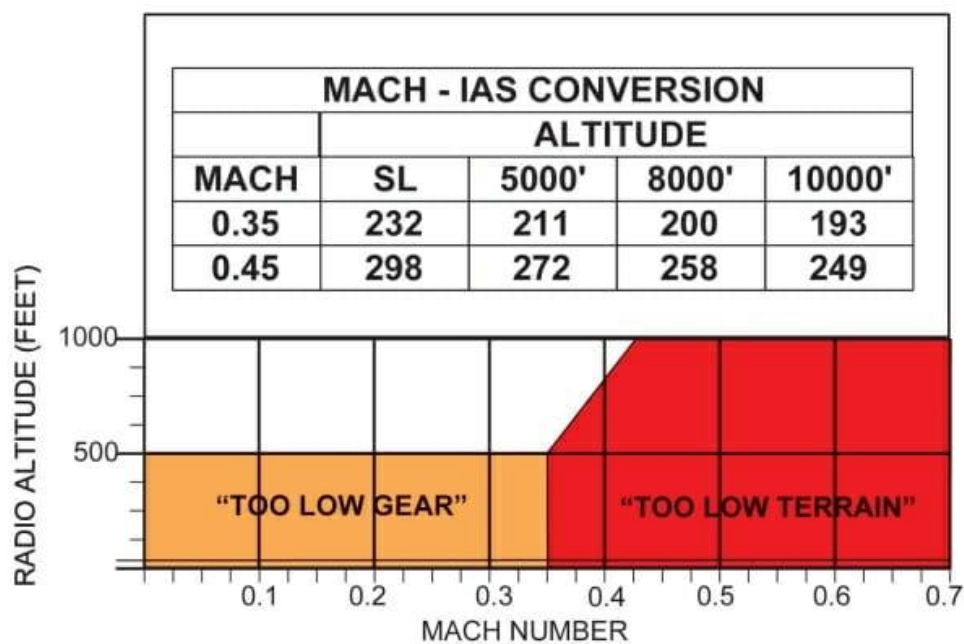
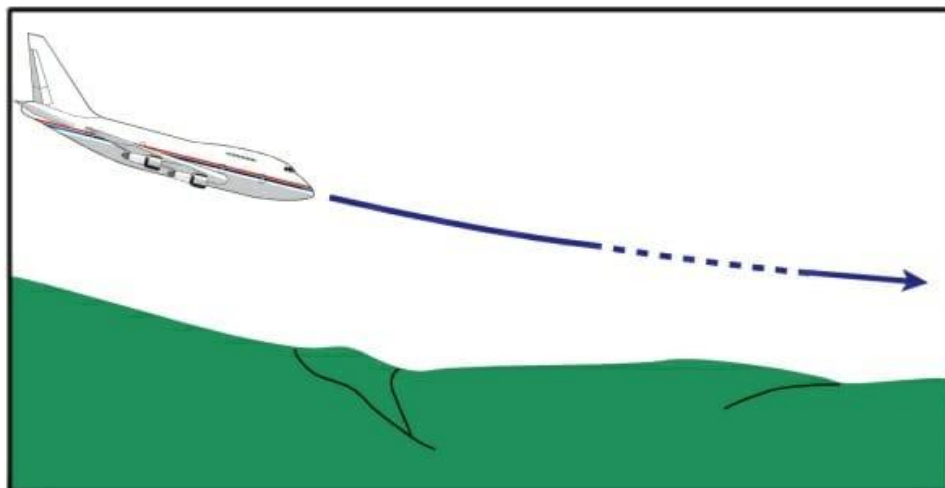


MODE 4A

AURAL ALERT - "TOO LOW GEAR"
"TOO LOW TERRAIN"

VISUAL -

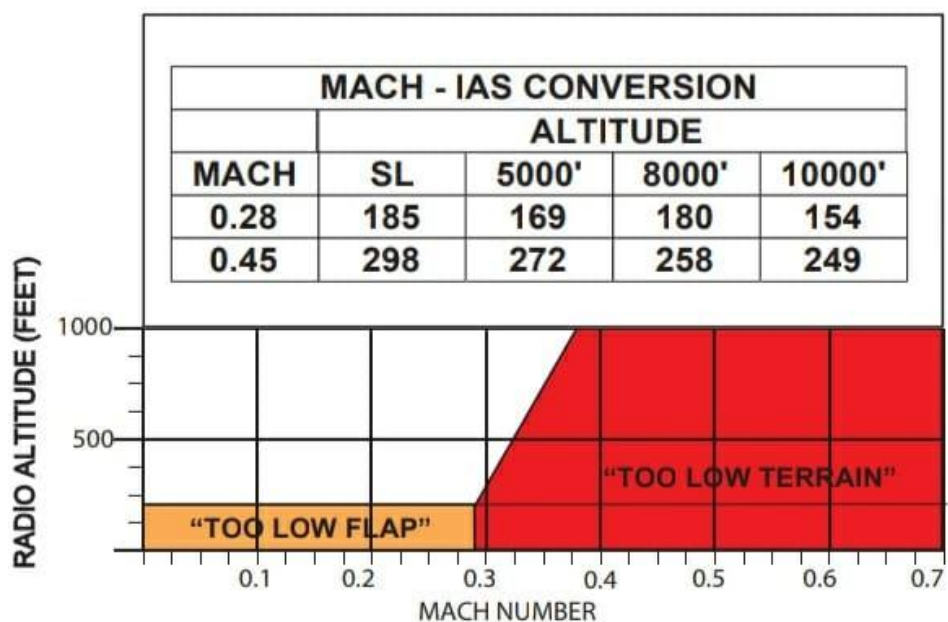
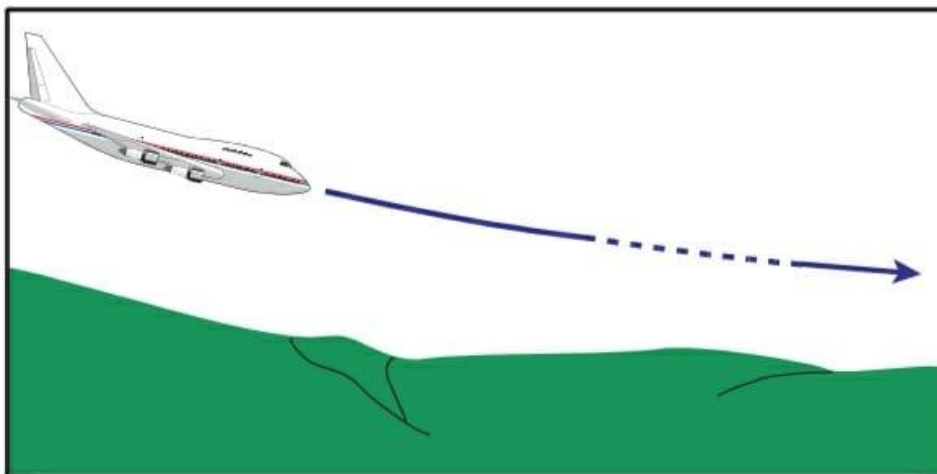
PULL UP



MODE 4B

AURAL ALERT - "TOO LOW FLAPS"
"TOO LOW TERRAIN"

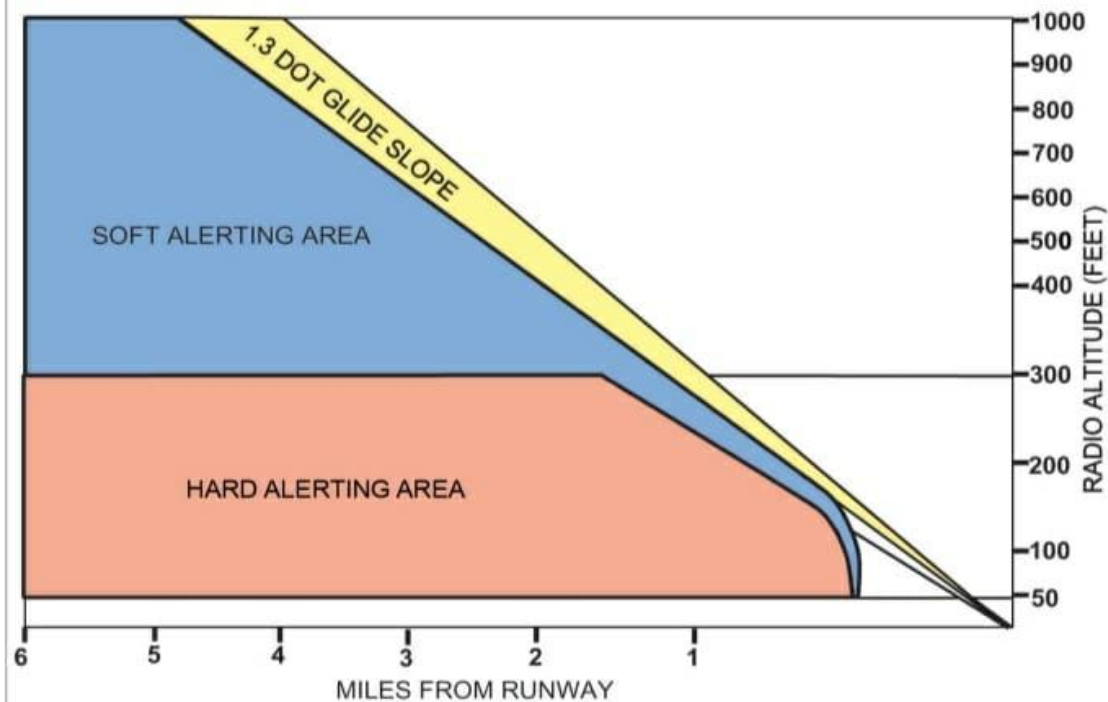
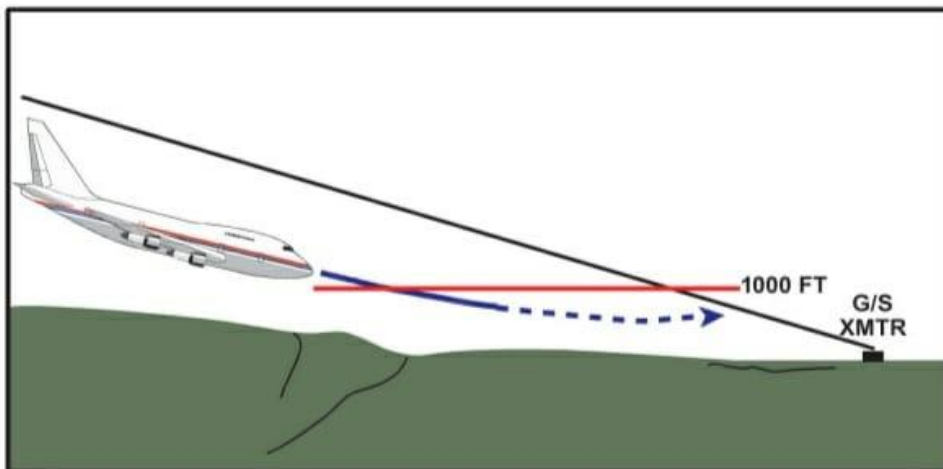
VISUAL -



MODE 5

AURAL ALERT - "GLIDE SLOPE"

VISUAL -

**BELOW G/S
P TO INHIBIT**

Note:

1. GPWS gives alert 60 sec before the impact.
2. GPWS has largely reduced accident associated with CFIT.
3. Mode 3 is activated when aircraft loses approximately 10% of current height.
Example - If aircraft is at 300' after takeoff, so mode C will activate at 270'.
4. When mode 5 is operational, mode 3 is inhibited.
5. Too low gear takes priority over too low flaps.
6. Mode 1 to 4 has a priority alert on mode 5 alert.

PILOT'S ACTIONS

- A. **ALERT** : Reduce ROD or level off or turn to safe area or lower U/C flaps
- B. **WARNING**: To be treated as command. Wings level, apply full T/O power, climb away at airspeed and angle for max climb gradient

BITE (BUILT IN TEST EQUIPMENT)

- ✓ Used only on ground, audio visual warning to indicate equipment serviceable

SERVICEABILITY:

- ✓ If GPWS is U/S, a/c should not fly. But if it can be repaired it is permitted to continue the flight to an aerodrome where it can be repaired, provided the flight consists of no more than 6 segments

DISADVANTAGE:

- ✓ The biggest disadvantage is that the GPWS has no look ahead facility, provided the flight consist no more than 6 segments.

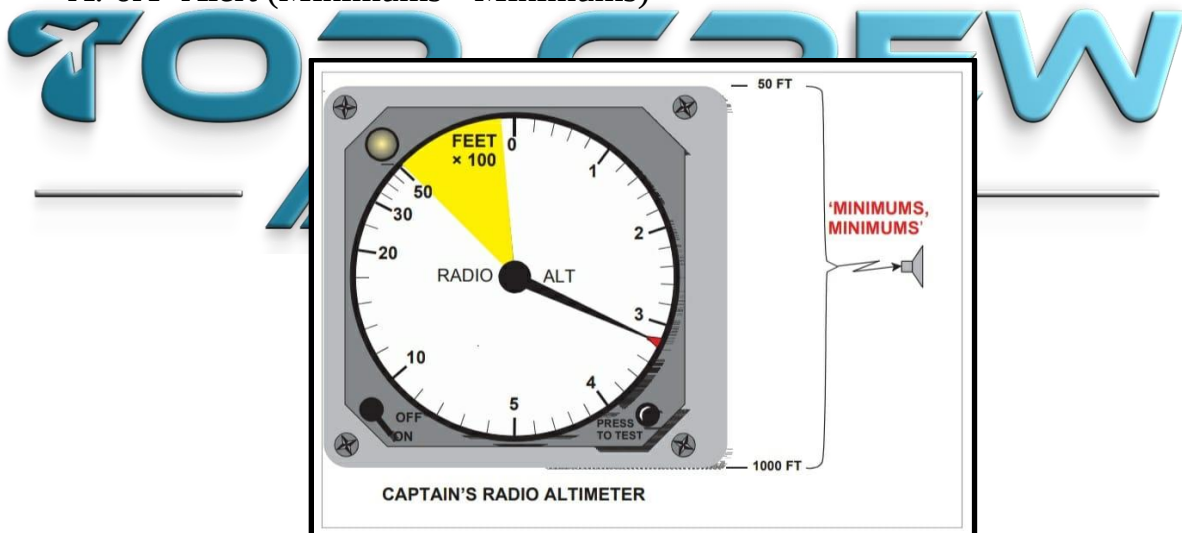
EGPWS (ENHANCED GROUND PROXIMITY WARNING SYSTEM)

- ✓ This is also called TAWS (terrain awareness and warning system) compare the position in 3D with that of the ground around it and then generates a warning. The info of ground around is available in the form of a data base. This data base is provide for general terrain worldwide, for obstacles in certain areas.

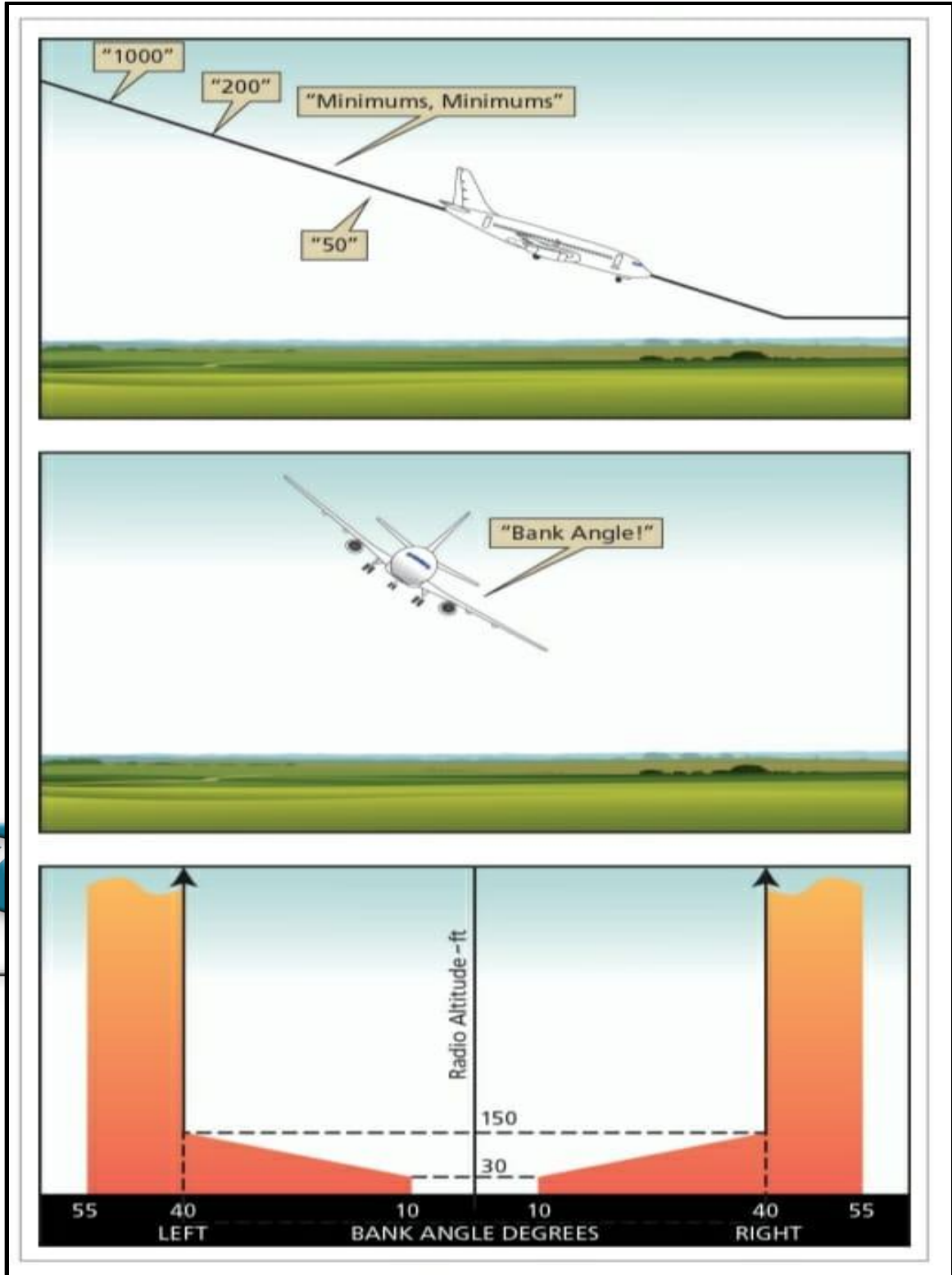
ADDITIONAL MODES OF EGPWS

- ✓ Mode 6

A. 6A- Alert (Minimums - Minimums)

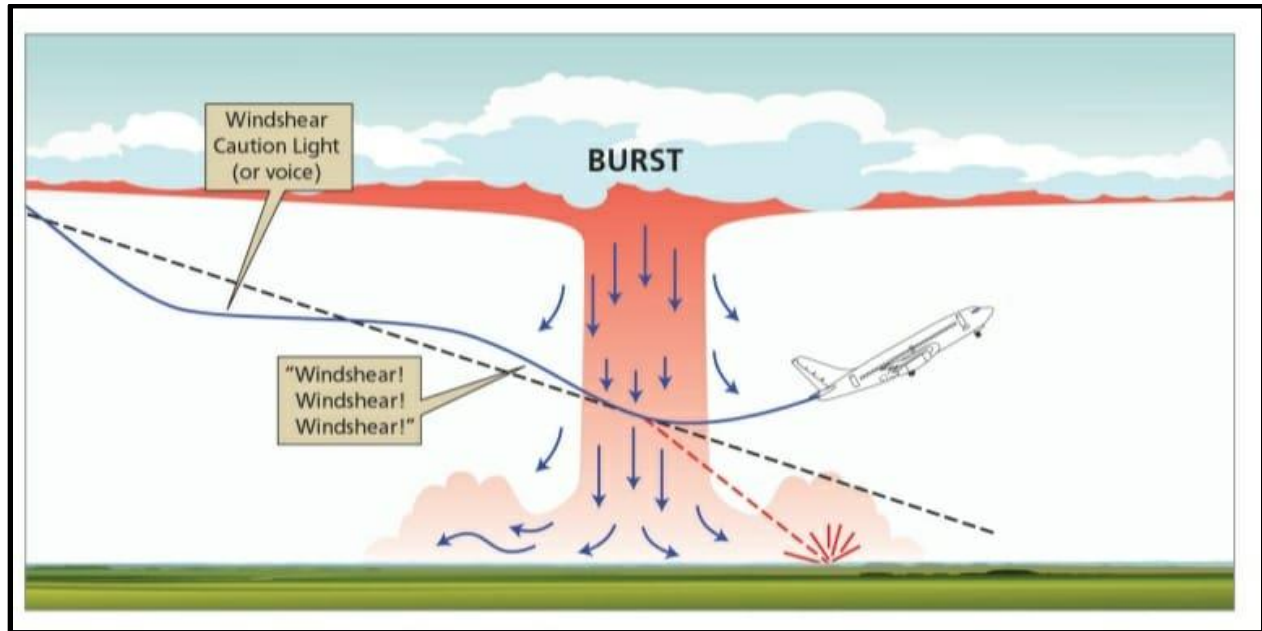


B. 6B – Alert (Bank angle – Bank angle)



✓ Mode 7 – Wind Shear

- ✓ Alert – Wind Shear
- ✓ Warning – Wind Shear



Note: Mode 5

Soft Alert – Glide slope – Glide slope

Alert sound at a low pitch when a/c is deviated 1.3 dot below the G/S.

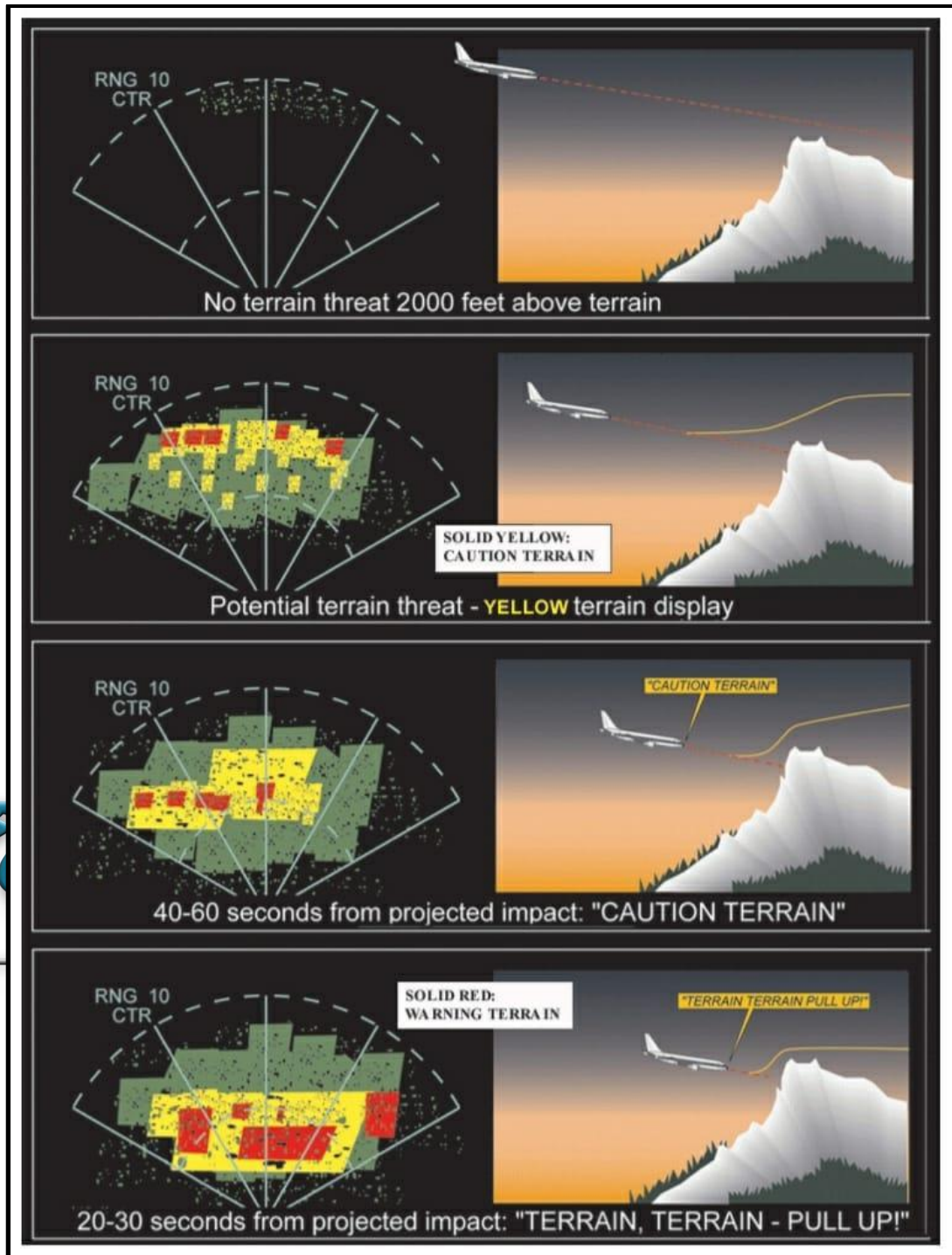
Hard Alert – Glide slope -2 sounds with high pitch when a/c is deviated below the half of full scale deflection.

EGPWS DISPLAY:

On the EFIS the display is given in different colours. The colour codes are:

- ☞ Red - Danger
- ☞ Yellow - Caution
- ☞ Green - Safe

☞ Magenta - Unknown Terrain



- ✓ Terrain display is up to the ranges selected on the display. All terrain above the ac is necessarily displayed.

- ✓ Terrain below up to 2000 ft below is displayed. Other display features are:
 - A. Terrain more than 2000' above the ac is shown as red dots
 - B. Terrain between 1000' and 2000' above ac is shown as close yellow dots.
 - C. Terrain between 1000' above and a pre-programmed ht below the ax is shown as widely spread yellow dots
 - D. Terrain up to 1000' below the ac is shown as green dots
Terrain between 1000' and 2000' below the ac is shown as widely spread green dots.

- ✓ **ALERT** comes 1 min before collision and terrain display changes to solid yellow.
- ✓ **WARNING** comes 30 secs before collision and terrain display change to solid red. Pilot action is to get wings level, apply max T/O power and climb at airspeed and angle for max climb gradient till warnings is eliminated.



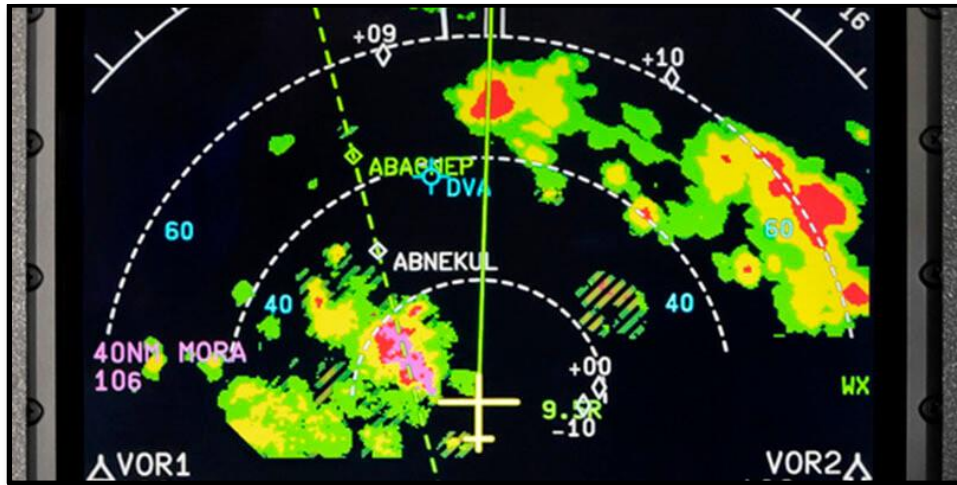
WIND SHEAR DETECTION:

- ✓ Only below 1500' from ground level
- A. **Alert:** given when increasing head winds or decreasing tail winds cross a given threshold. A CAUTION light will illuminate. Audio alert is “caution wind shear”.
- B. **Warning:** given when decreasing head winds or increasing tail winds cross a given threshold. A red warning light will illuminate. Audio alert is “wind shear” three times.

- ✓ In case you have a weather radar, wind shear warning will come on this as expanding bands of colour across the screen. Wind shear causes the pressure instruments to misread (especially ASI).
- ✓ In case HW shears (CHANGE) to TW, the IAS will reduce, the pitch attitude will decrease and the ac will go down. Pilot will increase power to counter the situation.
- ✓ In case TW shears (CHANGE) to HW, the IAS will increase, the pitch attitude will reduce and the ac will go above GP. Pilot will reduce power to correct the situation. This is more dangerous than the first situation as when you get out of the situation you are left at low power situation.



AIRBORNE WEATHER RADAR (AWR)



PRINCIPLE – Echo & Search Light

Echo – Range

Search Light – Bearing of the object

Frequency – 9375 MHz

Band – SHF

Wave length – 3 cm

Range – 150 NMs and on EFIS 300 NMs

- ✓ AIRBORNE WEATHER RADAR (AWR) is used to provide the pilots with information regarding both weather ahead as well as navigation. It requires interpretation by the pilot and its use is enhanced by the skill of the user.

FUNCTIONS

- ✓ Detect the size of water droplets and hence deduce where the areas of turbulence are within the cloud.
- ✓ Determine the height of cloud tops by tilting the radar beam up or down.
- ✓ Map the terrain below the aircraft to provide navigational information and high ground avoidance.
- ✓ Provide a position fix (range and bearing) from a prominent feature.
- ✓ **AWR** is primary radar and both of its functions, weather detection and ground mapping use the echo principle to depict range and the searchlight principle to depict relative bearing of the targets.

OPTIMUM FREQUENCY

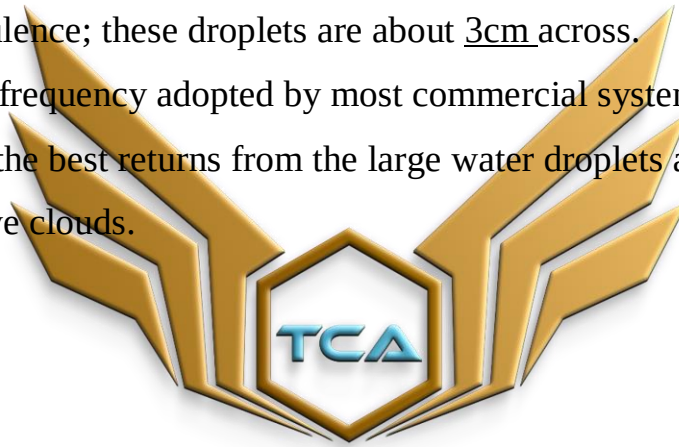
- ✓ Is the frequency which produces wave length equal to the size of the target which needs to be detected.
- ✓ 3 cm W.V or 9375 MHz is used in AWR to ensure that large water droplets as well as small water droplets are detected.
- ✓ Types of Beams
 1. Pencil beam or conical beam –Is used for weather and longer range (>60nm) mapping.
 2. Fan Shaped beam or (Cosecant)²- Is used for short range mapping.

AWR AERIALS

- ✓ The radar beam is produced by antenna in the nose of the aircraft. It is usually necessary to tilt the antenna down when using the radar in the mapping mode.
- ✓ Aerials are fitted in the random and stabilized in pitch & roll axis of the a/c.

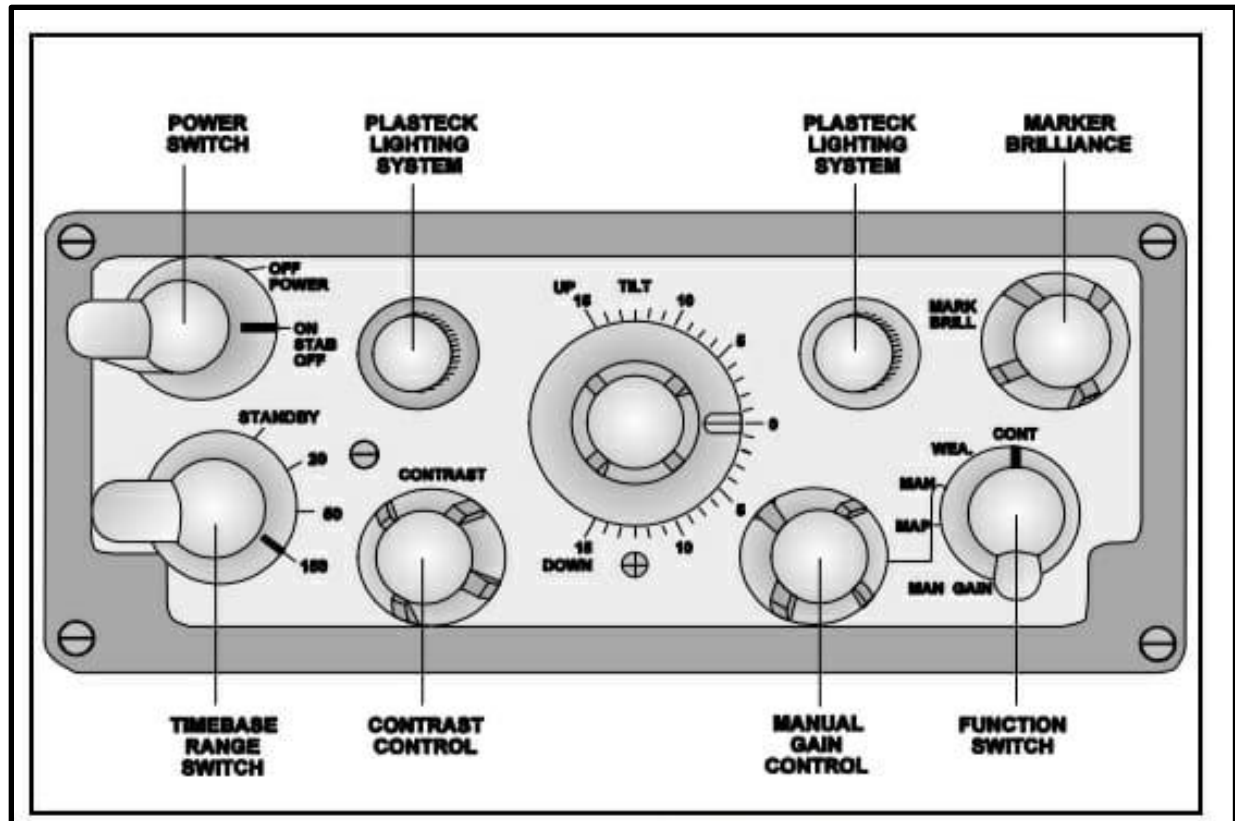
RADAR FREQUENCY

- ✓ The large water droplets and wet hail which in turn are associated with severe turbulence; these droplets are about 3cm across.
- ✓ The typical frequency adopted by most commercial systems is 9375 MHz, as it produces the best returns from the large water droplets and wet hail found in convective clouds.



CONTROLS





POWER SWITCH

- ✓ In the ON position the system is energized and the aerial is automatically stabilized in PITCH and ROLL.
- ✓ Switching to the STAB OFF position will lock the scanner to the pitch and roll axes of the aircraft.

RANGE SWITCH

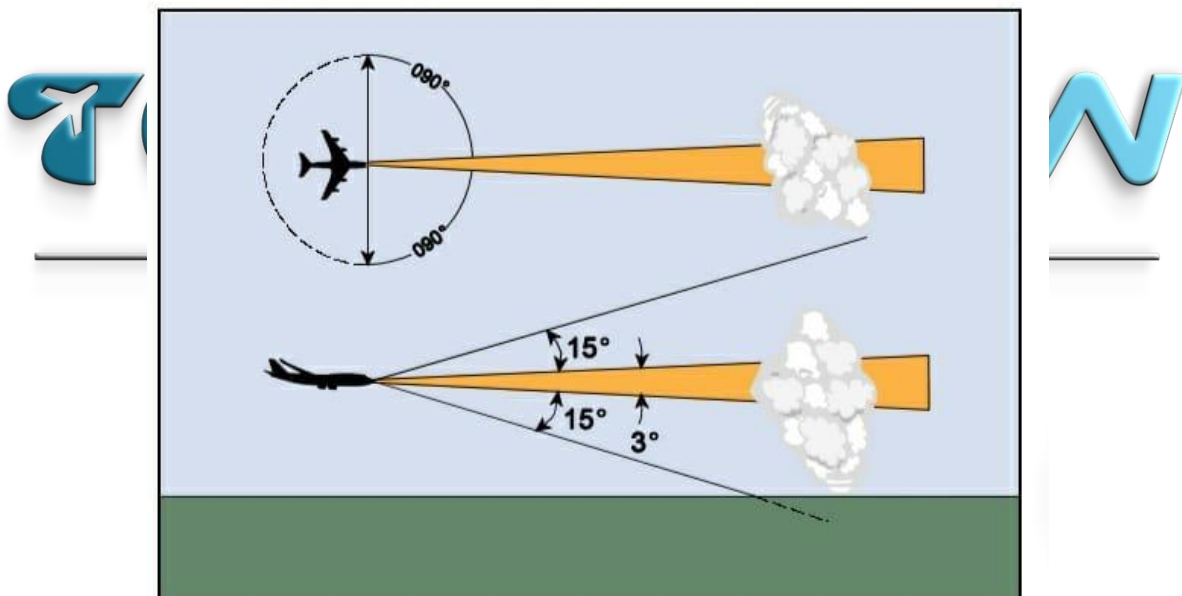
- ✓ Selection of a range position energizes the transmitter.
- ✓ AWR should be kept on STANDBY position till the time a/c is clear of personals, trees, and hanger because the radiation from AWR aerial is harmful for human being.

- ✓ The radiation can damage health and the reflections from adjacent structures can damage the equipment. It can be turned on with a uplift selected on Weather mode for ground testing
- ✓ The maximum practical range for weather and for navigation is in the region of 150nm. Now up to 300 NM in modern equipment's.

Note- AWR can be turned on the ground at the time of lineup with adequate safe guard.

BEAM TILT CONTROL

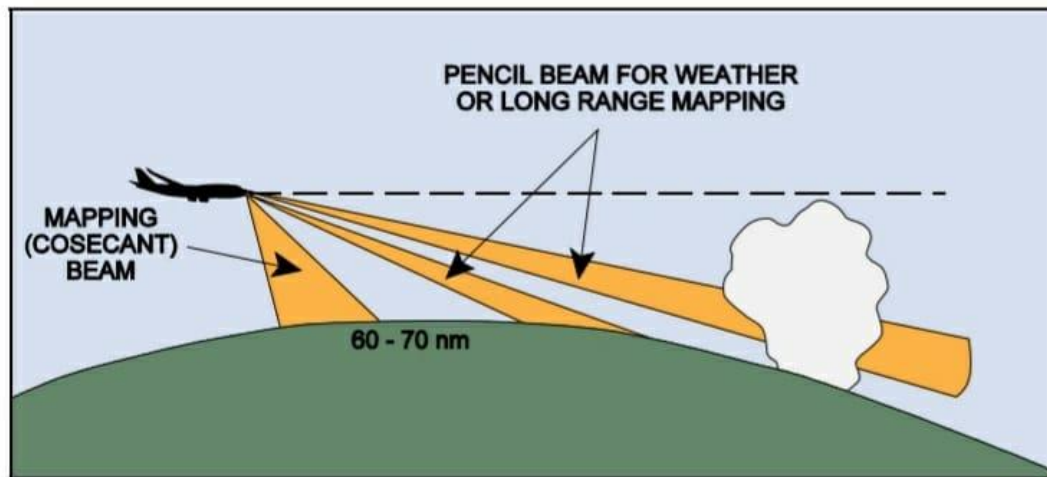
- ✓ This control enables the radar beam to be tilted from the horizontal within 15° up (+) and 15° DOWN (-).
- ✓ +/- 90° lateral sweep.



Uses of AWR

1. Weather mapping
2. Ground mapping
3. To find a cloud top
4. Plotting a fix

1. MAP MODE



- ✓ The radar produces a mapping beam. Presentation of surface features the transmitted power is progressively reduced as decreases so that the power directed to the closed object is minimum so the return doesn't destroy antenna.
- ✓ This reduction in power with decreasing range is a function of the cosecant of the depression angle – hence the name cosecant beam; another description is “fan-shaped” beam. “Minimum (15 nm) and maximum (60 to 70 nm)”.
- ✓ Signal amplification is adjustable via the adjacent manual gain knob.

2. MAN MODE

- ✓ To map beyond 60 to 70 NMs the conical pencil beam should be used by selecting the MAN MODE.
- ✓ PENCIL BEAM FOR WEATHER OR LONG-RANGE MAPPING.
- ✓ This is used for cloud detection and mapping between about 70 and 150nm and selects the conical pencil shaped beam.
- ✓ MANUAL GAIN for signal amplification is operative with this selection.

3. WEAMODE

- ✓ This selects the conical pencil beam and is the usual position for observing cloud formations; **MANUAL GAIN control is now INOPERATIVE.**

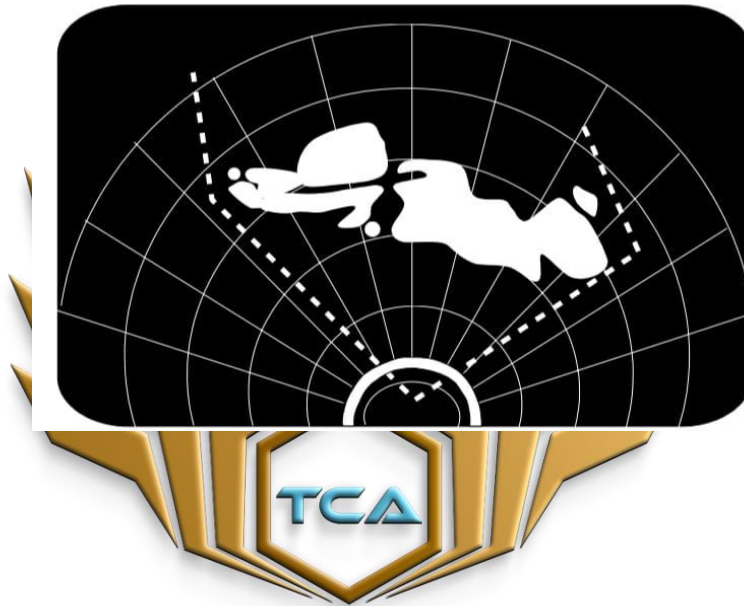


SWEPT GAIN TOP CREW AVIATION

- ✓ Sensitive Time Control or Automatic Gain Control (AGC) is automatically available. This system of circuits decreases the gain of echoes received from the ever-decreasing range of clouds.
- ✓ Operates up to about 25nm and ensures (brilliance) of display of a particular cloud is independent of range.
- ✓ Thus, a small cloud at 5 nm does not give an increasingly stronger return than a larger and more dangerous cloud at 20 nm; all clouds up to about 25 nm are thus compared on equal terms.

4. CONTOUR

- ✓ Is a cloud formation presentation with CONT (CONTOUR) selected for a black and white display?
- ✓ The black holes indicate dangerous areas of concentrated Rainfall and potential turbulence.



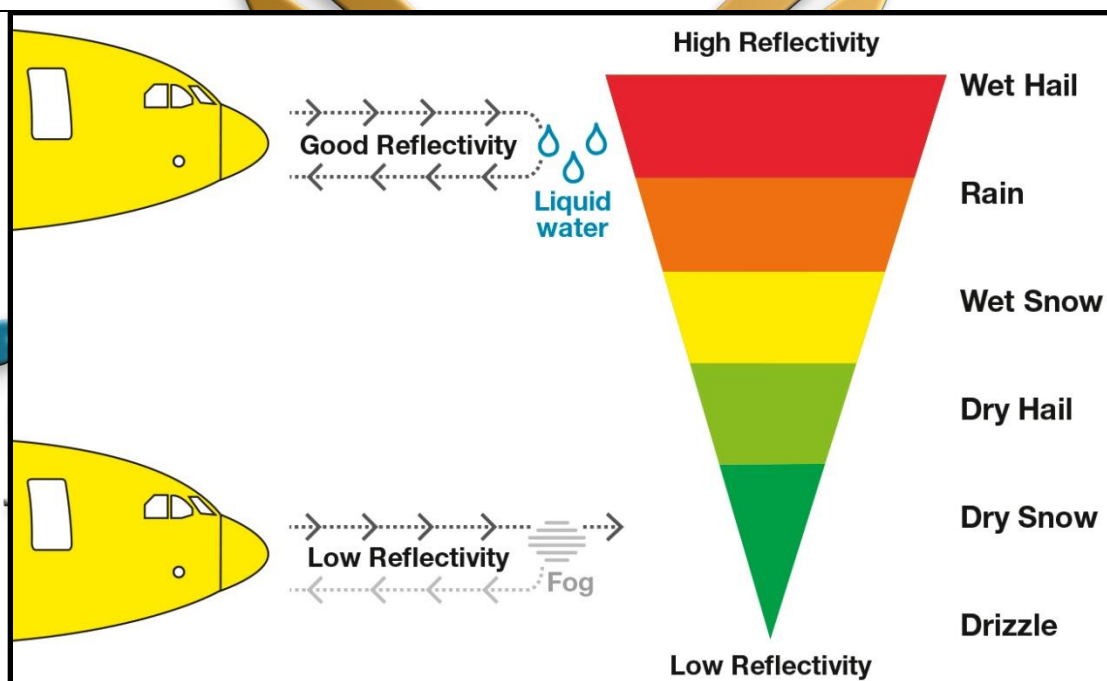
- ✓ Is used to indicate storm intensity, turbulence and areas to avoid.

AVOIDING THUNDERSTORMS

- ✓ Below 20,000ft - min 5 nm
- ✓ 20-25,000 ft - 10 nm
- ✓ 25-30,000ft - 15 nm
- ✓ Above 30,000 ft -20 nm
- ✓ Vertically avoid by 5000 ft

AWR COLOUR DISPLAY OF EFIS AS BELOW:

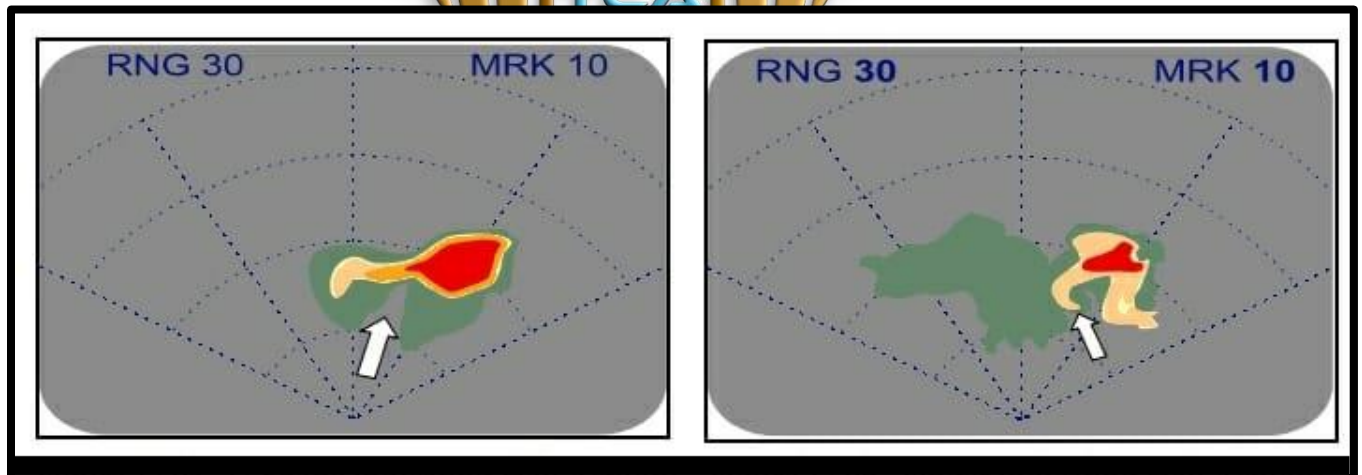
COLOUR	PILOT INTERPRETATION	AMOUNT OF RAINFALL
BLACK	Very light or no returns	Less than 0.7 mm/hr.
GREEN	Light returns	0.7 – 4mm/hr.
YELLOW	Medium returns	4 – 12mm/hr.
RED	Strong returns	Greater than 12mm/hr.
MAGENTA	Turbulence	Due to rainfall intensity

**Note:**

1. Rapid change in colour, steep colour gradient also signifies presence of severe turbulence.

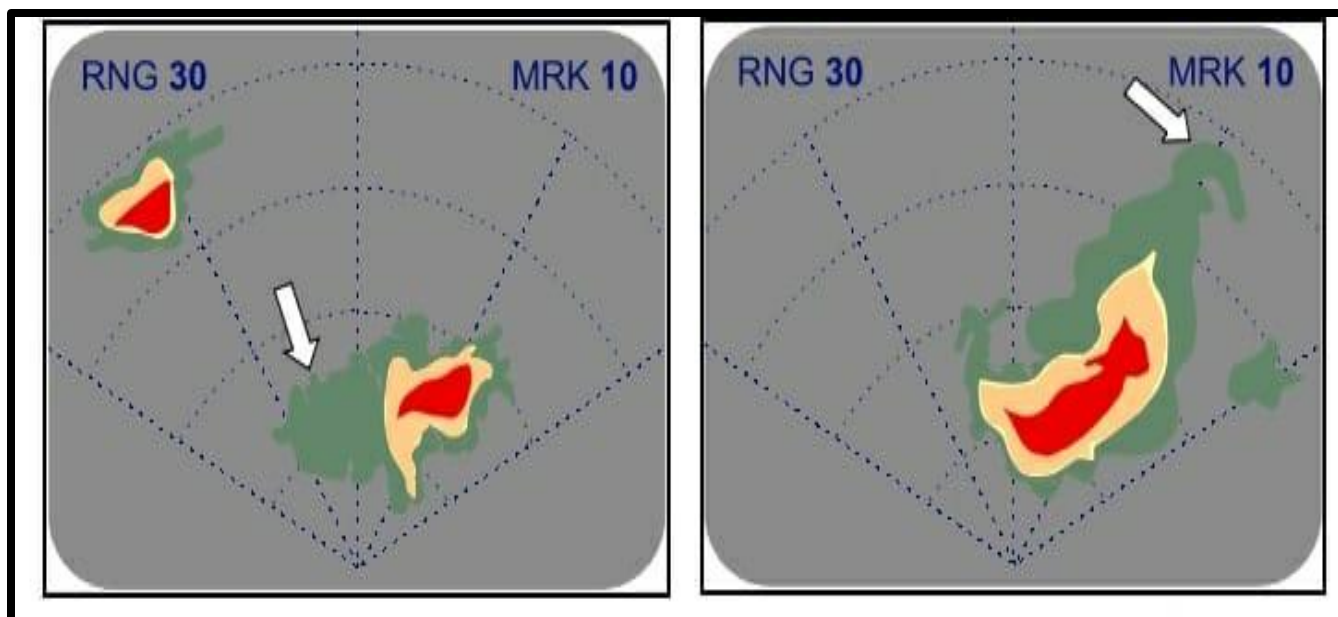
2. Black holes on a white background also signifies severe turbulence on a CRT.(Cathode rays tube)
3. U shape, Finger Shape, Hook shape also indicates presence of severe turbulence on a black and white screen.

The areas of greatest potential turbulence occur where the colour zones are closed together i.e. the steepest rainfall gradient.



Scalloped Edges

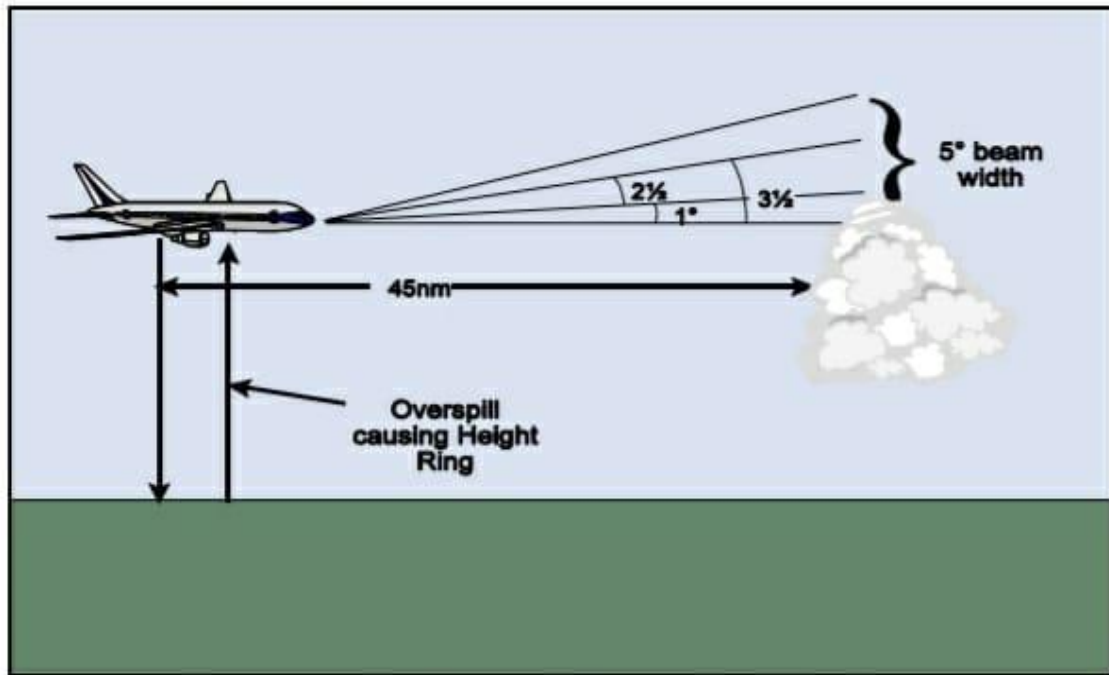
Hooks



HOW TO FIND THE CLOUD TOP



- ✓ For ht of cloud top above the ac: the angular elevation of the cloud top can be calculated by aerial tilt – $\frac{1}{2}$ BW.



- ✓ Less energy gets reflected from sea. This makes the coastline very distinct. Bright echoes come from built up area and tall buildings. The area behind a hill cost a shadow behind it giving it weak returns. This gives a false image of a water body behind the hill.

$$\text{Relative Height of the cloud} = \frac{\text{Range} \times (\text{Tilt} - \frac{\text{Beam Width}}{2}) \times 100}{2}$$

$$\text{Cloud Top} = \text{P.A.} \pm \text{Relative height of cloud}$$

ELECTRONIC FLIGHT INSTRUMENT SYSTEM

Electronic flight instrument system is basically combination of two LCD screens where all the information related to navigation and altitude of aircraft can be find, which reduces pilot scanning time.



COMPONENT OF EFIS

- ✓ Presents information to pilot on 2 electronic displays:
 1. Attitude information (EADI) – PFD
 2. Navigation information (EHSI) – ND
- ✓ EFIS Computer
- ✓ Symbol generator

EFIS has 2 displays, one for the captain and the second one for the first officer.



PFD (Primary Flight Data) OR EADI

All the information related to altitude or piloting is available on PFD, as below:

1. Airspeed
2. Altitude
3. Vertical Speed
4. Decision Height
5. Auto Thrust mode
6. Auto Pilot Mode
7. Flight Director Bar



ND (Navigation Display) OR EHSI

- ✓ All the information related to navigation is displayed.
- ✓ Modes of ND:
 1. Full VOR Mode
 2. Full ILS Mode
 3. Full Expanded VOR Mode

4. Full Expanded ILS Mode
5. MAP Mode
6. Plan Mode



- ✓ In expanded mode only 90% of compass ARC is available white triangle at the bottom represents the aircraft.

1. Current Heading (Colour White)
2. Selected Heading (Magenta)

- ✓ VOR deviation bar/ILS deviation bar/Glide slope deviation bar are in magenta colour.
- ✓ Current track in white and selected course would be in magenta.
- ✓ Active way points of PLN in magenta.

Note: AWR information is available only in all expended mode and NAV or MAP mode.

- ✓ Wind information is also available on navigation display.
- ✓ The plan mode is only mode which is true north oriented.
- ✓ Wx radar is not available in full VOR, full NAV, full ILS and plan mode.
- ✓ EHSI compass rose can be performed true or magnetic north between 73 N and 60 S but above this latitude it can be referred true north.
- ✓ Wind is not displayed in plan mode.
- ✓ Any malfunction in VOR or ILS it will come magenta deviation bar to disappear from display.

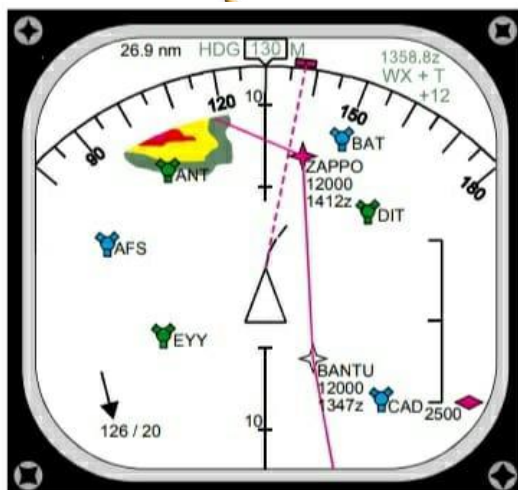
SYMBOL GENERATOR

- ✓ Works as an interface between EFIS computers and display it converts the analog information into digital information. There are three symbol generator.

1. For ND
2. For PFD
3. Other is STAND BY

EFIS COLORS

- ✓ Present situation (Current Heading) - WHITE
- ✓ Flight to command (Selected Heading) – MAGENTA
- ✓ ARM Mode – BLUE
- ✓ Engage Mode - WHITE





TOP CREW AVIATION

FLIGHT MANAGEMENT SYSTEM

1. IMPROVED NAVIGATION
2. IMPROVED FUEL EFFICIENCY
3. REDUCED CREW WORK LOAD

FMC produces automatic navigation guidance and performance management. The flight management computer basically generates two types of guidance:



LNAV (LATERAL GUIDANCE)

- ✓ Normally great circle tracks between way points. It includes takeoff SID, Transition SID, To top of climb than transition from TOD to STAR, STAR to Instrument app, GO around and diversion.

V-NAV (VERTICAL GUIDANCE)

- ✓ It includes vertical profile ROC/ROD, Speeds etc.
- ✓ FMS contains NAV database and performance database.

NAV DATABASE

- ✓ It includes NAV aid frequency, elevation, runway length, SID, STAR, company routes, airspace, designated reporting point, holding pattern, missed app procedure and approach procedure.

Note:

1. The period of validity for navigation database is 28 days and this validity cycle is called as AIRAC.
2. NAV data for crew is READ ONLY.
3. There are 2 shades of database one is for first 28 days and other is for next 28 days.

PERFORMANCE DATABASE

- ✓ It includes climb PERF, thrust limits, fuel flow, and optimum altitudes.

CDU (CONTROL DISPLAY UNIT)

✓ It is an interface between crew and FMC.

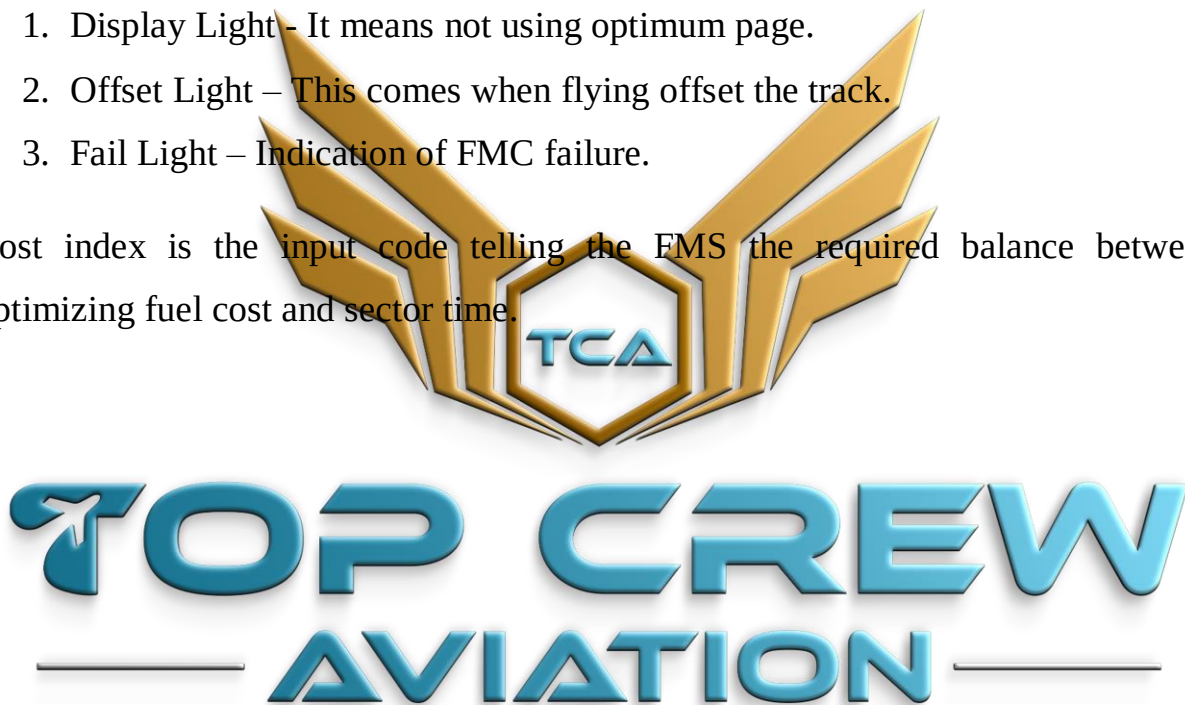
STARTUP SEQUENCE

IDENT	POS INIT	ROUTE	PER INIT
-------	----------	-------	----------

CDU Enunciator Light

1. Display Light - It means not using optimum page.
2. Offset Light – This comes when flying offset the track.
3. Fail Light – Indication of FMC failure.

Cost index is the input code telling the FMS the required balance between optimizing fuel cost and sector time.



RNAV (AREA NAVIGATION)

PBN (Performance based Navigation)

- ✓ Is a method of navigation in which the accuracy is determined on the basis of centricity, functionality, integrity, availability.
- ✓ PBN has two types:
 1. RNP (Required navigation performance)
 2. RNAV (Area Navigation)

RNP

- ✓ It is a method of navigation bases of Area Navigation in which on board performance mandatory & alerting system is mandatory.

RNP-RNAV Designators

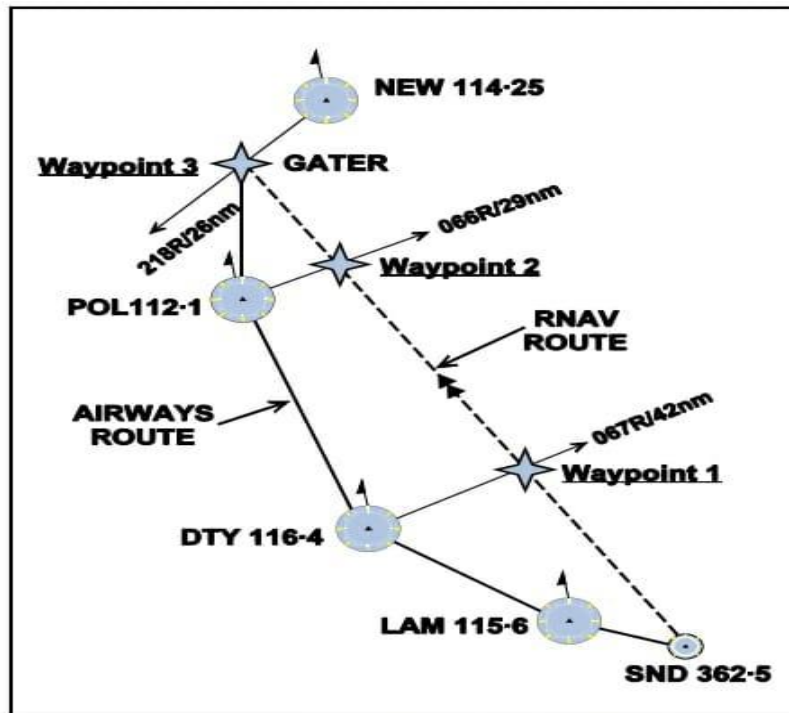
- ✓ This is in numbers; let's say RNP-5 which means a/c must be with 5 NMs longitudinally and laterally from its position 95% of the time.

R-NAV

- ✓ This method of navigation permit aircraft operation on any desired route (shorter route) with in coverage of radio navigation without over flying them.



**TOP CREW
AVIATION**



There are three levels of RNAV:

1. **2D** – ONLY HORIZONTAL PLANE
2. **3D** – HORIZONTAL AND VERTICAL PLANE
3. **4D** – HORIZONTAL, VERTICAL AND TIME BASED

Types of RNAV:

B-RNAV (Basic RNAV) – In basic RNAV accuracy should be 5 NM 95% TIMES.

P-RNAV (Precision RNAV) – In this accuracy should be 1 NM 95% TIMES.

APPROACH .5

Note: For RNAV onboard performance monitoring and alerting system is not mandatory

RNAV CALCULATES POSITION WITH REFERENCE TO

1. VOR-VOR
2. VOR-DME
3. DME-DME (BEST FIX)
4. ADF-VOR
5. INS-IRS
6. GPS

Note: - The most accurate RNAV position achieved by using DME-DME

ADVANTAGE OF RNAV

- ✓ Less distance, less time, direct route, less fuel, less cost.

PHANTAM STATIONS- way points defined with respect of radial and distance.

Example- GATER

In case the aircraft is outside coverage RNAV shift to DR mode (dead reckoning mode) i.e. (last speed, position)

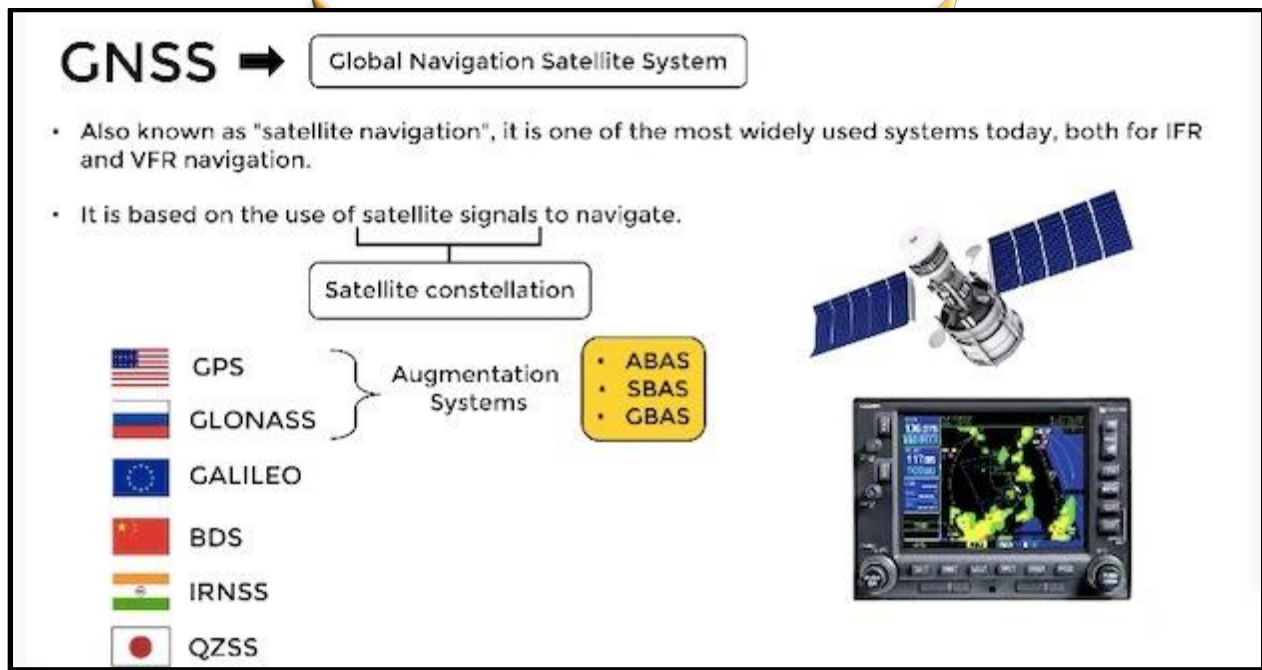
HIS (Horizontal Situation Indicator)

- ✓ Full scaled deflection is 5NMs during en route
- ✓ Full scaled deflection is 2.5NMs during approach

SATELLITE NAVIGATION SYSTEM

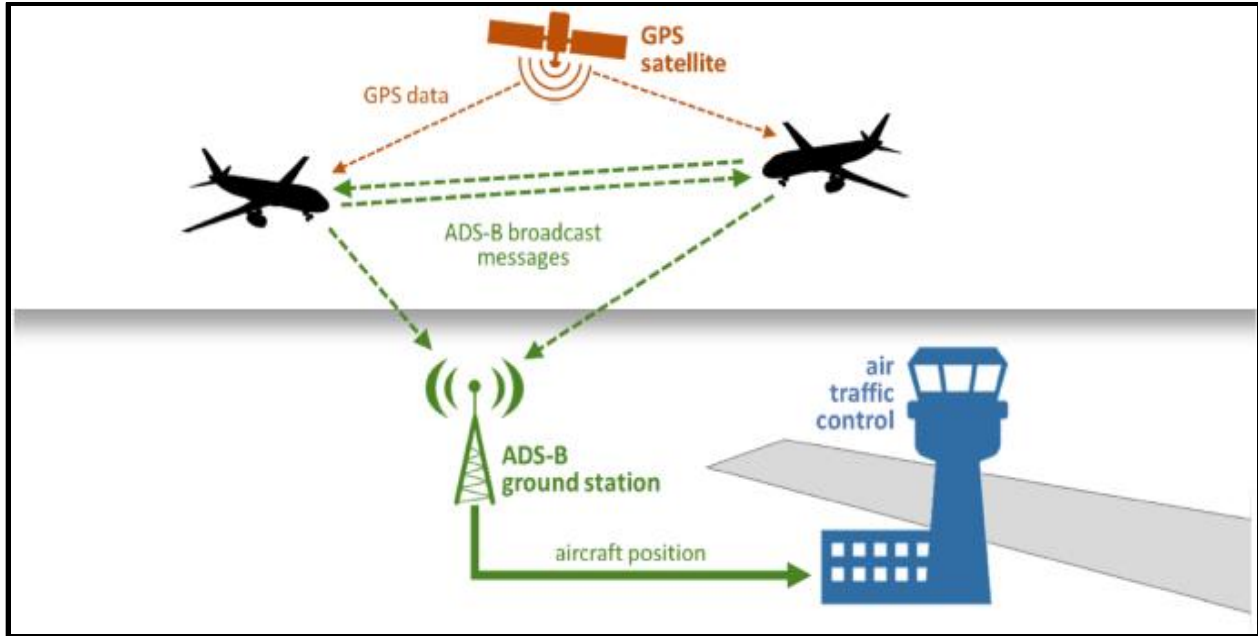
- ✓ To calculate a 3D position in space you need at least 4 satellites giving you range. In case only 3 satellites are available then only latitude and longitude can be derived and the ht can be manually fed. Using a barometric altimeter.

USA – (GPS) Global Position System



ICAO (Europe) – (GNSS) Global Navigation Satellite System

RUSSIA – GLONASS – Global Orbiting Navigation Satellite System



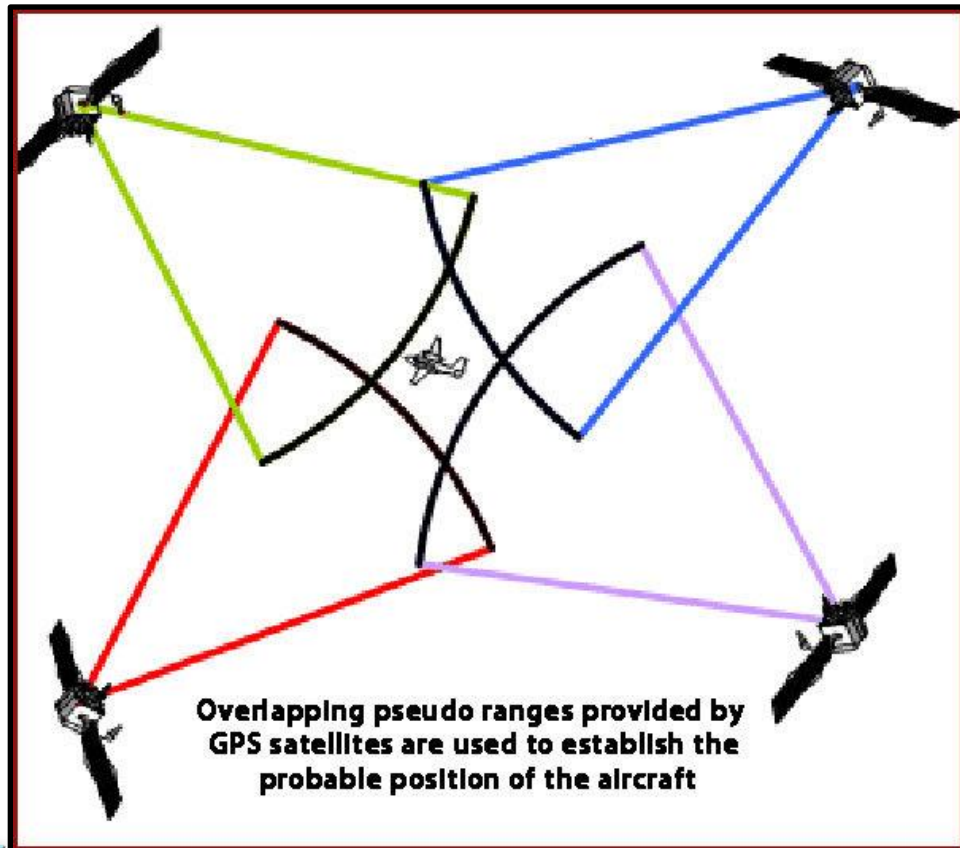
GPS

- ✓ 21 active satellite systems. Plus 3 spare
- ✓ Transmission UHF (L BAND)
- ✓ Position fixed by range position line
- ✓ Satellite send their position and signal
- ✓ Receivers at aircraft note time difference between transmission and reception
- ✓ Distance measured with time difference and thus range position lines are obtained

PRINCIPLE OF OPERATION:

- ✓ The global navigation satellite system (GNSS) is satellite based and uses principle similar to DME.
- ✓ The U.S.A system is called Global Positioning System (GPS) and the Russian system is called Global Navigation Satellite System (GLONASS)

- ✓ The Rx measures the time taken for a radio signal to travel from a Tx in a satellite at a known point in space.



With this time distance can be calculated by the Rx computer.

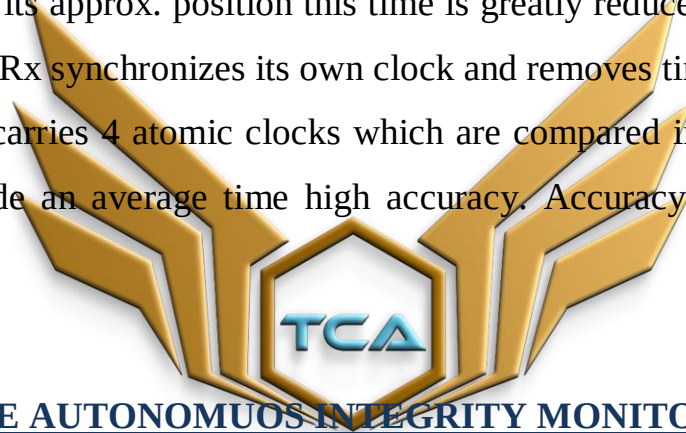
SPACE SEGMENT:

- ✓ The GPS also called NAVSTAR uses 21 main and 3 spare satellite. They fly in 6 different orbits at 20,200 kms (10900nm) above the earth taking 12 hours for each orbit.
- ✓ The orbit is inclined at 55 deg to the earth's equatorial axis.
- ✓ The satellite transmits on 2 different frequencies.

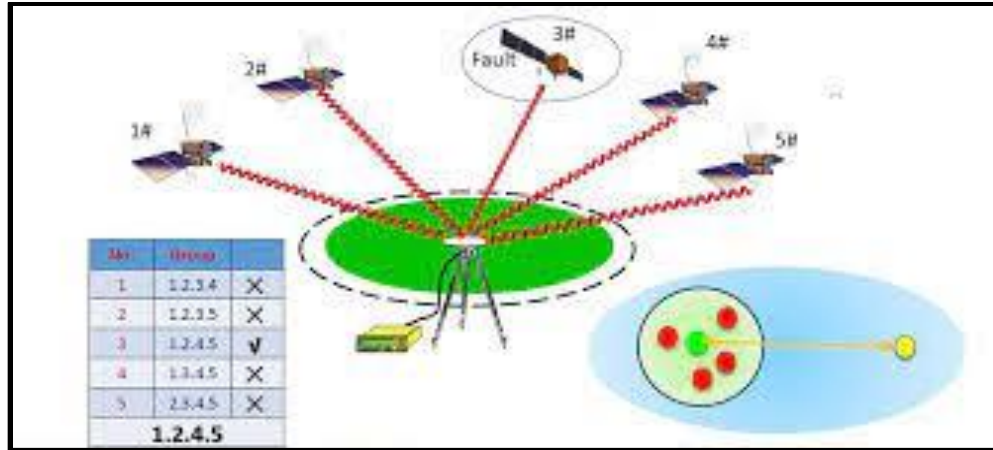
- ✓ The first one is called the L1 band transmitted at 1517.42 MHz called the coarse acquisition or CA code. This provides the SPS or standard positioning service and can be used by all users.
- ✓ The freq also transmits at 90 deg phase difference a P code for military uses. The precise code or P code is called the L2 band transmitted at 1227.6 MHz providing the PPS or precise positioning service. The P code can be encrypted and is then called the Y code.
- ✓ The transmission from each satellite includes the following: identification, orbit and the position in the orbit. (The orbital position is called 'almanac'). From which satellites to expect for getting ranges.
- ✓ But the ranges that you get are called pseudo ranges and are not accurate. This is due to the fact that the satellite uses an accurate atomic clock while the Rx uses an inaccurate quartz clock. So the ranges have to be made more accurate and the time bias error to be removed.
- ✓ To remove this Rx takes fix from 3 satellites at a time (say satellite no. 1, 2, 3). Then the set of next three (2, 3, 4), and then the next three (3, 4, 1) and finally the last three (4, 1, 2) with this the Rx makes changes to its clock and removes the time bias. This gives an accurate position.
- ✓ Each satellite carries 4 atomic clock which are compared internally with each other to provide an average time high accuracy. Accuracy is of the order of Nano-seconds.

USER SEGMENT (RECEIVER)

- ✓ When Rx is switched on, it searches for signals on the correct freq. Rx which has no knowledge of its own position must download full set of almanac of all satellite.
- ✓ Once that is done it loads its computer with this info. When the receiver searches it selects those satellites whose signal it can receive.
- ✓ Then it generates its own pseudo random code and searches for the satellite with similar code. When both codes match the Rx locks on to the satellite and now locks for the next satellite and then the next and so on. If the satellite already knows its approx. position this time is greatly reduced. Once it receives 4 satellites the Rx synchronizes its own clock and removes time bias.
- ✓ Each satellite carries 4 atomic clocks which are compared internally with each other to provide an average time high accuracy. Accuracy is of the order of Nano-seconds.

**RAIM (REMOTE AUTONOMOUS INTEGRITY MONITORING)**

- ✓ It requires 5 satellites to determine if one of the satellites is wrong and with 6 satellites the satellite giving wrong info can be excluded (called FDE-failure detection and exclusion). This capability is called RAIM.
- ✓ If 5 satellites are not received a warning light will glow informing no RAIM.
- ✓ This is required as there is no warning in the receiver light will glow informing no RAIM if there is a degradation in one of the satellite leading to inaccurate position due to wrong calculation.



RAIM THUS PROVIDES A WARNING SYSTEM

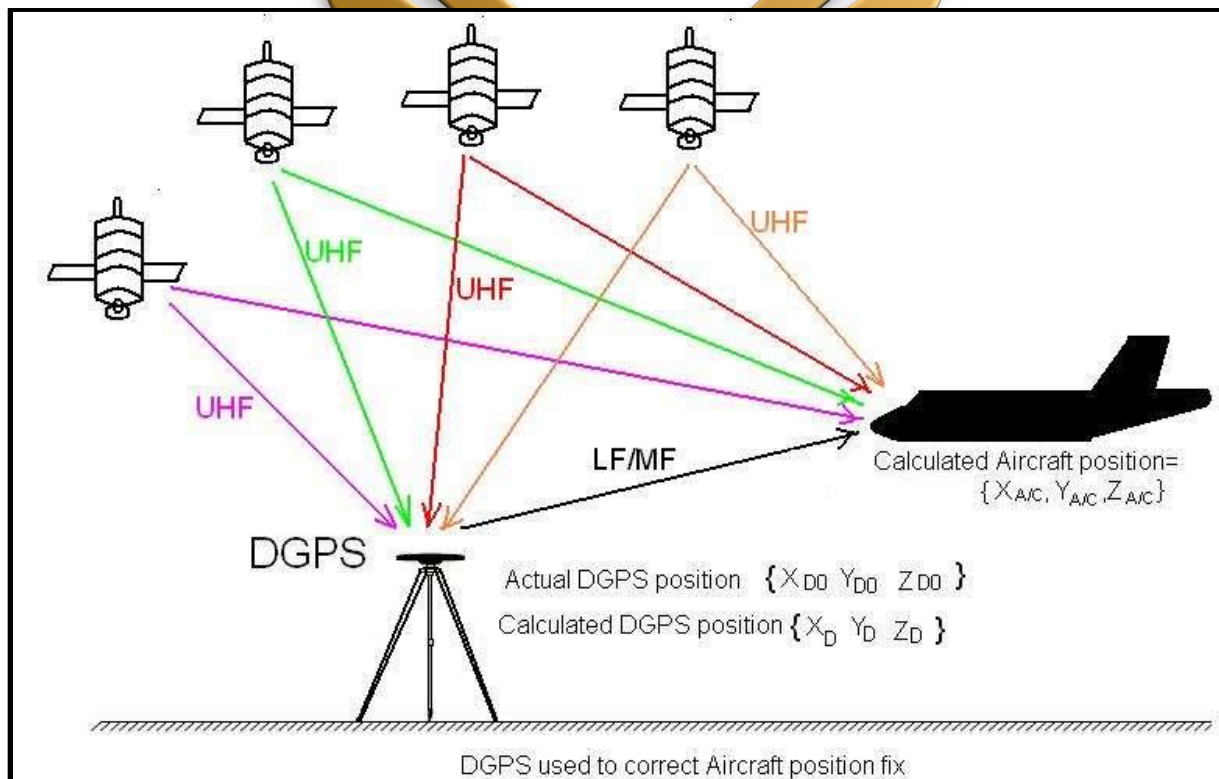
ERROR OF GPS:

- ✓ **Clock time bias error** (or time bias error)
- ✓ **Satellite clock error:** error in the atomic clock of the satellite. Gives inaccuracies up to 1.5 m
- ✓ **Satellite ephemeris error:** satellite may not be where it is saying it is due to disturbance from the planned orbit
- ✓ **Selective availability:** the error deliberately introduced by the US to protect its own interests
- ✓ **Ionosphere refraction:** there is a small amount of refraction in the ionosphere causing ionosphere group delay.
- ✓ **Interference:** caused by interference from signals on similar freq as the freq received by Rx from the satellite is very weak (due to distance between the sat and the Rx)
- ✓ **Geometric dilution of position:** between satellites giving position is around 60 deg. But when the satellites give a poor cut at a very large angle, there is a dilution in the accuracy of the derived position.

- ✓ Best cut is when one is overhead and other 3 are at 120° apart.

DGPS (DIFFERENT GPS)

- ✓ These are ground stations that know their own position very accurately. When they receive the GPS derived position, they know the error that is there of this GPS derived position in comparison to the real position.
- ✓ This correction is then transmitted on a known freq. Rx thus receives this correction and feed it to the FMS which corrects the GPS derived position suitably.
- ✓ The range is 70 nm.



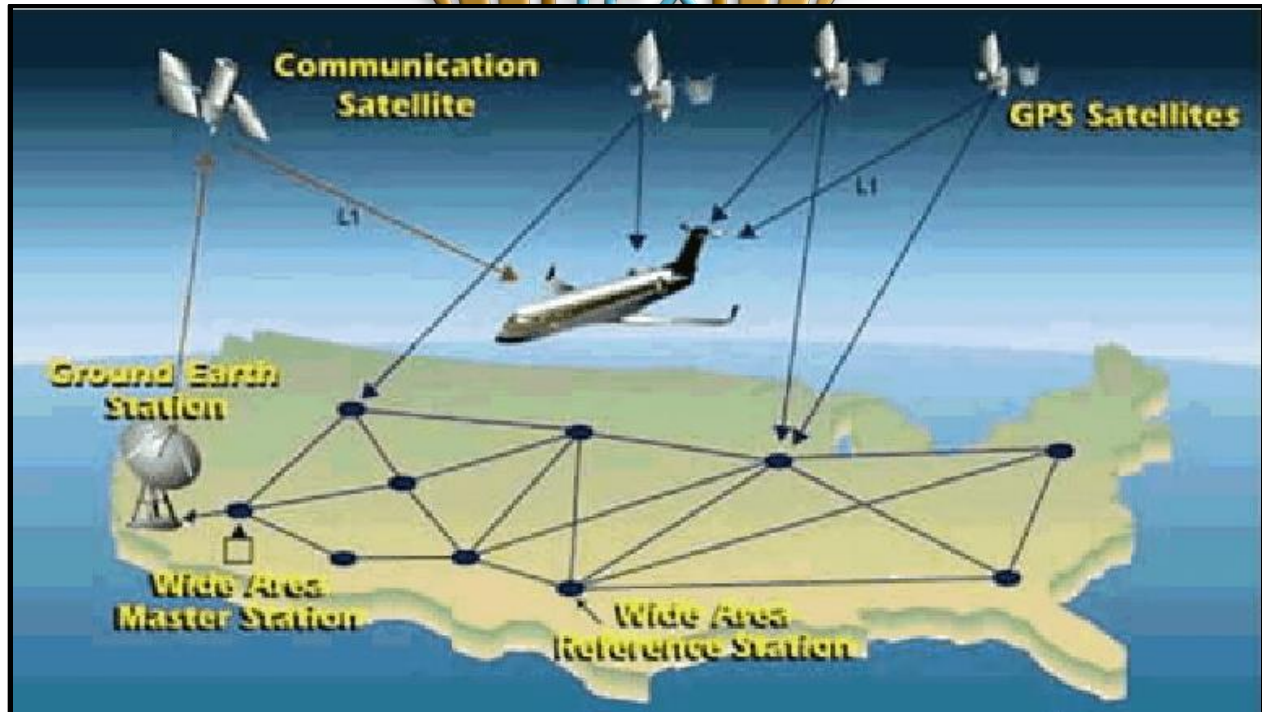
So the Dgps can remove selective availability error and group delay. But it cannot remove GDOP error. A major advantage is that it can provide external integrity monitoring and can supplement RAIM.

ACCURACY:

- ✓ In CA code accuracy is 30 m on 95% of the occasions and 1600 ft on 99.95% of the occasions.
- ✓ There are 2 types of DGPS system for providing certain services. There are:

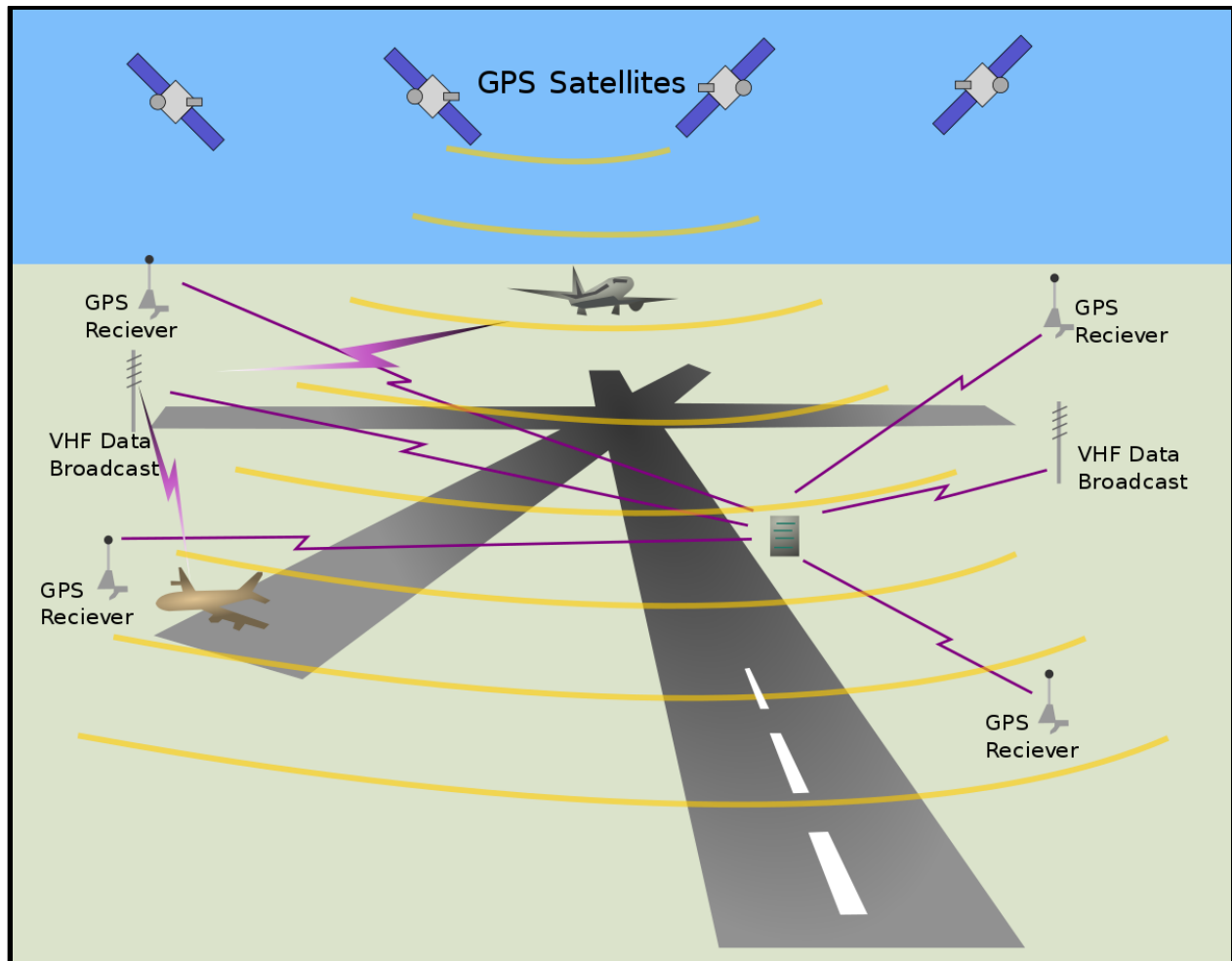
1. WAAS (WIDE AREA AUGMENTATION SERVICE):

- ✓ Meant for giving accurate enroute navigation info.



2. LAAS (LOCAL AREA AUGMENTATION SERVICE):

- ✓ Meant for giving accurate navigation info for approach and landing. Presently gives info up to cat I ILS. Planned to upgrade to cat III ILS in the near future. This is holding the sale of MLS.



GLONASS: this is the Russian satellite system. It uses separate freq for each of its satellite inclined at 60 deg. They orbit the earth every 12 hrs at an altitude of 19,000 kms. There is no selective availability facility in the system making it globally available.